# Computational Supplement:
Frame covariance of CMBR lensing in the exact $1+3$ covariant radiation hierarchy

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Supplement v1.3.20 — 30 October 2026
Paper 1 verification ledger and chapter layout; computational v1.3.0-dev

**Abstract**

This supplement accompanies *Frame covariance of CMBR lensing in the exact* $1+3$ *covariant radiation hierarchy*, Paper 1 (Gebbie, 2026), and documents an independent Python reconstruction of the almost-Friedmann–Lemaître calculation of Gebbie et al. (2000). The calculation is developed from the $1+3$ equations rather than adapted from an existing Boltzmann code. Part I sets out the purpose, current situation and future situation. Part II, *The Physics*, develops the physics chain from the hierarchy and source terms to the acoustic oscillator, free streaming, matching, the line-of-sight solution and the angular spectrum. Part III records numerical resolution, parameter choices and interpretive limits. The accompanying evidence identifies fifteen findings concerning the antecedent and forty-eight concerning the implementation, without treating an inventory of findings as a certification of the physical calculation. The first-order frame law requires agreement of the two predicted skies in a common observer frame. The integration-by-parts reconstruction of the published line-of-sight solution identifies four discrepancies in the released assembly. Acoustic oscillations are present, but the first-peak position, low-$\ell$ Sachs–Wolfe plateau and secondary-peak structure are not yet recovered. CAMB and CLASS provide external comparisons, not inputs or calibration targets. The supplement makes no claim of new physics. The identifier `astro-ph/9912072v4` refers to Paper 1; the earlier versions of that identifier concern different documents, as recorded in the scientific scope.

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# Part I

# Purpose of This Document

The purpose of this supplement is to document and independently verify the mathematical and computational provenance of *Paper 1*, *Frame covariance of CMBR lensing in the exact* $1 + 3$ *covariant radiation hierarchy* (Gebbie, 2026), specifically `astro-ph/9912072v4` and the approved manuscript `TG-FrameCovariantCMBR-v1.1.6`. Paper 1 is the *provenance pivot*: its equation references, Table III conventions, source corrections and claims determine the sequence of verification. The published antecedents of Gebbie and Ellis, Gebbie–Dunsby–Ellis, and Maartens–Gebbie–Ellis are checked as evidence for those mappings, not used to silently replace them. A discrepancy in Paper 1 itself is recorded as a Paper 1 finding. Mathematical identities, independently verified claims, numerical implementation choices and unresolved physics remain distinguished throughout.

The repository's Python scripts and tests provide the existing independent verification of key derivations and claims of Paper 1. An implemented check, a passing isolated test, and a physically correct end-to-end spectrum are not interchangeable statements. Where the implementation does not yet meet an acceptance criterion, this document states the limitation at the point of use and retains its provenance in the accompanying audit files.

The software repository's version policy is *physics-facing*: releases are numbered by the successive physical terms admitted to the calculation, not by editorial, testing or formal-proof work. The scientific sequence and development statuses below follow the repository README at commit `70f0f44ff8f7 1f0313a97c17df7fcf0b83d6b753` (25 October 2026). Supplement editions such as v1.3.13 record documentation gates and are not assertions that the repository physics release has advanced.[1]

## 1 Current situation

At the pinned README revision the repository is `v1.3.0-dev`; `v1.2.0` is the last tagged release named there. The released Python reconstruction has progressed from free-streaming and PSTF mode recursions in v1.0.0, through a finite-width scattering visibility in v1.1.0, to the full linear source and frame-specialised integral assembly in v1.2.0. The present v1.3.0 scientific work concerns the acoustic oscillator, its source weighting and line-of-sight projection. Oscillations have been recovered; the physical spectrum has *not*: the primary peak placement, ordinary low-$\ell$ plateau and secondary peak structure remain unresolved, in agreement with the repository's own recorded qualification. The existing Python files verify distinct algebraic and numerical steps, rather than certifying every assumption or the complete observed sky.

The following table records the principal claims in the order of the seven-step calculation. Each is paired with its evidence and the limits of that evidence. Claims without a completed test are listed separately in §1.4. The status terms distinguish analytical statements, numerical measurements and unresolved implementation tests.

[1] Policy authority: `README.md`, "The release sequence, and what v1.5.0 needs from v1.3.0", "Current situation: v1.3.0-dev", and "Future situation: the route to the full spectrum"; repository snapshot `70f0f44ff8f71f0313a97c17df7fcf0b83d6b753`. The supplementary editorial number and the physics release number are deliberately distinct.

Table 1: Claims, evidence and limitations of the seven-step calculation.

| Step | Claim and status | Evidence and limitation |
|---|---|---|
| 1 | the exact brightness hierarchy has a covariant $1+3$ form; the displayed temperature transport equation requires reduction and time-convention checks<br>*Status:* source established; reduced form open | *Instrument:* Paper 1, Eq. (16); Gebbie et al. (2000)<br>*Does not establish:* whether the reduced equation retains all relevant couplings and factors of $a$ |
| 2 | the gravitational source closes at rank two<br>*Status:* established | *Instrument:* analysis<br>*Does not establish:* that the closure is implemented, which is §6 |
| 3 | $\ell \geq 2$ multipoles are frame invariant at first order in $v$<br>*Status:* established | *Instrument:* analysis, GDE99 (37)<br>*Does not establish:* invariance beyond first order |
| 4 | the oscillator's period is the antecedent's own, on Einstein–de Sitter where Appendix G is exact<br>*Status:* established | *Instrument:* peak spacing, 191.7 vs 192.7<br>*Does not establish:* **the phase**: it differences consecutive turning points, so the first one's position never enters |
| 5 | free streaming is reconstructed faithfully<br>*Status:* established | *Instrument:* **G1**, four external hierarchies, $\lesssim 10^{-13}$<br>*Does not establish:* anything about the spectrum; the source is switched off |
| 6 | (102) is not the exact by-parts of its own integral<br>*Status:* measured, failing | *Instrument:* **c3**, C3b, 2.55227<br>*Does not establish:* the weight error, which c3 holds fixed — B-17 (i) |
| 6 | the identity's $j'_\ell$ counterpart is absent from (102)<br>*Status:* measured, failing | *Instrument:* analysis, **T-124**<br>*Does not establish:* that it moves the peak: it adds power *below* it |
| 7 | the two routes to $C_\ell$ agree<br>*Status:* established | *Instrument:* criterion 5b, $9.5 \times 10^{-6}$<br>*Does not establish:* the weight: both sides carry $e^{-\kappa}$, and A-1/A-2 are a factor of eighty apart across the plateau |
| 7 | the first acoustic peak sits where a matched spectrum puts it<br>*Status:* measured, failing | *Instrument:* **D24**, 0.635 vs $0.757\,\ell_A$<br>*Does not establish:* its mechanism; and the two numbers use different peak estimators |
| 7 | the Sachs–Wolfe plateau is present<br>*Status:* measured, failing | *Instrument:* figure F9 (spectrum morphology); late-share diagnostic 0.0000 on both backgrounds<br>*Does not establish:* the late-share test detects the missing late-time source, not the cause of the entire plateau shape |

## 1.1 Verification layers

The checks behind Table 1 sit at four layers, and the layers differ in what kind of physical statement they are able to carry at all. Keeping them apart is not administrative bookkeeping. A count of checks that pass, taken across layers, is not a statement about the spectrum, and the clearest way to overstate this work is to report one as though it were.

A build or portability gate carries no physical claim whatever. It reads a build log and a file set and it fails on an undefined reference, a float that could not be placed, an overfull box, or a path that cannot be resolved on the machine that owns the repository. It exists because a green build that was never asked the question will be just as green next time. It settles nothing about a sign, a weight or a coefficient, and it is not represented in Table 1 at all.

A check that imports the project's own code can establish that the implementation does what the theory says. It cannot establish that the theory is right, and it is not independent of what it tests: it inherits the solver's conventions, its grids and its normalisations by construction, so a convention that is wrong in the code is wrong in the check as well. These are the repository's implementation tests. They belong to the physics release progression, and the two-frame and spectrum pipelines are still moving within them.

An instrument that imports no project code can establish a condition on the functional — a sign, an $\ell$-weight, a matching row, the exactness of an integration by parts — precisely because it builds its own kernel, visibility and boost potential and therefore cannot agree with the solver by construction. What it cannot establish is anything about the assembled spectrum. Each condition is verified in isolation, and a functional that satisfies every one of them separately may still be assembled wrongly. Table 1's step-six and step-seven rows record exactly that. §1.2 is this layer.

A manuscript check can establish that a paper states its own results correctly: that a column of a table follows from the algebra the paper cites for it, that a heading describes the body beneath it, that a cross-reference says something true about its target. It cannot establish the result. §1.3 is this layer, and it is the only one whose object is a document rather than a calculation.

None of the four, in any combination, reaches the observed sky. The sky is the subject of Part II; what is open in it is §1.4; and Table 1 above is the record of which steps of the chain are established, which are measured and failing, and what each piece of evidence does not reach.

## 1.2 Testing the physics

The statements these instruments carry are the steps of the chain Table 1 records, and the correspondence is not one instrument to one step. Steps one to three are settled by inspection and index algebra and have no instrument in this directory at all. Step four is gated by the oscillator's peak spacing and step five by **G1** against four external hierarchies, both of them repository diagnostics rather than frame checks. Exactly one row of Table 1 names an instrument from the set below — step six, where **c3** measures that (102) is not the exact integration by parts of its own integral. What the remaining eight establish sits beneath the table rather than in it: the conditions the matching coefficients must satisfy for the frame law to hold, which are prior to every row and are the reason a failing row can be localised at all. The inventory below should be read in that light, and not as a count of certified steps.

The source tree at the pinned `v1.3.0-dev` commit also contains independent, claim-facing Python instruments that are *absent* from the `v1.0.0`, `v1.1.0` and `v1.2.0` tags. This is part of the *current* repository situation, not a future Lean deliverable and not an increment in the physics-release sequence. The eleven instruments in `tests/frame/` use self-contained kernel, visibility and frame constructions rather than importing the spectral solver, and `tests/frame/run_all.py` invokes each in an isolated subprocess under `-I`. `tests/test_frame_checks.py`, which is the bundle's own harness, invokes seven of the eleven: `c8` through `c11` are reproducible and documented but are not gated there, so a regression in any of them would not fail the suite. That gap is recorded in `tests/frame/README.md` and is open. The file–to–claim mapping below is an inspected *source inventory*; this document has not rerun the pinned repository suite as part of this editorial gate.

Table 2: New `v1.3.0-dev` claim-facing Python checks. The inventory is distinct from software implementation tests. The status is what the source asserts or refutes, not independent certification of Paper 1 or of the observed spectrum.

| Python file | Scope and Paper 1 connection | Critical qualification |
|---|---|---|
| `c1_identity.py` | First-order threading identity and Sachs–Wolfe combination; Paper 1's frame transformation (Secs. II, VI) | Synthetic Einstein–de Sitter potential model; does not establish the full-sky, all-rank nonlinear claim. |

*Table 2 continued: v1.3.0-dev Paper 1 verification*

| Python file | Scope and Paper 1 connection | Critical qualification |
|---|---|---|
| `c2_weight.py` | Rank-two $k^{-2}$ weight and integration-by-parts boundary pair; line-of-sight provenance | Analytic and quadrature control on the stated kernels; not a spectrum calculation. |
| `c3_byparts.py` | Exact weighted bulk-source product rule and comparison with the printed *Annals II* expression | Detects and pins discrepancies in antecedent algebra; defects are not solver repairs. |
| `c4_rank_slot.py` | Conditional rank-two matching coefficients required under frame transformation; Paper 1's common-endpoint condition | A residual Thomson term is outside the simplified functional; no blanket invariance claim. |
| `c5_monopole.py` | Monopole matching and dipole-shift comparison | **Superseded conclusion** (finding T-107); the measurements are retained, but its proposed monopole restoration double-counts a source. Read with C8. |
| `c6_thomson.py` | Thomson contribution carried with visibility; differentiation of a coefficient versus its weight | Distinguishes the exact identity $W\kappa' = \mathcal{V}$ from the erroneous *extra* derivative; no new weighting is assumed. |
| `c7_kappa_ledger.py` | Integrating-factor and opacity ledger for the line-of-sight source | Separates $W$-weighted gravity and $\mathcal{V}$-weighted scattering; distinguishes legitimate from spurious opacity-rate terms. |
| `c8_rank0_partner.py` | Complete rank-zero integrand partner, correcting the C5 interpretation | Includes the $-C_1'$ partner and records the double-counting negative control; does not itself demonstrate the correct spectrum. |
| `c9_byparts_admissibility.py` | Added 2026-10-26. The condition under which the integration by parts of C6 is admissible at all. $W'C_1$ and $WC_1'$ sum to the total derivative $\mathrm{d}(\mathcal{V}\hat{v}_B)/\mathrm{d}\eta$; (102) carries $\mathcal{V}C_1'$ in place of $WC_1'$, larger by exactly $\kappa'$, and drops $W'C_1$. The split assembly — gravitational half by-parted, scattering half left at $K'$ — reproduces the ancestor | **Companion to C6, not a replacement**. C6 measures the frame obstruction the bundling in (90) creates, C9 the $\kappa''$ artefact that *differentiating* it creates, and the remedy is the same in both. It does *not* reinstate the rule C7 withdrew at T-113/S48: a coefficient may hold a rate; it may not be differentiated while it does. The two pieces do *not* cancel pointwise — the measured ratio of either to their sum is $\simeq 8$ — and the sum is bounded because it is a total derivative. |
| `c10_oscillator_phase.py` | Added 2026-10-26. The acceptance criterion for the acoustic step, scored against the driving it was meant to gate. Eliminating $\tilde{\tau}_1$ between *Annals II* (152) and (153) returns the oscillator with $c_s^2 = 1/[3(1+R)]$ exactly and the standard baryon-drag coefficient. With $\Phi$ constant the boundary value $(\delta\tilde{T}+\Phi_A)(\eta_*)$ is a pure cosine of amplitude $\Phi/3$ and its phase is zero; with $\Phi$ time-dependent the phase is non-zero | A criterion asking that the oscillator track $\cos(kr_s)$ in exact phase *accepts* the undriven solution and *rejects* the driven one, so the step that should have caught **A-3** certifies its absence. The decaying potential here is *not* a model of the radiation era: it falls at $x = k\eta/\sqrt{3}$, so the measured phase falls as $1/k$ where the real decay is tied to equality. What is established is the mechanism and its sign, not the magnitude in this bundle. |
| `c11_kernel_partner.py` | Added 2026-10-26. A by-parts taken against the kernel has three terms, not two: $\mathrm{d}(WCK)/\mathrm{d}\eta = \mathcal{V}CK + WC'K + WCK'$, whose integral is a pure boundary term. The boundary at $\eta_0$ is exactly zero for $\ell \geq 1$ because $j_\ell(0) = 0$, and the boundary at $\eta_i$ is bounded by $e^{-\kappa}(\eta_i)$, so $\int WC'K = -\int \mathcal{V}CK - \int WCK'$ to $5.4 \times 10^{-8}$ across $2 \leq \ell \leq 600$ | **Generalises C9, which holds the kernel fixed.** The Sachs–Wolfe term $\mathcal{V}CK$ places $5.5 \times 10^{-7}$ of its mass outside the visibility; the differentiated term $WC'K$ places 0.75 to 0.99 of its mass outside it. Keeping $WC'K$ without its $WCK'$ partner replaces a recombination-confined source with one that accumulates to today, and the discarded partner is 2.7 to $1.9 \times 10^3$ times the Sachs–Wolfe term it leaves behind. A constant $\Phi$ does not exempt this, which is why the Einstein–de Sitter null is the control that sees it. The monopole is a stand-in and the late fractions state where the terms live in $\eta$, not their amplitude in this bundle. |

The group is executed by `tests/frame/run_all.py` and integrated into the repository test harness

by `tests/test_frame_checks.py`; the generated `docs/THEORY-AUDIT.md` is produced by `tests/frame/gen_theory_audit.py` from their reported evidence. A harness or generated report is *not* an additional independent proof. In particular, the supersession recorded inside C5 must travel with any claim about how many checks "pass". Implementation tests, including the evolving two-frame and spectrum pipelines, remain part of the *physics* release progression rather than this inventory.

### 1.3 Verifying the manuscripts

A paper can be wrong in two ways that no check of the reconstruction reaches, because the reconstruction and the manuscript are independent accounts of the same physics and neither is evidence about the other. Its algebra can be wrong where the bundle's is right. And its algebra can be correct while the document misstates it: a heading attached to the wrong body, a label absent so that every later reference to a section drifts to the subsection whose title happens to describe that content, a sentence that ends by telling the reader an independent derivation shares no step with itself. On 2026-10-16 the submitted version of Paper 1 was found to carry all three, with Sec. V C holding no label at all. A heading is not a TEX error, a reference, a float or a box, so the build gate of the layer above was blind to every part of it; and re-reading does not catch it either, because a reader who already knows what a paragraph says stops reading the heading above it.

Each paper in this project therefore carries two standalone checks of its own, one for each failure, and their object is that paper and no other. Paper 1's are `paper/checks/independent_checks_v1.1.6.py` for the algebra and `paper/checks/coherence_gate.py` for the structure. Paper 2 is expected to carry its pair, written to the same scope rather than extended to cover both papers.

**What the algebra check re-derives, and what it only records.** `independent_checks_v1.1.6.py` takes Paper 1's coefficients and identities and re-establishes them in sympy from constructions of its own, rather than reading them back from the paper. Most of what it does is angular algebra: the non-aligned Cartesian PSTF identities of Eqs. (27)–(29) to rank six, evaluated off any symmetry axis and with no spatial harmonics anywhere in the construction; the $\Lambda$ recursion and the aligned generators to rank fourteen; the residuals of Eqs. (83)–(85) with their $\ell \to \infty$ weights; and the harmonic normalisations, the orthogonality relation and the $2\ell + 1$ dimension of the symmetric trace-free projector. Beneath that sit 1485 exact rational comparisons of five coefficient identities at every rank from $\ell = 4$ to $\ell = 300$, in `Fraction` arithmetic rather than in floating point.

The physical steps are fewer and each stands alone. The boost generator is decomposed from the kinematics and the $+Hv$ acceleration term derived from the expansion of $\nabla_b u_a$, which is the coefficient the frame law turns on. The conformal normalisation cancellation and the Gauss step (65)→(66) are carried through; so is the growth factor (49)→(50) and the two efficiency figures; and the quartic power expansion of Appendix C is reproduced in a flat-sky average whose convention the file now asserts, because the same symbol carries the three-dimensional polar cosine three sections earlier.

Section A can fail on the physics rather than on the bookkeeping. This is because Section A's general-rank energy integration by parts is scored against a direct integral arbiter evaluated at $\ell = 2, 4, 9$ — an explicit integration that does not use the symbolic rule it is testing. Sections B and I cannot. Table I's coefficients are transcribed from the paper, so Section B establishes that the table is internally consistent under the $4 + \text{rank}$ rule and does not derive it; and Section I is four counterexamples and qualifications, which bound the reading of a result rather than verify one.

*What makes it usable as evidence is that it does not report all of that as one number.* Its verdicts are four kinds and they are counted apart, which they were not before 2026-10-21:

Table 3: Verdict classes in `independent_checks_v1.1.6.py`, run 2026-10-26 under sympy 1.14.0 and Python 3.11.15, exit 0.

| Verdict | n | What it means |
|---|---|---|
| PASS | 187 | an identity re-derived here and found to hold, residual exactly zero in sympy or in exact rational arithmetic |
| FAILED AS REQUIRED | 3 | a deliberately wrong expectation that the suite must reject. Without these nothing would demonstrate that the comparison can fail at all, and a suite with no failing control cannot distinguish *every identity holds* from *the comparison is vacuous* |
| FINDING | 1 | a discrepancy against the paper. Not a pass, and not a note |
| RECORDED | 8 | a statement the file carries but does not verify — a qualification, a counterexample, or a summary of comparisons made elsewhere in it |
| total records | 199 | *of which 187 are verified checks.* The file reported all 199 as passes until 2026-10-21; a recorded statement is not a verified check, and reporting both as PASS overstates what ran |

**Open finding: Figure 3's signed residual.** The signed residual (83)−(84)−(85) tends to $-1/(2\ell^2)$. If Figure 3 or its caption prints $+1/(2\ell^2)$ as the signed value rather than as the magnitude, that is a discrepancy against the paper. Sec. 6.3's text states $O(\ell^{-2})$ without a sign, so the text is unaffected. The figure itself has not been checked.

**Withdrawn 2026-10-21: the variance-conservation check.** It read $A \cdot B/2 = (B/2) \cdot A$, which is commutativity of multiplication and cannot fail. Conservation requires the lensed spectrum's full second moment, including the linear–quadratic cross term, which this file does not compute.

**The structure check reads the document rather than the algebra.** `paper/checks/coherence_gate.py` reads the LaTeX source only and makes no claim about the rendered page. It tests two things: that every heading's own words appear somewhere in the body beneath it, and that every locating cross-reference — one whose sentence carries a verb such as *derived*, *obtained*, *stated* or *settled* — names the section that actually carries what the sentence claims for it. Exemptions are declared in the file with a reason, never passed silently, so a heading that legitimately does not repeat its own words is visible as a decision rather than as a gap. Paper 2 is expected to carry its counterpart, written to the same scope rather than extended to cover both papers.

**The coherence gate's qualification, which is the same shape as C5's.** The file was written on 2026-10-16 and then placed in a directory the Paper 1 repository's `.gitignore` excludes, where it remained until 2026-10-26. It was never committed, never reached the public repository, and no stage of the build script called it. Run on placement against both frozen sources — the arXiv line at `v1.1.6`, 47 headings, and the submitted version at `v1.4.4`, 46 headings — it reports no failures on either. Every heading carries a label, no heading has every word absent from its body, no section refers to itself, and every locating claim names the right section. The defect it was written for was therefore repaired before the freeze. That is a retrospective reading of a frozen document and not a gate that stood over it while it was being written, and the distinction must travel with the result in the same way C5's supersession does. Three warnings are raised on each line, identically. Twelve labels echo their own headings and eighteen section labels are never referenced, both cosmetic. Two headings each carry two section labels — `sec:hierarchy` with `sec:formulation`, and `sec:content` with `sec:physics` — which is the ambiguity the 2026-10-16 defect exploited, is present in both frozen lines, and belongs in the Paper 1 revision queue rather than in any change to a frozen manuscript.

### 1.4 Remaining problems

Two kinds of thing are open, and they are recorded together because they fail in the same place. The first is a physical question with an instrument pointed at it and an answer that does not satisfy it. The second is a claim with no instrument at all. Neither belongs in Table 1, which pairs a claim with its evidence and has no row shape for a claim whose evidence does not exist.

**Late-time weighting and the low-$\ell$ spectrum.** The late-time share diagnostic is 0.0000 on the Einstein–de Sitter and `PLANCK_MODEL` backgrounds. Einstein–de Sitter is a null control for the *late* integrated Sachs–Wolfe (ISW) term when its potential is constant. On the Planck background the released line-of-sight bracket is weighted by $\mathcal{V}\mathcal{D}$. Since $\mathcal{V}$ vanishes after recombination, it suppresses the late-time gravitational contribution. The benchmark expectation of order 0.1 recorded in the original diagnostic is model- and diagnostic-dependent; it is not a consequence of frame covariance.

The distinction between the source and the power spectrum matters. The integrating factor in (46) gives

$$W = e^{-\kappa}, \qquad W' = \kappa' W = \mathcal{V}, \qquad W(\eta_0) = 1, \qquad \mathcal{V}(\eta_0) \simeq 0. \tag{1}$$

Thus a late-time gravitational term carried with $W$ can survive after visibility has closed, whereas the same term carried with $\mathcal{V}$ vanishes in that limit. This explains the absence of the *late ISW* contribution in the released transport assembly. It does not, by itself, explain the missing ordinary Sachs–Wolfe plateau: a matter-dominated Einstein–de Sitter model has the ordinary plateau without a late ISW effect (Sachs and Wolfe, 1967; Gebbie et al., 2000).

Finally, $C_\ell$ is quadratic in the temperature transfer. When the transfer is separated into ordinary and integrated contributions, their cross term remains:

$$|\tau_\ell^{\mathrm{SW}} + \tau_\ell^{\mathrm{ISW}}|^2 = |\tau_\ell^{\mathrm{SW}}|^2 + 2\,\mathrm{Re}\Big(\tau_\ell^{\mathrm{SW}} \tau_\ell^{\mathrm{ISW}*}\Big) + |\tau_\ell^{\mathrm{ISW}}|^2. \tag{2}$$

An accumulation fraction for a line-of-sight *amplitude* is therefore not, without further definition, a fraction of $C_\ell$. The absence of the plateau remains a separate spectrum-level finding. Approximation A-1 is established as the missing late-time weight; its contribution to the total low-$\ell$ morphology still requires a separate test.

**Asserted, not gated.** This list is the half of *every gate has a claim and every claim has a gate* that a gate-driven generator cannot supply, because it enumerates precisely the claims no gate reaches. It is MIL-STD-498's rule that a requirement which cannot be verified is not a requirement: a statement that cannot be scored does not become established by being repeated, and it is recorded here, with the reason, rather than in the table above.

The following statements are not independently established by a completed acceptance test. They remain in the scientific record, with the qualification or measurement required to resolve them.

- That the solution computed here is exact. It is not claimed. The approximations entering after the transport equation are named — the scattering-time expansion, A-1, A-3, A-5, the diffusion treatment — and §25 carries what is not costed.
- The rank-one extra term $(\star) = -H\mathrm{D}_{\langle a}\tilde{\mathrm{D}}_{b\rangle} \ln T$: two readings, and this bundle chooses neither.
- The residual peak displacement has no established mechanism. A-3 is ruled out for the $\kappa''$-suppressed configuration and its standing elsewhere is a property of the configuration, not of the project.
- The overall low-$\ell$ plateau shape and the secondary peaks have no morphology gate. Two of the three morphology failures are presently diagnosed from the spectrum figure. The late-share diagnostic addresses one physical contribution, not the plateau morphology as a whole.
- Six is a modelling choice. $\eta_{\mathrm{couple}} = \min(\eta_0, 6\eta_*)$ is not derived and is not claimed to be; Q16 is what would close it.

## 2 Future situation

**Physics-facing development sequence.** The v1.3.0 calculation must first resolve the projection and source assembly without relaxing the common-physical-endpoint frame test. As the README specifies, `v1.4.0` is reserved for radiation-era driving (A-3), the tight-coupling transition and slip/quadrupole terms (A-5). `v1.5.0` then restores the nonlinear shear terms retained in the exact $1 + 3$ hierarchy but absent from the linear oscillator, with the intended endpoint being an implementation of the full *in-scope* physics introduced in Paper 1, using antecedent equations subject to the corrections documented here; `v2.0.0` is designated for almost-EGS bounds. These are *planned physical stages*, not results claimed by the current supplement. The scheduling rule is physical: an envelope correction is not advanced while the source assembly it depends on remains disputed.

**Source-resolved lensing is a stream and is not scheduled against a version.** It should be noted that CAMB and CLASS apply the standard CMB lensing transformation to the complete unlensed temperature spectrum using one CMB lensing-potential spectrum. This loses the source-distance information of late secondary temperature contributions. The numerical bias in specific observables remains to be measured. The transformation referred to is the published lensed-spectrum method (Lewis and Challinor, 2006); neither code's source has been read for this edition, and the statement is about that method rather than about an inspected implementation. The kernels are derived at §24.2 and the assembly-order argument is §27. No version in the sequence above carries this work, and it is recorded here so that the line-of-sight assembly is not frozen in a form that discards the source distance before anyone has decided whether it is wanted.

**The order of operations is part of the result, and nothing here shows the residue is zero.** A possible and presently unquantified contribution to the Hubble tension is that standard low- and high-redshift inference pipelines may not preserve the complete observer-defined observable, because a physical frame is fixed before free streaming, source projection, lensing and higher-order mode coupling have been completed. Four operations are involved — frame specialisation, perturbative truncation, source integration and propagation — and nothing in this document establishes that they commute. Where they do not, the order in which a pipeline performs them is part of what it reports, and two pipelines agreeing on every parameter can still disagree on the observable. The paragraph above is one instance: collapsing the source integral before the lensing operation is applied destroys a distinction the lensing would otherwise have acted on. No estimate of the residue is offered here and none is implied. What is claimed is that the residue is not known to be zero, and that a calculation carried along a single physical threading, with the order of operations declared, is what would settle it.

**The Lean certification is a separate repository and is not yet public.** The machine-checked certification of Paper 1 is being prepared at `timgebbie/frame-covariant-cmb-lean`, which is a repository in its own right and not a directory of this bundle. It is private as this edition is written, so no address is given, nothing in this document cites it, and no claim here rests on it. Editions before v1.3.19 described a `lean/` directory inside the v1.3.0 workstream; the sources are not there. The certification supplements, and does not replace, the Python instruments or the Paper 1-to-antecedent provenance network. Lean is *not a separate physics milestone* and cannot change the repository's version sequence. A claim is recorded as machine-certified only after its exact theorem statement, hypotheses, compilation and correspondence to Paper 1 have been checked. Proposed certificates and unverified pilot accounts confer no certification, and a repository a reader cannot open is a proposed certificate.

**Final editorial verification, after scientific closure.** After the scientific and provenance gates, a separate final review of this supplement will first assess global cohesion, consistency, terminology and

flow, then examine each section for directness and avoidable repetition. The review will identify proposed stylistic adjustments and their precise locations for approval before any changes are made. It will not shorten, compress, rederive or silently alter the mathematics, physical assumptions, tables, figures or Paper 1 provenance network. This is editorial assurance, not a new physics release.

**Final document deliverables, after editorial acceptance.** The final approved document edition will be distributed in two archives carrying the same supplementary-document version and identical final scientific source. The complete *project bundle* will retain the reproducibility evidence, provenance, tests, checksums, accepted baselines, comments and gate records needed to audit the work. The separate *Overleaf-ready document bundle* will contain only the main LaTeX source, its required bibliography and the figure, table, TikZ and other files actually used to compile it; no audit, version-governance, provenance-register, test, historical recovery or intermediate build files will be included. Both are to be built and verified from the same accepted version: the thin bundle must compile independently from a clean extraction, produce the same scientific document and have matching source and artifact hashes for every shared file. This is a distinction between distribution purposes, not a change to the repository's physics-facing release policy.

**Open scientific dependencies.** The unresolved checks are ordered by dependency. P1-T denotes a theoretical derivation and P1-B denotes an implementation or numerical test.

| **open item** | **what would close it** | **blocked by** | **owner** |
|---|---|---|---|
| carry the late-time sources to the observer under $W$ rather than $\mathcal{VD}$ | the late share at $\ell = 2$ becomes $O(0.1)$ on `PLANCK_MODEL` and stays exactly 0 on Einstein–de Sitter | — | P1-B |
| a peak repair that holds the two frames together | CDM and Newtonian peaks within the released 15%, not 3.1× apart | — | P1-B |
| whether A-3 is the remainder once such a repair exists | wire A-3 in: it must move the peak down and must land short of $0.757\,\ell_A$ | the row above | both |
| whether the $j'_\ell$ counterpart moves the peak | restore that slot alone; it must move the peak down or not at all | — | P1-B |
| the rank-one extra term ($\star$) | two readings exist and this bundle chooses neither | — | P1-T |
| helium, **Q15** | $\lesssim 0.5\%$ at the first peak; material only in the damping tail | scoped to v1.2.0 | P1-B |
| the tight-coupling window, **Q16** | assert $\mathcal{V}(\eta_{\rm couple})/\mathcal{V}(\eta_*)$ below a stated threshold on every background | — | P1-B |

# Part II

# The Physics

## 3 Scientific scope

Paper 1 v4 (Gebbie, 2026) is the provenance pivot for the scientific claims, sign conventions and equation-reference network. In particular, its Table III records which antecedent equation supplies each convention and which corrections the paper makes to earlier work. The journal versions of Gebbie and Ellis (2000) and Gebbie et al. (2000), together with Maartens et al. (1999), are the primary evidence against which those mappings must be tested. Their own equation numbers are quoted where the Paper 1 network calls for them; they do not supersede the Paper 1 mapping. A reference or mathematical error found in Paper 1 is reported explicitly and left traceable, never silently repaired by substituting an antecedent or an alternative gauge description.

This supplement records the analytical companion to *Frame covariance of CMBR lensing in the exact* 1+3 *covariant radiation hierarchy*, Gebbie, Paper 1, University of Cape Town, 2026, `arXiv:astro-ph/9912072v4`, doi:10.48550/arXiv.astro-ph/9912072, submitted 2026-10-16, CC BY 4.0. Cited versioned throughout: the bare identifier has carried three documents — `v3` is the correction and `v1`/`v2` the legacy 1999 work, neither of them this paper. The bundle is an independent Python reconstruction, from first $1+3$ covariant principles, of the almost-Friedmann–Lemaître results of Gebbie, Dunsby & Ellis, *Annals of Physics* 282, 321–394 (2000), `doi:10.1006/aphy.2000.6034`. It is archived at `doi:10.25375/uct.34069509`, live since v1.0.0. That is the concept DOI and always resolves to the latest version, so it is the one to cite.

It is written bottom-up: not a port of, and not a wrapper on, an existing Boltzmann code, and it inherits no other code's conventions. It reproduces; it does not argue. No new effect is claimed.

**Criterion 5b compares against CAMB and CLASS. It does not port them.** From v1.2.0 this bundle ships a `reference/` directory holding TT spectra generated by CAMB and CLASS, and that is the only form in which either code appears. Nothing under `functions/` imports `camb` or `classy`, and nothing ever may: a reconstruction that imported the codes it is checked against would not be an independent one, and a reader reproducing figure F1 should not need a Fortran toolchain. `scripts/make_reference_spectra.py` is the single caller, it is not a figure stage, and what enters the repository is data with a stamp — the spectra, the versions that produced them, every setting, and the derived parameters — never vendored source. Where a published result is carried on this side so that a comparison is like for like, it is taken from the literature and named, not read out of either code. The external codes are a check on this reconstruction, never an input to it.

The background expansion is *derived* from the $1+3$ covariant energy constraint rather than transcribed from a printed expansion law, so that the antecedent's own Eqs. (185) and (G.3) are checks on this bundle and not its sources. That ordering is what makes a reconstruction an audit, and it is load bearing: a port would have inherited the misprint recorded below.[2]

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[2]The Gauss constraint at (G.3) prints $+\frac{2}{3}\Theta^2$ where the companion thesis and the Cargèse lectures Eq. (55) carry $-\frac{2}{3}\Theta^2$. In the Friedmann limit the printed sign gives $H^2 = K/a^2 - (\mu+\Lambda)/3$, which for a flat section is $-(\mu+\Lambda)/3$ and so has no solution at positive energy density. A misprint, recorded; the derived background already carries the correct sign, and a test now pins it so that a later edit towards the printed form fails rather than passes.

# 4 Theory-to-computation map

Table 4 links each published equation the numerics implement to its provenance and to the `.py` file and object implementing it, together with the test that checks it. Table 5 gives the notation and the names the quantities carry in the code.

Both are *generated* from `config/implementation-register.toml` by `scripts/make_tables.py`, and `tests/test_register.py` resolves every module, object and test named in them. Their identifiers are drawn from the implementation register; the repository's register tests check that the named modules, objects and tests exist. The reformatted Table 4 retains all six generated fields and all 56 records, but displays the module, object and check on separate lines. The generated source is preserved unchanged for comparison.[3]

The harmonic normalisation is $Q_{A_\ell} = (-k_{\mathrm{phys}})^{-\ell} \mathrm{D}_{\langle A_\ell \rangle} Q$. Both candidate conventions are run end to end; only this one reproduces the real, opposite-sign form of the external hierarchies.[4]

Nomenclature is kept strictly separate: $\tau_\ell$ are mode coefficients and $\tau_{A_\ell}$ are multipole coefficients. *Annals II* notes that the distinction is not made in Bardeen-variable treatments. Every released figure plots a multipole mean-square, which holds for general geometries, rather than a mode mean-square, which holds only for almost-Robertson–Walker ones.

[3] **Confirmed provenance error in the implementation register, not in Paper 1.** The first two Annals I rows of the generated equation table cite (24) and (25) for $\beta_\ell$ and its recurrence. Published Annals I, p. 294 (PDF page 10), instead prints $\beta_\ell$ as Eq. (23), its recurrence as Eq. (24), and an angular-correlation expansion as Eq. (25). Paper 1 Table III C5.2–C5.3 correctly cites (23)–(24). The original generated table cells remain unchanged for source–render provenance; the wrong references are corrected *here*, not silently rewritten in the generated registry. The software-register fix is a separate governed repository change.

[4] *Annals I* prints $(-1)^\ell$ at the corresponding step where its own definition gives $i^\ell$. The covariant multipole $\tau_{A_\ell}$ is invariant under the choice, so the misprint never produced a wrong result; the mode coefficients are not invariant, and the two statements are distinct. Both are asserted as tests. A misprint, recorded in `provenance/ANNALS-II-ANTECEDENT-AND-CORRECTIONS-v1.md`.

Table 4: Equations implemented in the numerics, their published-paper provenance and the code that implements them. All six original fields are retained. Rows without code or tests are explicitly marked.

| **Quantity or relation** | **Source and equation** | **Implementation and validation** |
|---|---|---|
| $\beta_\ell = O^{A_\ell} O_{A_\ell} = (\ell!)^2 2^\ell/(2\ell)!$ | Annals I (24) | *File:* `functions/harmonics/weights.py` *Object:* `beta` *Check:* `test_beta_module_matches_symbolic_exactly` |
| $\beta_\ell = \frac{\ell}{2\ell-1}\beta_{\ell-1}$, $\beta_{\ell+1} = \frac{\ell+1}{2\ell+1}\beta_\ell$ | Annals I (25) | *File:* `functions/harmonics/weights.py` *Object:* `beta` *Check:* `test_beta_recurrence_descending_general_ell` |
| $\Delta_\ell = 4\pi\beta_\ell/(2\ell+1)$ | Annals I (119) | *File:* `functions/harmonics/weights.py` *Object:* `delta_over_four_pi` *Check:* — |
| $\alpha_\ell$, defined by $\beta_\ell = \alpha_\ell^{-1}(2\ell+1)$ | Annals II App. F, p. 380 | *File:* `functions/harmonics/weights.py` *Object:* `alpha` *Check:* `test_f4_is_direct_substitution_into_f3` |
| free-streaming weight $w_\ell = (\ell+1)^2/[(2\ell+3)(2\ell+1)]$ | Annals II (F.1), p. 379 | *File:* `functions/harmonics/weights.py` *Object:* `free_streaming_weight` *Check:* `test_f1_weight_is_the_mode_recursion_weight_shifted` |
| covariant recursion $-\dot{\tau}_\ell = (k_{\rm com}/a)[w_\ell\tau_{\ell+1} - \tau_{\ell-1}]$ | Annals II (F.1), p. 379 | *File:* `functions/harmonics/free_streaming.py` *Object:* `free_stream` *Check:* `test_the_three_normalisations_are_one_solution` |
| $\beta$-normalised recursion, conformal time | Annals II (F.3), p. 380 | *File:* `functions/harmonics/free_streaming.py` *Object:* `free_stream` *Check:* `test_the_three_normalisations_are_one_solution` |
| $\alpha$-normalised recursion | Annals II (F.4), p. 380 | *File:* `functions/harmonics/free_streaming.py` *Object:* `free_stream` *Check:* `test_f4_variable_is_exactly_the_spherical_bessel_function` |
| free-streaming solution $\alpha_\ell^{-1}\tau_\ell = j_\ell(k_{\rm com}\Delta\eta)$ | Annals II App. F, p. 380 | *File:* `functions/harmonics/free_streaming.py` *Object:* `analytic_projection` *Check:* `test_f4_variable_is_exactly_the_spherical_bessel_function` |
| $\dot{\Theta}_\ell = k[\frac{\ell}{2\ell-1}\Theta_{\ell-1} - \frac{\ell+1}{2\ell+3}\Theta_{\ell+1}]$ | Hu & Sugiyama (6), p. 2601 | *File:* `functions/harmonics/external.py` *Object:* `WEIGHTS` *Check:* `test_external_hierarchies_match_the_bundle_at_machine_precision` |
| $\dot{F}_{\nu\ell} = \frac{k}{2\ell+1}[\ell F_{\nu(\ell-1)} - (\ell+1)F_{\nu(\ell+1)}]$ | Ma & Bertschinger (49), (50) | *File:* `functions/harmonics/external.py` *Object:* `WEIGHTS` *Check:* `test_external_hierarchies_match_the_bundle_at_machine_precision` |

*Table 4 (continued)*

| Quantity or relation | Source and equation | Implementation and validation |
|---|---|---|
| $\dot{\Delta}_{T\ell} = \frac{k}{2\ell+1}[\ell\Delta_{T(\ell-1)} - (\ell+1)\Delta_{T(\ell+1)}]$ | Seljak & Zaldarriaga (3d) | *File:* `functions/harmonics/external.py` *Object:* `WEIGHTS` *Check:* `test_external_hierarchies_match_the_bundle_at_machine_precision` |
| relative-sign control: the $c^2 = -1$ normalisation, run as a wiring check | Annals II (F.1), p. 379 | *File:* `functions/harmonics/free_streaming.py` *Object:* `free_stream` *Check:* `test_sign_control_fails_in_a_known_shape` |
| $\int_0^\infty z^{-m} j_\ell^2(z)\, dz$, corrected to carry $[(m/2)!]^2$ | Annals II (199), p. 372 | *File:* `scripts/derive_ell_range.py` *Object:* `bessel_integral_corrected` *Check:* `test_corrected_bessel_identity_matches_quadrature` |
| $\ell_A = \pi\Delta\eta_*/r_s$, bounding criterion 5's range | Annals II §8.3.2, p. 372 | *File:* `scripts/derive_ell_range.py` *Object:* `acoustic_scale` *Check:* — |
| $(a'/a)^2 = H_0^2[\Omega_r a^{-2} + \Omega_m a^{-1} + \Omega_k + \Omega_\Lambda a^2]$ | this work the $1+3$ energy constraint | *File:* `functions/background/flrw.py` *Object:* `Background` *Check:* `test_the_gauss_constraint_sign_is_the_corrected_one_and_the_printed_one_has_no_solution` |
| $\mathcal{P}(k) = \mathcal{A}(k\eta_0)^{n-1}$ | Annals II (J.1) | *File:* `functions/spectra/transfer.py` *Object:* `power_spectrum_j1` *Check:* `test_j1_is_constant_at_n_equals_one_and_eta0_drops_out` |
| $\mathcal{P}(k) = \mathcal{B}k^n T^2(k)$ | Annals II (J.3) | *File:* `functions/spectra/transfer.py` *Object:* `power_spectrum_j3` *Check:* `test_j1_and_j3_differ_by_one_power_of_k` |
| $T(k) = [1 + (ak + (bk)^{3/2} + (ck)^2)^\nu]^{-1/\nu}$ | Annals II (J.5) | *File:* `functions/spectra/transfer.py` *Object:* `transfer_function` *Check:* `test_b10_the_printed_transfer_function_diverges_where_it_must_vanish` |
| $\dot{\delta}_\ell = -n_e\sigma_T\delta_\ell - ikT\{\frac{\ell}{2\ell-1}\delta_{\ell-1} + \frac{\ell+1}{2\ell+3}[1 - \frac{\ell(\ell+2)K}{k^2}]\delta_{\ell+1}\}$ | Wilson (8), p. 4 | *File:* `functions/harmonics/external.py` *Object:* `integrate_wilson` *Check:* `test_wilson_reproduces_the_bundle_and_tests_the_phase_convention` |
| $x_e(z)$, equilibrium Saha | this work standard; not in the antecedent | *File:* `functions/spectra/decoupling.py` *Object:* `saha_ionisation` *Check:* `test_saha_is_monotone_and_spans_the_full_range` |
| $\mathcal{V}(k,\eta) \simeq \kappa' e^{-\kappa}$ | Annals II §5.2 | *File:* `functions/spectra/decoupling.py` *Object:* `RecombinationHistory` *Check:* `test_the_visibility_is_a_normalised_probability_density` |

*Table 4 (continued)*

| Quantity or relation | Source and equation | Implementation and validation |
|---|---|---|
| $\mathcal{D}(\eta_0, k) = \int d\eta\, \mathcal{V} e^{-(k/k_D)^2} \simeq e^{-(k/k_D)^2}$ | Annals II (101), §5.2 | *File:* `functions/spectra/decoupling.py` *Object:* `RecombinationHistory` *Check:* `test_the_visibility_is_a_normalised_probability_density` |
| the integral solution with $e^{-\kappa}$ on gravitational and $\kappa' e^{-\kappa}$ on scattering sources | Annals II (98), §5 | *File:* `functions/spectra/decoupling.py` *Object:* `Weighting` *Check:* `test_the_two_weightings_are_distinct_objects` |
| the same solution with both weights replaced by $\mathcal{V}$ | Annals II (102), §5 | *File:* `functions/spectra/decoupling.py` *Object:* `Weighting` *Check:* `test_the_visibility_weighting_cannot_carry_a_late_isw` |
| $\gamma \simeq \frac{k^2}{6\kappa'} \frac{R^2 + \frac{4}{5}(1+R)}{(1+R)^2} = k^2/k_D^2$ | Annals II (168), §6 | *File:* `functions/spectra/decoupling.py` *Object:* `diffusion_scale` *Check:* `test_damping_suppresses_small_scales_and_leaves_large_ones` |
| $-C_0 = \frac{5}{2} a\mathcal{B}_2$, $-C_1 = \frac{1}{k}(a\mathcal{B}_1 + \kappa' v_B)$, $-C_2 = \frac{1}{k^2} a\mathcal{B}_2$ | Annals II (90), §5 | *File:* — *Object:* — *Check:* — |
| $\tilde{\tau}_1' = -\frac{R'}{1+R}\tilde{\tau}_1 - \frac{k}{1+R}\delta\tilde{T} - k\Phi_A$; $\delta\tilde{T}' = -aH\Phi_A - \Phi_H' + \frac{k}{3}\tilde{\tau}_1$ | Annals II (152), (153), §7.1 | *File:* `functions/spectra/acoustic.py` *Object:* `solve_acoustic` *Check:* `test_the_pair_reduces_to_the_printed_oscillator_exactly` |
| $\delta\tilde{T}'' + \frac{R'}{1+R}\delta\tilde{T}' + k^2 c_s^2 \delta\tilde{T} = -\Phi_H'' - \frac{R'}{1+R}\Phi_H' - \frac{k^2}{3}\Phi_A$ | Annals II (154), §7.1 | *File:* `functions/spectra/acoustic.py` *Object:* `solve_acoustic` *Check:* `test_the_pair_reduces_to_the_printed_oscillator_exactly` |
| $c_s^2 = 1/(3(1+R))$, $R = 3\rho_M/4\rho_R$ | Annals II §6 | *File:* `functions/spectra/acoustic.py` *Object:* `sound_speed_squared` *Check:* `test_sound_speed_limits` |
| $\delta\tilde{T} + \Phi_A \approx [\delta\tilde{T}(0) + (1+R)\Phi_A]\cos(kr_s) - R\Phi_A$ | Annals II (180), §7.1.2 | *File:* `functions/spectra/acoustic.py` *Object:* `sachs_wolfe_at_last_scattering` *Check:* `test_the_integration_converges_to_the_closed_form_180_as_r_is_held_constant` |
| $[\delta\tilde{T} + \Phi_A](\eta_*) \approx \frac{1}{3}\Phi_A(0)(1+3R_*)\cos(kr_s^*) - R_*\Phi_A(0)$ | Annals II (181), §7.1.2 | *File:* `functions/spectra/acoustic.py` *Object:* `sachs_wolfe_at_last_scattering` *Check:* `test_b8_equation_181_sign_is_fixed_by_its_own_limit` |
| $\frac{\beta_\ell \tau_\ell(\eta_0)}{2\ell+1} = S_P j_\ell(k\Delta\eta_*) + \int d\eta\, (S_{\rm DISW} + S_{\rm ISW}) j_\ell(k\Delta\eta)$ | Annals II (176), §7.1.1 | *File:* `functions/spectra/sources.py` *Object:* `temperature_transfer` *Check:* `test_176_with_an_impulsive_source_is_the_free_streaming_projection` |
| $S_P = D(\eta_0, k)\,[(\tilde{\delta}_T + \Phi_A) + \kappa' v_B]$ | Annals II (177), §7.1.1 | *File:* `functions/spectra/sources.py` *Object:* `source_primary` *Check:* `test_176_with_an_impulsive_source_is_the_free_streaming_projection` |

*Table 4 (continued)*

| Quantity or relation | Source and equation | Implementation and validation |
|---|---|---|
| $S_{\mathrm{DISW}} = V e^{-(k/k_D)^2}[\frac{1}{3}k\tilde{\tau}_1 + (\kappa'\tilde{v}'_B + \kappa''\tilde{v}_B)]$ | Annals II (178), §7.1.1 | *File:* `functions/spectra/sources.py` *Object:* `source_doppler` *Check:* `test_the_printed_doppler_bracket_equals_the_contracted_form` |
| $S_{\mathrm{ISW}} = V e^{-(k/k_D)^2}[(\Phi'_A - \Phi'_H) - aH(\tilde{\delta}_T + 3\Phi_A)]$ | Annals II (179), §7.1.1 | *File:* `functions/spectra/sources.py` *Object:* `source_integrated` *Check:* `test_b6_sign_alternative_is_not_a_no_op` |
| $C_\ell = \frac{2}{\pi}\frac{\beta_\ell^2}{(2\ell+1)^2}\int \frac{dk}{k} k^3 \lvert\tau_\ell\rvert^2$ | Annals II (186), p. 368 | *File:* `functions/spectra/angular.py` *Object:* `cl_mode_route` *Check:* `test_the_harmonic_normalisation_cancels_out_of_186_exactly` |
| $\langle \tau_{A_\ell} \tau^{A_\ell} \rangle = \frac{\beta_\ell}{2\pi^2}\int k^2 dk \lvert\tau_\ell\rvert^2$ | Annals II (187), p. 368 | *File:* `functions/spectra/angular.py` *Object:* `multipole_mean_square` *Check:* `test_188_is_delta_over_2l_plus_1_times_187` |
| $C_\ell = \Delta_\ell (2\ell+1)^{-1} \langle \tau_{A_\ell} \tau^{A_\ell} \rangle$ | Annals II (188), p. 368 | *File:* `functions/spectra/angular.py` *Object:* `cl_covariant_route` *Check:* `test_the_two_routes_agree_to_machine_precision_on_real_input` |
| $C_\ell^{\mathrm{SW}} = \frac{1}{2\pi} H_0^4 \Omega_0^{1.54} \int \frac{dk}{k^2} P(k) j_\ell^2(k\Delta\eta_*)$ | Annals II (198), p. 371 | *File:* `functions/spectra/sachs_wolfe.py` *Object:* `cl_sachs_wolfe_quadrature` *Check:* `test_the_analytic_reduction_matches_the_quadrature` |
| large-scale closed form, $n = 1$ | Annals II (200), (201), p. 372 | *File:* `functions/spectra/sachs_wolfe.py` *Object:* `cl_large_scale_reduced` *Check:* `test_b5_printed_201_is_short_by_the_comoving_distance` |
| closed-form k-integral kernel at complex $m$, $\int z^{-m} j_\ell^2(z) dz$ | Annals II (199), p. 372, corrected | *File:* `functions/spectra/kernels.py` *Object:* `bessel_square_kernel` *Check:* `test_complex_kernel_reduces_to_the_real_identity` |
| primary-term $k$-integral by power-law decomposition | Annals II (176) primary term, (198) | *File:* `functions/spectra/kernels.py` *Object:* `fftlog_primary_integral` *Check:* `test_fftlog_matches_quadrature_on_a_non_power_law_integrand` |
| integrated terms, $\int d\eta\, S(k,\eta)\, j_\ell(k\Delta\eta)$ | Annals II (176), p. 365 | *File:* `functions/spectra/bessel_table.py` *Object:* `line_of_sight` *Check:* `test_line_of_sight_matches_direct_evaluation` |
| truncation condition: four consecutive vanishing harmonics force $\sigma_{ab} = 0$ | Annals II App. E, (E.1), p. 378 | *File:* — *Object:* — *Check:* — |
| $C_0, C_1, C_2$ from the rank-by-rank match of the integral solution | Annals II (89) | *File:* `functions/spectra/coefficients.py` *Object:* `match_coefficients` *Check:* `test_all_three_of_90_are_reproduced_by_the_match` |

*Table 4 (continued)*

| Quantity or relation | Source and equation | Implementation and validation |
|---|---|---|
| $-C_0 \simeq +\frac{5}{2}a\mathcal{B}_2$, $-C_1 \simeq \frac{1}{k}(a\mathcal{B}_1 + \kappa' v_B)$, $-C_2 \simeq \frac{1}{k^2}a\mathcal{B}_2$ | Annals II (90) | *File:* `functions/spectra/coefficients.py` *Object:* `c0_from_printed_89` *Check:* `test_b11_is_a_sign_in_89_not_in_90` |
| $-\tilde{C}_1 \approx \kappa' \tilde{v}_B + (\delta\tilde{T} + \Phi_A)$ | Annals II (109) | *File:* `functions/spectra/coefficients.py` *Object:* `newtonian_coefficients` *Check:* `test_the_newtonian_specialisation_is_equation_109` |
| the slow-decoupling integral solution carried with a generic $u^a$: two boundary terms and one integrated term | Annals II (102) | *File:* `functions/spectra/integral_solution.py` *Object:* `integral_solution` *Check:* `test_the_specialisation_recovers_what_the_pipeline_computes` |
| $(a\tilde{\mathcal{B}}_1)' \approx -a^2 H\tilde{\mathcal{B}}_1 + k(\Phi_A' - \Phi_H') - 2Hak\Phi_A + \frac{1}{3}k^2\tilde{\tau}_1$ | Annals II (106) | *File:* `functions/spectra/coefficients.py` *Object:* `newtonian_c1_prime` *Check:* `test_t23_splining_the_amplitude_is_wrong_and_the_test_can_tell` |
| $\dot{\sigma}_{\langle ab\rangle} + 2H\sigma_{\langle ab\rangle} - \mathcal{D}_{\langle a}A_{b\rangle} + E_{\langle ab\rangle} \simeq \frac{1}{2}\pi_{\langle ab\rangle}$, and its second derivative | Annals II (G.12) | *File:* `functions/spectra/coefficients.py` *Object:* `generic_shear_derivatives` *Check:* `test_the_rank_two_control_passes_symbolically` |
| $C_2$, $C_2'$ and $C_2''$ from $\mathcal{B}_2 = \sigma_{ab}$ | Annals II (90) | *File:* `functions/spectra/coefficients.py` *Object:* `c2_derivatives` *Check:* `test_no_numerical_derivative_survives_in_the_c2_chain` |
| $\sigma_{ab} = \mathcal{D}_{\langle a}v_{b\rangle}$ in the CDM frame, with $\dot{\sigma}$ and $\ddot{\sigma}$ | Annals II (31) + (G.12) | *File:* `functions/spectra/coefficients.py` *Object:* `cdm_frame_shear` *Check:* `test_B13_the_printed_n_t1_does_not_close_and_the_general_rule_does` |
| $C_0, C_1, C_2, C_1', C_2''$ in the CDM frame, $A_a = 0$ | this work (89) + (31)-(37) + (106) + (152) | *File:* `functions/spectra/coefficients.py` *Object:* `cdm_set` *Check:* `test_the_frame_shift_of_every_coefficient_is_the_closed_form_one` |
| $\tilde{\tau}_a = \tau_a - v_a$, as $\tilde{\tau}_1 \to \tilde{\tau}_1 - v/k$ | Annals II (37) | *File:* `functions/spectra/frame_transform.py` *Object:* `FrameShift` *Check:* `test_T27_the_dipole_carries_the_same_inverse_k_as_the_monopole` |
| $(\rho, p, \mathcal{H}')$ from the $\Omega$'s, consistent with $\dot{H} = -H^2 - (\rho + 3p)/6$ | Annals II Appendix G | *File:* `functions/spectra/coefficients.py` *Object:* `background_fluid` *Check:* `test_the_frame_shift_of_every_coefficient_is_the_closed_form_one` |

Table 5: Notation, and the names the quantities carry in the code. The mode/multipole distinction is the one that matters most: *Annals II* notes that it is not made in Bardeen-variable treatments.

| Symbol | In code | Meaning | Note |
|---|---|---|---|
| $\Phi_A$, $\Phi_H$ | `phi_a, phi_h` | the two gauge-invariant potentials entering (177) and (179) | — |
| $\tilde{\delta}_T$ | `delta_T` | covariant temperature perturbation | frame-dependent; $\tilde{\delta}_T + 3\Phi_A$ is the combination in dispute (B-6) |
| $\tilde{\tau}_1$ | `tau_1` | the dipole, which supplies the Doppler term of (178) | — |
| $\tilde{v}_B$ | `v_b` | baryon velocity | — |
| $\kappa'$, $\kappa''$ | `kappa_prime` | differential optical depth and its conformal-time derivative | supplied, never manufactured inside the source terms |
| $V$ | `visibility` | visibility function, weighting the integrated sources | — |
| $k_D$ | `damping_k` | diffusion damping scale, entering (178) and (179) as $e^{-(k/k_D)^2}$ | — |
| $aH$ | `conformal_hubble` | $a'/a$, the background combination (179) carries | derived from the $1+3$ constraint, not from a printed expansion law |
| $S_P$, $S_{\rm DISW}$, $S_{\rm ISW}$ | `source_primary, source_doppler, source_integrated` | the primary, Doppler and integrated temperature sources of (177)–(179) | individually frame-dependent |
| $\tau(x, e)$ | — | temperature anisotropy as a function of position and direction | — |
| $\tau_{A_\ell}$ | — | **multipole** coefficients — the covariant objects | valid for general geometries; what every released figure plots |
| $\tau_\ell$ | `tau, values` | **mode** coefficients | almost-Robertson–Walker only. Conflating these two is how a sign is lost |
| $O^{A_\ell}$ | — | $e^{\langle A_\ell \rangle}$, the PSTF basis on the sphere | Annals I (15) |
| $Q_{A_\ell}$ | — | harmonic basis tensor, $(-k_{\rm phys})^{-\ell} \mathrm{D}_{\langle A_\ell \rangle} Q$ | the adopted normalisation; see criterion 3 |
| $\beta_\ell$ | `beta` | $O^{A_\ell} O_{A_\ell}$ | Hu & Sugiyama's normalisation |
| $\alpha_\ell$ | `alpha` | $(2\ell+1)/\beta_\ell$ | Ma & Bertschinger's and Seljak & Zaldarriaga's normalisation |
| $k_{\rm com}$ | `k_com` | comoving wavenumber | **never a bare `k` in this codebase** |
| $k_{\rm phys}$ | `k_phys` | physical wavenumber, $k_{\rm com}/a$ | Appendix F's $k$ is comoving; the conventions sheet's is physical |
| $\eta$ | `eta` | conformal time, $dt = a\, d\eta$ | — |
| $\Delta\eta_*$ | `d_eta` | $\eta_0 - \eta_*$ | — |
| $j_\ell$ | `spherical_jn` | spherical Bessel function | the free-streaming solution |
| $u^a$ | — | the frame 4-velocity | carried generic; frames are specialisations made at the end |
| $\sigma_{ab}$ | — | shear | $\sigma$ is the shear and nothing else |
| $A_a$ | — | acceleration, $\dot{u}_a$ | — |
| $\Theta_\ell$ | — | Hu & Sugiyama's multipole variable | $\beta$-normalised |

# 5 The acceptance spine

Appendix F of Gebbie et al. (2000), pp. 379–380, Eqs. (F.1)–(F.4), links the covariant mode recursion to external treatments written in real coefficients. Reproducing that chain tests the harmonic normalisation

without introducing a representation, and it is where this bundle's independence lies.[5]

Five external treatments are read and four are drawn — Hu & Sugiyama Eq. (6), Wilson (1983) Eq. (8), Ma & Bertschinger Eqs. (49) and (50), and Seljak & Zaldarriaga Eq. (3d) — each transcribed from its own paper.[6] The fifth is Wilson & Silk (1981) Eq. (7), the earliest of them and likewise read from its own page: at $K = 0$ it *is* Wilson (1983) Eq. (8), so it is credited and not drawn, four panels carrying four independent numerical checks. Hu & Sugiyama and Wilson write the recursion in the $\beta$ normalisation — the normalisation their Eq. (2) fixes — and the other two in the $\alpha$ normalisation, which is why Appendix F prints both (F.3) and (F.4). A basis phase leaking into the couplings would break at least one of the two matches.

Wilson is the strongest of the four, and not because of his weights, which are Hu & Sugiyama's. He writes the recursion in the *imaginary* convention, with both bracket terms positive under an overall $-ikT$, where the others carry opposite signs under a real $k$. The two are related by $\Theta_\ell = i^\ell \delta_\ell$, which turns one form into the other exactly. He is integrated complex and as printed, and the phase removed only afterwards; transforming the equation first would assume the identity under test. That the transformed solution is *real* to $10^{-10}$ is a separate assertion from that it agrees. Wilson therefore tests the phase convention — the subject of §2's footnote and of acceptance criterion 3 — from a source published before any of the others.[7]

## 6 Conventions and their limits

The same hierarchy is written four different ways in the literature this bundle checks against, and the differences are not cosmetic. Table 6 is the translation, generated from `config/implementation-register.toml` so that it cannot drift from the code, with the test that checks each row.

**"RW", "FL" and "FLRW" are three different claims, and this bundle makes the middle one.** The names are used interchangeably often enough that it is worth separating them once.

- Robertson–Walker (RW) is a statement about *geometry*: the spacetime is spatially homogeneous and isotropic. It says nothing about what drives the expansion, and it follows from symmetry alone — no field equation has been used.
- Friedmann–Lemaître (FL) is that geometry *with its dynamics*: the Einstein equations imposed on it, giving the Friedmann and Raychaudhuri equations and hence $a(\eta)$ for a given matter content. FL is RW *plus* the field equations.
- FLRW is the two together, and is the usual compound name.

*Annals II* and this bundle say almost-Friedmann–Lemaître, and the choice is not stylistic. The background here is integrated from the $1 + 3$ energy constraint rather than assumed, so the dynamical

[5] The comparison applies the $(2\ell + 1)^{-1}$ division that the text preceding (F.4) specifies and the display omits; (F.4) is otherwise correct as printed. A misprint, recorded.

[6] Ma & Bertschinger print the same recursion twice. Eqs. (49) and (50) are the *massless neutrino* hierarchy, synchronous and conformal Newtonian, and are collisionless; their Eqs. (63) and (64) are the photon hierarchy, carrying the identical recursion plus a Thomson term $-an_e\sigma_{\rm T}F_{\gamma\ell}$. Appendix F's (F.4) is a free-streaming recursion, so (49) and (50) are the comparison and (63) and (64) are not. Published and arXiv numbering agree for this paper, which is not the case for the Annals papers.

[7] Appendix F's citation pairs the name of one paper with the key of another. Both papers are real and both numbers are right for the paper they belong to: Wilson & Silk (1981) Eq. (7) *is* the $\ell > 2$ hierarchy, and Wilson (1983) Eq. (7) is that paper's quadrupole equation while its Eq. (8) is the hierarchy. The prose names Wilson & Silk, the key is Wilson alone and consistent across every citation, and the load-bearing $i^{-\ell}$ remark names Wilson alone — so the match and the phase remark are about Wilson (1983), and the number beside them is correct for the 1981 paper the prose names. One mismatched pairing, not two independent misprints; recorded, and it moves no result.

half is in use from the start: Eqs. (185) and (G.3) are checks on this bundle rather than inputs to it. A treatment that only assumed almost-RW would have the geometry and not the expansion history, and could not perform that check.

The distinction is also what the almost-EGS theorem is about, and Sec. 26 depends on reading it the right way round. Almost-isotropy of the measured radiation bounds the *geometry* — an almost-RW conclusion. Getting from there to almost-FL requires the field equations as a second step. Collapsing the two names collapses a two-step argument into an assertion, which is precisely the move that paper is written to avoid.

**There are exactly two conventions, and they differ by $c^2 = -1$.** Changing the harmonic normalisation $Q_{A_\ell} = \mu^{-\ell}\mathrm{D}_{\langle A_\ell\rangle}Q$ rescales the mode coefficients as $\tau_\ell \to c^\ell\tau_\ell$ with $c = \mu'/\mu$. A recursion $\tau'_\ell = k[A_\ell\tau_{\ell-1} + B_\ell\tau_{\ell+1}]$ then carries $(A_\ell/c,\, B_\ell c)$, so the ratio $B/A$ picks up $c^2$. At $c = i$ an opposite-sign real bracket becomes a same-sign bracket under an overall $i$. Hu & Sugiyama, Ma & Bertschinger and Seljak & Zaldarriaga print the first; Wilson prints the second. That is the whole of the difference, and it is asserted symbolically rather than described.

**What the match establishes.** The $\ell$-weights, which are convention-independent, are confirmed by five sources in two normalisations, four of them drawn and numerically independent. The bundle is shown to sit at the same $c^2$ as Hu & Sugiyama, Ma & Bertschinger and Seljak & Zaldarriaga, and Wilson is shown to sit at the other. *Both* Wilson papers are in the imaginary convention, which settles a standing hypothesis in the negative: no author published this recursion in both conventions, so the split is between authors and not within one.

**One source fixes its own phase, and that matters.** Ma & Bertschinger define their Legendre expansion with an explicit factor $(-i)^\ell(2\ell+1)$ and state why: it is "chosen to simplify the expansion of a plane wave", so that $\exp(-i\mathbf{k}\cdot\mathbf{x})$ has coefficients $F_\ell = j_\ell(kr)$. Their coefficients therefore sit at $(-i)^\ell$ from the plane wave *by their own definition*, are real, and satisfy the opposite-sign real recursion. For this source the phase is read off the paper rather than assumed, which is exactly the standing the next paragraph says three of the four lack.

**What it does not establish, and this release says so.** It does not establish that $Q_{A_\ell} = (-k_{\mathrm{phys}})^{-\ell}\mathrm{D}_{\langle A_\ell\rangle}Q$ is the right reading of *Annals I*'s definition. Rephasing *both sides* of a comparison by $i^\ell$ is a change of variable and cannot fail — measured at $6.2\times10^{-15}$ either way — so a match constrains the convention only where the phase applied to the external source is fixed from outside the comparison. For two of the four, no phase is applied, and that is a choice rather than a derivation; Ma & Bertschinger and Wilson are the two that are fixed from outside the comparison.

**The exception, and it is the useful one.** *Annals II* states, in the line following its flat-case mode functions, that the covariant form "differs by a factor of $i^{-\ell}$ from Wilson since we are using plain mode functions instead of plane waves". That phase is specified by the antecedent, not chosen here. So Wilson's panel in Figure 1 is the one comparison carrying information about the convention, and it is checked in two independent ways: the transformed solution agrees with the reconstruction, *and* it is real to $10^{-10}$, which the agreement alone would not show.

Acceptance criterion 3 was therefore reopened and restated to the claim the evidence supports. Settling it needs *Annals I*'s definition of $Q_{A_\ell}$, not a better match. Nothing computed here depends on it: the covariant $\tau_{A_\ell}$ is invariant under the choice, the mode coefficients are not, and every released figure plots the invariant quantity.

Table 6: How each treatment in the foundational literature writes the same hierarchy, and how this bundle checked it. The two conventions in the literature are related by $\tau_\ell \to c^\ell \tau_\ell$ with $c^2 = -1$ and nothing else. **Where the transformation applied is "none" that is a choice, not a derivation** — which is why acceptance criterion 3 was reopened; see §6.

| **Treatment** | **Variable** | **Bracket weights** | **Phase convention** | **Transformation applied here** | **How it was checked** |
|---|---|---|---|---|---|
| Annals II (F.1), Appendix F | $\tau_\ell$ | $\frac{(\ell+1)^2}{(2\ell+3)(2\ell+1)}$ on $\tau_{\ell+1}$, 1 on $\tau_{\ell-1}$ | covariant; $Q_{A_\ell} = (-k_{\rm phys})^{-\ell} \mathrm{D}_{\langle A_\ell \rangle} Q$ | —, this is the reference | arithmetic on $\beta_\ell$ alone; no representation introduced |
| Annals II (F.3), Appendix F | $\beta_\ell \tau_\ell$ | $\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$ | real, opposite-sign bracket | none | (F.1)→(F.3) by multiplying through by $\beta_\ell$, as exact rationals |
| Annals II (F.4), Appendix F | $\alpha_\ell^{-1} \tau_\ell$ | $\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$ | real, opposite-sign bracket | the $(2\ell+1)^{-1}$ the prose states and the display omits | **finding B-2**: printed (F.4) is not yet in the Ma & Bertschinger form |
| Hu & Sugiyama Eq. (6), p. 2601 | $\Theta_\ell$ | $\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$ | real, opposite-sign bracket | none — *a choice*, see Q1 | integrated as printed from its own paper, never through Appendix F |
| Wilson Eq. (8), p. 4 | $\delta_\ell$ | $\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$ | **imaginary**; both bracket terms + under an overall $-ikT$ | $\Theta_\ell = i^\ell \delta_\ell$ — **fixed by the antecedent**, which states the $i^{-\ell}$ itself | integrated *complex and as printed*; the phase removed only afterwards, and the result asserted real to $10^{-10}$ separately from that it agrees |
| Wilson & Silk Eq. (7), p. 15 | $\delta_\ell$ | $\frac{\ell}{2\ell-1}$, $\frac{\ell+1}{2\ell+3}$ | **imaginary**, as Wilson 1983 — *not* the real form | none required: at $K = 0$ it is Wilson 1983 (8) exactly | read from its own page. The **earliest** of the five and a fifth source for the weights, but **not** a fifth numerical check — the same equation cannot check itself |
| Ma & Bertschinger Eqs. (49), (50) | $F_{\nu\ell}$ | $\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$ | real, opposite-sign bracket | none — *a choice*, see Q1 | integrated as printed from its own paper; synchronous and conformal Newtonian gauges |
| Seljak & Zaldarriaga Eq. (3d) | $\Delta_{T\ell}$ | $\frac{\ell}{2\ell+1}$, $\frac{\ell+1}{2\ell+1}$ | real, opposite-sign bracket | none — *a choice*, see Q1 | integrated as printed; also the method CMBFAST implements, cited for criterion 5b |

**The second table, and what its two thesis columns are for.** Table 6 translates between the treatments this bundle is *checked against*. Table 7 does something different: it writes each coupling of this reconstruction in the conventions of Maartens, Gebbie & Ellis (1999), Challinor (2000), the thesis in both of the conventions it uses, Beneke & Fidler (2010) and Pitrou (2009). Every cell is derived from a named line of the normative sheet, and no cell is blank: where a treatment has no established counterpart the cell says so.

Two things in it are worth reading rather than skimming. The thesis takes two columns and they differ on exactly the kinematic rows — its $\dot{\mathcal{T}}$ form moves every kinematic term to the right-hand side, flipping the overall sign against its K/Π form. Elsewhere the two agree and the table says so in as many words, because a difference in wording where there is none in substance is precisely how two conventions get conflated while appearing distinct. And the Beneke & Fidler and Pitrou columns are mostly "not established": the sheet records both as unsettled at the level of hierarchy coefficients, Beneke & Fidler working in $(\ell, m)$ tetrad form and Pitrou in Poisson gauge, and neither has had a hierarchy coefficient reproduced here. That is the honest content of those columns and it is printed rather than left to inference.

Table 7: The couplings of this reconstruction written in each external convention. Every cell is derived from the line of `provenance/conventions.md` named in the *Sheet* column; no cell is blank, and a treatment with no established counterpart says so rather than being left to inference. **The thesis takes two columns and they differ on exactly the kinematic rows** — the $\dot{\mathcal{T}}$ form moves every kinematic term to the right-hand side, flipping its overall sign against the K/Π form. On the remaining rows the two agree, and the table says *as Th-K* rather than restating them: a difference in wording where there is none in substance is how two conventions get conflated while appearing distinct.

| **Coupling** | **Sheet** | **MGE99** | **CL** | **Th, K/Π** | **Th, $\dot{\mathcal{T}}$** | **BF** | **P09** |
|---|---|---|---|---|---|---|---|
| Rank raising | `C4.2` | identity | identity in form; $\mathrm{D}_a$ unchanged, the sign sits in $k$ | identity | as Th-K | not established: BF work in $(\ell, m)$ tetrad form, no coefficient reproduced | not established: Pitrou works in Poisson gauge, no hierarchy coefficient checked |
| Rank lowering | `C4.4` | identity | identity in form; CL (2.25) prints $+\mathrm{D}^b I_{bA_\ell}$, reproduced from MGE99 (71) | identity; Th App. D footnote to (D.44) at $\ell = 1$ agrees | as Th-K | not established | not established |
| Free-streaming mode recursion | `C4.5` | identity | identity in form; CL (2.25) free streaming reproduced from MGE99 (71) | identity | as Th-K | not established | not established |
| PSTF contraction on the sphere | `C6.1` | identity (one source: EMT 1983, via C6.4) | identity; CL expands in the same $O^{A_\ell}$ | identity, Th (1.20) | as Th-K | not established: BF use ${}_sY_{\ell m}$, not a PSTF basis | identity at the basis level: Pitrou (1.28) expands in $n^{A_\ell}$ |
| PSTF self-contraction $\beta_\ell$ | `C5.2` | identity | identity; CL absorbs it into $\Delta^C_\ell$, see the row below | identity, Th (1.23) | as Th-K | no counterpart: $(\ell, m)$ normalisation carries $\sqrt{4\pi/(2\ell+1)}$ instead | identity, Pitrou (1.29) in the equivalent $4\pi\ell!/(2\ell+1)!!$ form |
| Angular measure $\Delta_\ell$ | `C5.5` | identity, MGE99 (47), same value | $\Delta^C_\ell = 4\pi(-2)^\ell(\ell!)^2/(2\ell+1)! = (-1)^\ell\Delta_\ell$, CL (2.23) | identity | as Th-K | no counterpart in this form | identity, Pitrou (1.29), verified symbolically |

| **Coupling** | **Sheet** | **MGE99** | **CL** | **Th, K/Π** | **Th, $\dot{\mathcal{T}}$** | **BF** | **P09** |
|---|---|---|---|---|---|---|---|
| Shear coupling at $\ell+2$ | `C9, T-MGE and T-Th-T` | $-(\ell+2)$ as printed in (71), (88), (90), (99) — **defect E9**; $+(\ell+2)$ is correct, and MGE99 (89) at $\ell=2$ prints $+\frac{8}{15}\rho\sigma$, agreeing with the correction | $-\frac{8}{15}\delta_\ell^2 I\sigma$ gives $+(\ell+2)$, independently of MGE99 | $-(\ell+2)$ in Th (2.73) — carries E9 | $+(\ell+2)$ **on the RHS** in Th (2.84): every kinematic term flips overall sign against (2.73) because it is moved across. **This row is where the two thesis columns differ** | not established: BF put the observer at rest, so they carry no shear coupling | not established |
| Acceleration coupling at $\ell\pm1$ | `C9, T-CL and T-Th-T` | identity | $A_a \to -A_a$ under $g \to -g$ at fixed $u^a$ (CL footnote to §2), the signature difference and nothing more | identity | overall sign flipped with the rest of the kinematic terms — **differs from Th-K** | not established: observer at rest | not established |

# 7 The covariant hierarchy and its reduction

*The hierarchy is stated with its exact and approximate components explicitly distinguished.*

## 7.1 The expansion, and why it is PSTF

The temperature anisotropy is expanded in projected, symmetric, trace-free multipoles about a chosen 4-velocity $u^a$. This is distinct from the exact distribution-function and brightness hierarchy; the conversion from brightness to temperature multipoles is linear in the anisotropy (Paper 1, Eqs. (12)–(15)):

$$\boxed{\tau(x,e) = \sum_{\ell\geq 0} \tau_{A_\ell}(x)\, O^{A_\ell}(e)}, \qquad O^{A_\ell} = e^{\langle A_\ell\rangle}, \tag{3}$$

with $A_\ell$ the multi-index $a_1 \dots a_\ell$, $e^a$ a unit direction orthogonal to $u^a$, and $\langle\,\rangle$ the PSTF part. The sum includes $\ell = 0$ only with the convention $\tau_0 = 0$; Paper 1, Eq. (12), accordingly starts at $\ell = 1$. This convention must be retained in the rank-zero matching condition below.

PSTF is not a choice of basis among several. On the 2-sphere orthogonal to $u^a$ the PSTF tensors are the *irreducible* representations of the rotation group: each rank carries exactly the information no lower rank carries, and the decomposition is unique. Any other basis — spherical harmonics, Legendre polynomials in $\hat{k}\cdot e$ — is a representation of the same content that first requires a direction to be singled out. The PSTF form singles out nothing but $u^a$, which is the whole reason a $1+3$ treatment can carry the frame generically and choose it at the end.

**Two objects, and conflating them is how a sign is lost.** $\tau_{A_\ell}$ are multipole coefficients and exist for a general geometry. $\tau_\ell$ are mode coefficients, obtained by expanding $\tau_{A_\ell}$ on harmonics $Q_{A_\ell}$, and exist only for an almost-Robertson–Walker one. *Annals II* notes the distinction is not made in Bardeen-variable treatments. Every released figure plots the multipole quantity, and §17 says why that is not a presentational preference.

## 7.2 The transport equation and its reduction

The reconstruction uses the following compact rank-$\ell$ transport relation. It separates streaming, the gravitational source and Thomson scattering. The attribution to Eq. (68) of Gebbie et al. (2000) is retained as the project's published-paper cross-reference, but its equality with the printed journal equation remains subject to direct page-level verification.[8]

$$\dot{\tau}_{A_\ell} + \underbrace{\tfrac{\ell+1}{2\ell+3}\mathrm{D}^a\tau_{A_\ell a} - \mathrm{D}_{\langle a_\ell}\tau_{A_{\ell-1}\rangle}}_{\text{free streaming}} + \underbrace{\kappa'\tau_{A_\ell}}_{\text{out-scattering}} = \underbrace{a\mathcal{B}_{A_\ell}}_{\text{gravitational}} + \underbrace{\kappa'\mathcal{S}_{A_\ell}}_{\text{in-scattering}} . \tag{4}$$

This is *not* the full exact brightness hierarchy of Paper 1, Eq. (16), in which acceleration, shear and vorticity multiply anisotropy multipoles and Thomson scattering is itself truncated in baryon velocity. Paper 1, Eqs. (16)–(20), and the published nonlinear dynamics of Maartens et al. (1999) distinguish the kinematic Liouville operator from the collision approximation.[9] The present equation omits those explicit

[8] Direct check of the published *Annals II*, p. 339, finds the reduced angular transport relation in Eq. (67); its Eq. (68) gives the gravitational source $\mathcal{B}$ and Thomson source $\mathcal{C}$, not the displayed rank-$\ell$ multipole equation verbatim. Paper 1, Eqs. (16)–(20) remain the governing exact-hierarchy references. This is a citation correction to the supplement, not an emendation of Paper 1 or a proof of the reduced equation from *Annals II*.

[9] Published *MGE 1999*, Eq. (71) (and its higher-multipole Eq. (90)), prints $-(\ell+2)$ for the rank-lowering shear coefficient. Paper 1, Eq. (16) and its Sec. III D, identifies the sign as a misprint and gives $+(\ell+2)$ on the left-hand side. A separate rank-two check sets the brightness to an isotropic $\Pi_\emptyset$: Paper 1's Liouville Eq. (20) has the redshift source $-4\sigma_{ab}e^a e^b\Pi_\emptyset$ on the right-hand side, which contributes $+4\sigma_{ab}\Pi_\emptyset$ to the left-hand quadrupole equation, consistent with Paper 1 at $\ell = 2$ and inconsistent with the printed $-4$ of MGE Eq. (71). This local check does not by itself prove the sign at every rank.

nonlinear kinematic products, so its role here is a *reduced transport equation*. The source term and the ordering of the reduction must be checked against the published *Annals II* hierarchy before crediting an exactness claim.

A separate consistency requirement concerns time. With overdots along $u^a$ and primes in conformal time, $dt = a\,d\eta$ gives

$$\tau'_{A_\ell} = a\dot{\tau}_{A_\ell}, \qquad \Gamma_{\rm T} = n_e\sigma_{\rm T}, \qquad \kappa' = a\Gamma_{\rm T}, \qquad k_{\rm phys} = k_{\rm com}/a. \tag{5}$$

These identities do not by themselves license mixing $\dot{\tau}_{A_\ell}$ with $\kappa'$ in one unrescaled equation. All terms must be in the same time variable. The displayed relation is therefore kept as a source object for audit; its factors of $a$ have not been certified here.[10]

Four terms, and each later section is about one of them. Free streaming is §12; the gravitational source closes at rank two and is §*The generic source derivatives*; the two $\kappa'$ terms are the two halves of one collision term and their separation is what §14.1 turns on.

**Both $\kappa'$ are the same $\kappa'$.** The out-scattering on the left removes photons from the beam and the in-scattering on the right returns them; they are one Thomson process written on two sides. That is why multiplying through by the integrating factor assigns both weights at once rather than by choice, which is (46)'s content.

### 7.3 Scope of the reduction

The exact starting point is the $1+3$ Liouville and brightness hierarchy, not the reduced temperature equation (4). The former requires no coordinate-gauge choice or fixed metric specification. The latter is used for the almost-Friedmann–Lemaître reconstruction, with its ordering and proper-/conformal-time conversion still under audit. The computed solution is not exact: the scattering-time expansion, A-3's static transfer, A-5's collision term and the diffusion treatment are separately qualified in §25.

**The sources are gauge-invariant because they vanish in the background.** $\mathcal{B}_0$, $\mathcal{B}_a$ and $\mathcal{B}_{ab}$ all vanish in the background, so by Stewart–Walker they are gauge-invariant at first order — which is the paper's own *"note that these are CGI with respect to $u^a$"*. Gauge-invariance here is a consequence of the objects chosen, not a construction imposed on them, and that is the difference between a covariant treatment and a gauge-fixed one with covariant notation.

**What the collision term contains, and what A-5 drops.** *Annals II* writes $C[\tau] \simeq \sigma_{\rm T} n_E(\bar{f}_1 e^a v_a^B + \bar{f}_2 e^a e^b \tau_{ab} - \tau)$. This release takes $\bar{f}_1 = 1$ and $\bar{f}_2 = 0$, so $\mathcal{S}_2 = 0$: no polarisation feedback and no quadrupole in-scattering. That is approximation A-5 stated at the collision term rather than at the oscillator, and §14.1 shows it cannot produce a frame disagreement at linear order.

[10] The displayed reduced equation combines an overdot (proper-time derivative) with $\kappa' = an_e\sigma_{\rm T}$, a conformal-time scattering rate. These terms require a whole-equation rescaling. The published source and the code normalisation have not yet been reconciled; the equation is retained for audit, not certified as written.

### 7.4 Where the hierarchy is gated

| gate | what it tests | what it does not |
|---|---|---|
| **G1**, criterion 1 | the free-streaming part of (4) against four external hierarchies, two normalisations, two phase conventions | anything with the source on — §12 |
| `test_register` | that every module, object and test named in the equation and notation tables resolves | that the equation implemented is the one printed |
| **C2** round trip | covariant → mode → covariant, identically zero in exact arithmetic | that either representation is the right one — that is criterion 3, reopened |

The listed gates test selected consequences of the reduced hierarchy, not the full brightness equation or its reduction. The comparison between the reduced form, Paper 1 Eq. (16) and the published *Annals II* equations is a separate mathematical acceptance test; it is not satisfied by agreement among numerical representations of the same reduced model.

## 8 The generic source derivatives

*The source derivatives are checked against* `provenance/GENERIC-SOURCE-DERIVATIVES-v1.md`*; open items are retained there.*

**Derivatives required by Eq. (102).** (102)'s integrand is $C_0 - C_1' + C_2''$ and its boundary terms carry $C_2'(\eta_*)$. With (90),

$$-C_0 \simeq +\tfrac{5}{2} a\mathcal{B}_2, \qquad -C_1 \simeq \tfrac{1}{k}(a\mathcal{B}_1 + \kappa' v_B), \qquad -C_2 \simeq \tfrac{1}{k^2} a\mathcal{B}_2,$$

Two entries of that display are corrected in §14. The $+\frac{5}{2}$ on $C_0$ is $\alpha_2$, the rank-*two* normalisation, applied to a monopole row where $\alpha_0 = 1$ belongs — finding T-104, a factor of $\frac{15}{2}$ — and the $\kappa' v_B$ inside $C_1$ is a bare $\kappa'$ in a *coefficient*. This is not prohibited by an identity, but it is a site to check for consistent integration by parts and scattering weight; §14.3 describes the structural test. Both are stated here at the point of use so a reader of this display does not meet the correction four hundred lines later.

so the solution needs $(a\mathcal{B}_1)'$, $(a\mathcal{B}_2)'$ and $(a\mathcal{B}_2)''$ for arbitrary $u^a$. *Annals II* prints these only after the Newtonian specialisation has removed the shear, which is exactly the specialisation this bundle declines to make early. The derivations are therefore assembled here, and this section records what is closed, what is assembled, and what is open.

**The three sources, by rank, with no frame chosen.** From §4's integrated Boltzmann equation $\dot{\tau} + e^a \mathrm{D}_a \tau + \mathcal{B} \simeq C[x, e]$,

$$\mathcal{B} = -\tfrac{1}{3}\mathrm{D}^a \tau_a + (\mathrm{D}_a \ln T + A_a) e^a + \sigma_{ab} e^a e^b,$$

giving $\mathcal{B}_0 = -\frac{1}{3}\mathrm{D}^a\tau_a$, $\mathcal{B}_1 = \mathrm{D}_a \ln T + A_a$ and $\mathcal{B}_2 = \sigma_{ab}$. The rank-two closure refers to this reduced gravitational source. It does not remove the acceleration, shear and vorticity couplings between anisotropy

multipoles in the exact brightness hierarchy (Paper 1, Eq. (16); Maartens et al. (1999)). All three vanish in the background, so all three are gauge-invariant at first order by Stewart–Walker — which is the paper's own *"note that these are CGI with respect to $u^a$"*.

**The coefficient-derivative ledger.** The following is an algebraic consequence of the three adopted matching coefficients, not an additional approximation to the full hierarchy. To keep the two types of source separate, write

$$\mathsf{A}(\eta) \equiv a\mathcal{B}_2, \qquad \mathsf{F}(\eta) \equiv a\mathcal{B}_1 + \kappa' v_B, \qquad k \equiv k_{\rm com}, \quad k' = 0. \tag{6}$$

These are scalar mode amplitudes in the conventions of the matching equation; $\mathsf{A}$ is not the acceleration $A_a$. For the corrected monopole matching, and *conditional on the source normalisation used in (89)*, the three coefficients are

$$C_0 = -\tfrac{1}{3}\mathsf{A}, \qquad C_1 = -k^{-1}\mathsf{F}, \qquad C_2 = -k^{-2}\mathsf{A}. \tag{7}$$

Since $k$ is a constant *comoving* label, direct differentiation gives

$$C_0' = -\tfrac{1}{3}\mathsf{A}', \qquad C_1' = -\tfrac{1}{k}\mathsf{F}', \qquad C_2' = -\tfrac{1}{k^2}\mathsf{A}', \tag{8}$$

$$C_2'' = -\tfrac{1}{k^2}\mathsf{A}'', \qquad \mathsf{F}' = (a\mathcal{B}_1)' + \kappa'' v_B + \kappa' v_B'. \tag{9}$$

The two boundary coefficients and the full bulk bracket in (102) are thus

$$C_2' - C_1 = -\frac{\mathsf{A}'}{k^2} + \frac{\mathsf{F}}{k}, \qquad kC_2 = -\frac{\mathsf{A}}{k}, \tag{10}$$

$$C_0 - C_1' + C_2'' = -\frac{\mathsf{A}}{3} + \frac{\mathsf{F}'}{k} - \frac{\mathsf{A}''}{k^2}. \tag{11}$$

Here the primes act on the complete coefficient, including $a$, the fields, $\kappa'$ and $v_B$. If instead $k$ denotes the physical wavenumber, $k_{\rm phys} = k_{\rm com}/a$, then $(k_{\rm phys}^{-1})' = \mathcal{H}k_{\rm phys}^{-1}$ and $(k_{\rm phys}^{-2})' = 2\mathcal{H}k_{\rm phys}^{-2}$; hence (9) is *not* valid unchanged in the physical-wavenumber convention. The supplementary conventions and the implementation must be compared on precisely this point. The independent symbolic calculation, inputs, expected outputs and negative control are in `tools/check_coefficients_v1.3.5.py` and `COEFFICIENT-PROVENANCE-v1.3.5.csv`. No numerical agreement among pipelines using the same $C_i$ certifies their physical origin.

**The three matching rows as a linear system.** The numerical prefactors in (7) may be checked without importing the conclusion. Let $h_\ell(x) = \alpha_\ell j_\ell(x)$, with $\alpha_\ell = (2\ell+1)/\beta_\ell$ and $\beta_\ell = \ell!/(2\ell-1)!!$ (Paper 1, Table III, C5.2–C5.3, citing the published *Annals I*). The ascending Bessel series gives

$$j_0(x) = 1 - \tfrac{x^2}{6} + O(x^4), \qquad j_1(x) = \tfrac{x}{3} - \tfrac{x^3}{30} + O(x^5), \tag{12}$$

$$j_2(x) = \tfrac{x^2}{15} - \tfrac{x^4}{210} + O(x^6), \qquad (\alpha_0, \alpha_1, \alpha_2) = (1, 3, \tfrac{15}{2}). \tag{13}$$

It follows by differentiation at $x = 0$ that $h_0(0) = 1$, $h_0''(0) = -1/3$, $h_1'(0) = 1$ and $h_2''(0) = 1$. These are four separately tested values; the sign of $h_0''$ and the choice of $\alpha_0$ rather than $\alpha_2$ in a rank-zero row are independent questions.

With $\mathsf{A}$ and $\mathsf{F}$ defined in (6), the *three rows as reconstructed in* `functions/spectra/coefficients.py`, with a vanishing independent rank-zero forcing, are

$$\boxed{\begin{pmatrix} 1 & 0 & -k^2/3 \\ 0 & k & 0 \\ 0 & 0 & k^2 \end{pmatrix} \begin{pmatrix} C_0 \\ C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 0 \\ -\mathsf{F} \\ -\mathsf{A} \end{pmatrix}.} \tag{14}$$

The last row gives $k^2 C_2 = -\mathsf{A}$; the middle one gives $kC_1 = -\mathsf{F}$; inserting the last into the first gives $C_0 = \frac{1}{3}k^2 C_2 = -\mathsf{A}/3$. The determinant is $k^3$, so this conditional solution is unique for a nonzero comoving mode. It reproduces (7) by elimination rather than by restating the coefficients.

**Independent alternatives, kept separate.** If the monopole row instead has a forcing $\mathsf{G}_0$, its right-hand side is $(\mathsf{G}_0, -\mathsf{F}, -\mathsf{A})^{\mathsf{T}}$ and the same elimination gives $C_0 = \mathsf{G}_0 - \mathsf{A}/3$. If one retains the rank-two factor $\alpha_2$ in the rank-zero projection, the upper-right matrix entry becomes $-5k^2/2$ and the zero-forcing solution is $C_0 = -5\mathsf{A}/2$. Changing the sign of $j_0''(0)$ is a *third*, distinct alternative. None of these alternatives is equivalent to adjusting an output normalisation of $C_\ell$. The rank-zero condition $\mathsf{G}_0 = 0$ must follow from the projected covariant monopole constraint, not from the success of an internal two-route test. The full published *Annals II* journal page containing Eq. (89), rather than an arXiv rendering, is still required to establish whether its definition has precisely the rows of (14).[11] The four series values, three row solutions and negative controls are reproduced by `tools/check_matching_boundary_v1.3.6.py`.

**The scale-factor rule, and why the footnote to (105) carries $2H$.** A rank-$n$ PSTF source's natural variable carries $a^n$: the paper writes $(a\tilde{\mathcal{B}}_1)'$ at (106) and $(a^2\sigma_{ab})^{\cdot}$ in Appendix G. So

$$\dot{\mathcal{B}}_1 + 2H\mathcal{B}_1 = \frac{1}{a^2}(a^2\mathcal{B}_1)^{\cdot},$$

and the rank-two object built from a rank-one source, $\mathrm{D}_{\langle a}\dot{\mathcal{B}}_{b\rangle}$, carries the rank-two power. It is not a commutator correction bolted on; it is the rank showing up in the measure. That is conventions row C3.1a read in the direction nobody had needed, and it is checkable: under C3.1, C3.2 and C4.2 the mode functions are time-independent at every rank, so *all* the $H$'s come from the $a^n$ and none from the $Q$'s — which predicts (106)'s $2H$ before (106) is consulted. Implementing it the other way, by splining a stored amplitude, is wrong by a median of 100%: finding T-23, and the reason `integral_solution` takes every derivative as an input rather than differentiating one.

**Rank 2: closed.** $\dot{\mathcal{B}}_2$ *is* in the paper generically, in Appendix G, and closes to second order with an exact symbolic control that fails when the rank-two term is removed — so the check is one that can fail, which S27's C rung requires and finding T-24 exists because an earlier one could not.

**Rank 1: assembled, and carrying one term more than (105) prints.** $\dot{\mathcal{B}}_1$ is assembled from printed equations. Three of its four terms close exactly against (105)/(106) — $\mathcal{B}_0$, $\dot{\Phi}_A$ and $-2H\Phi_A$, the last only because both of its contributions are present — and substituting the expansion–potential relation of §*newtg* reproduces every term (105) prints. The generic form then carries one extra:

$$(\star) \qquad -H\,\mathrm{D}_{\langle a}\tilde{\mathrm{D}}_{b\rangle}\ln T.$$

This is the temperature-gradient half of $-H\tilde{\mathcal{B}}_a$, whose acceleration half *does* survive into (105). Two readings, and this bundle chooses neither. It may be the small-scale approximation — §6 drops all terms of $O(\epsilon\epsilon_H)$ and $(\star)$ has that shape — except that so does $-2H\mathrm{D}_{\langle a}\mathrm{D}_{b\rangle}\Phi_A$, which is kept. Or it may be an omission, in which case (105), (106) and what descends from them are short a term of the same order as ones they keep. It is first order and it does not obviously vanish, which is enough to make the question real; settling it needs §2.5's $O(\epsilon\epsilon_H)$ bookkeeping read against each term of (105), not a judgement about which is negligible. Logged as open, not filed as a finding against the paper.

**Rank 0: not needed.** Un-withdrawn 2026-10-22: the claim stands, the reason does not. This paragraph read: *"$\dot{\mathcal{B}}_0$ is not required by (102) at all — $C_0$ carries $\mathcal{B}_2$, not $\mathcal{B}_0$."* It was withdrawn on 2026-10-21 on the authority of T-105, which held that (89)'s $a\mathcal{B}_0\delta_{\ell 0}$ row had been dropped in error. T-105 has since fallen, three times over, and the claim above is restored with a different reason.

[11] Unresolved source audit: the published *Annals of Physics* 282 (2000), Eq. (89), has not been verified directly from the journal typesetting. The matrix is an explicit reconstruction of the implementation's adopted matching, not a transcription certified against that page. The independent monopole forcing, including a possible $a\mathcal{B}_0$ contribution, remains a separate derivational question.

$\tau_0 \equiv 0$ is a *constraint*, not initial data: $T$ is *defined* as the monopole of $T(x, e)$, so differentiating $\int A_0 \, \mathrm{d}\eta' = 0$ puts zero on the right of (89)'s $\ell = 0$ row — $\mathcal{B}_0$ is a functional of the solution and not an external forcing. Solving that row with $\alpha_0$ in place of $\alpha_2$ and $h_0''(0) = -\frac{1}{3}$ gives $C_0 = -\frac{1}{3} a\mathcal{B}_2$ with *no* $\mathcal{B}_0$ in it, and the factor against (90)'s printed $-\frac{5}{2} a\mathcal{B}_2$ comes out as $\alpha_2/\alpha_0 = \frac{15}{2}$ exactly — finding T-104, which is *cleaner* standing alone. Independently, $a\mathcal{B}_0 = -(k/3)\tau_1$ is an identity and $-C_1'$ carries $+(k/3)\tau_1$ already, so restoring $a\mathcal{B}_0$ to $C_0$ counts the monopole source twice. The measurement agrees: criterion 5b doubles, to 98.5%, under the restoration — finding T-107.

So $\mathcal{B}_0$'s *content* is required and its *row* is correctly empty, and both statements hold at once. The released code has carried this since 2026-10-21: `MONOPOLE_ROW = "corrected"` is $C_0 = -\frac{1}{3} a\mathcal{B}_2$, and the restoration is kept only as the regenerable control that refuted it.

**Status.** The derivation is complete for what the integral solution needs. What remains open is ($\star$). The $a\mathcal{B}_0\delta_{\ell 0}$ row of (89) is not open and is not dropped in error — it is correctly absent, for the reason above. $\Delta C_0 = +kv/3$ is non-zero and its partner is cross-coefficient, in $-C_1'$ rather than in $C_2$, which is why the rank-0 partner check `c8` scores the bracket and the single-coefficient residual $\gamma$ cannot: M15. A $\gamma$ driven to zero by adding a term is a fit, not a gate; the $\ell = 1$ and $\ell = 2$ rows are controls because they come out at unity without adjustment.

## 9 Frames and specialisation

**"Frame" is being asked to do two jobs, and they are different objects.** The word is used in this literature for something specified at a single event and for something specified throughout a region, and the two are not interchangeable. The PI's enumeration, recorded here because the rest of this section depends on it:

| Object | What is specified | Where |
|---|---|---|
| Instantaneous observer rest frame | A unit timelike vector $u^a(p)$, determining a local time direction and an orthogonal rest space | At an event $p$ |
| Local orthonormal frame | A tetrad $e_{\hat{\alpha}}{}^a(p)$, with $e_{\hat{0}}{}^a = u^a$, also choosing three spatial axes | At an event |
| Observer worldline | A timelike curve with tangent $u^a$ | Along one observer's history |
| **Threading** | A smooth timelike congruence, represented by a unit tangent field $u^a(x)$ | Throughout a spacetime region |
| Adapted orthonormal frame field | A tetrad field whose timelike leg follows the congruence | Throughout that region |

Nothing in this bundle is a tetrad. No calculation here chooses spatial axes: every quantity is a PSTF tensor built from $u^a$ and $h_{ab} = g_{ab} + u_a u_b$, so rows two and five are named only to be set aside. What the bundle actually uses is rows one and four, and it uses both:

- the receiving observer's frame — a $u^a$ at one event, reception at $\eta_0$ — which is what the *measurement* is referred to, and in particular what the dipole is a statement about;
- the CDM threading and the Newtonian threading — unit tangent fields $u^a(x)$ on the whole region between last scattering and reception — which is what the *propagation* integral is carried along, and what $\sigma_{ab}$, $A_a$ and $q_a$ are properties of.

The general transformations (31)–(37) relate two *threadings*; the specialisations (40)–(45) stipulate one. $\tilde{\sigma}_{ab} = 0$ and $\tilde{A}_a = 0$ are conditions on a congruence throughout a region and are not statements about any single observer, which is why neither can be imposed "at the observer" and why the Newtonian one

is admissible only in restricted cases. Figure F7's two panels are exactly this pair kept apart: the $\ell \geq 2$ panel compares two *threadings* carrying the same observable, and the dipole panel is where the *receiving observer's frame* enters, because the dipole is the one multipole that is a statement about $u^a$ at reception rather than about the congruence that brought the radiation there.

**The words this bundle uses, and why none of them is retired.** *Threading* names the congruence as an object. *Receiving observer's frame* names the local $u^a$ at reception. *Frame* continues to name a choice of observers and, following *Annals II* §4.1's own usage, each named choice — the energy frame, the zero-acceleration (CDM) frame, the Newtonian frame. Every one of those is a threading in the first sense: the words sit at different levels and are not competing. Decision S29a.

No identifier is renamed. The code says `"cdm"` and `"newtonian"`, and *Annals II* prints "frame" for exactly these objects, so the code agrees with the source it reconstructs. The rule that a word naming two things should have one of them renamed is satisfied here by naming the two objects, which is what the table above does. And the rename operation is itself the hazard: a blanket substitution in this bundle inverted the meaning of diagnostic D7, whose subject is the CDM *threading* measured against the total-energy frame — the one place both senses appear in a single sentence, and therefore precisely what a substitution cannot see. Finding T-33. A rename a regular expression can perform is a rename whose exceptions nothing will catch.

**Specialisation.** The solution is carried with a generic $u^a$. The explicit frames appear as choices made at the end, not assumed at the start: the Newtonian frame $\tilde{\sigma}_{ab} = 0$, which is the specialisation *Annals II* itself makes at p. 365 in order to remove the shear terms, and the CDM frame $\tilde{A}_a = 0$. Carrying the generic frame and specialising late is the difference the $1 + 3$ covariant approach buys; a gauge-fixed treatment fixes the frame before the hierarchy is solved and cannot exhibit it.

**"CDM" names two different things, and this supplement keeps them apart.** *CDM the model* is a cosmology — flat, matter dominated, no $\Lambda$ — and it is one of the two models this release delivers a spectrum for. *The CDM frame* is a choice of frame $u^a$: the congruence in which the cold dark matter is at rest. The two are independent. The $\Lambda$CDM *model* is computed in the same *frame*, and a change of model is not a change of frame. In the code the models are named `CDM_MODEL` and `LCDM_MODEL` and the frames `"cdm"` and `"newtonian"`, so the two cannot be confused at a call site.

**The CDM frame is *not* the total-energy frame, and an earlier version of this supplement said it was.** The definition used here is exact in general relativity and needs no approximation: cold dark matter is pressureless and collisionless, so it is acted on by no non-gravitational force and its worldlines are geodesics. The CDM frame is therefore the frame with $\tilde{A}_a = 0$, which is what *Annals II* §4.1 calls the *zero acceleration frame*, listing it as the second of three useful choices — the first being the total-energy frame $q_a = 0$ and the third the Newtonian frame. They are three stipulations, not two, and the paper gives the reason this calculation wants the second: it is "geodesic right through tight-coupling, slow decoupling and into the free-streaming era", whereas the energy frame is recommended there for *two* coupled species and this is a three-species problem that crosses decoupling.

The two agree only in the limit of matter domination. Setting $q_a = 0$ in a mixture gives $\rho_C v_C^a + \frac{4}{3}\rho_R v_R^a + \ldots = 0$, so in the CDM rest frame the total flux is *not* zero — the baryons and the photons are moving — and the energy frame tilts away by $(\rho_b + \frac{4}{3}\rho_R)/\rho_C$. Measured in diagnostic D7, that is 0.24 at last scattering for this release's background and 0.49 for the stamped Planck column, with the baryons carrying 37% of it. It is largest exactly where the acoustic physics is, and falls afterwards on both counts, since $\rho_R/\rho_C$ decreases and the baryons stop moving with the photons once they decouple. Recorded as finding T-29; the amendment is decision S8a.

**The one-line reconciliation, and it is the PI's.** The CDM frame is the energy frame *of the CDM component*. For pressureless dust $T^C_{ab} = \rho_C u^C_a u^C_b$, so its flux relative to a frame $u^a = u^a_C + v^a$ is $q^C_a = \rho_C v_a$ to first order, and $q^C_a = 0$ gives $v_a = 0$ — the CDM at rest, exactly. So the two conditions are the same statement about one species, and the conflation this supplement previously carried was the dropped qualifier: "energy frame" unqualified means the *total*, and the total includes the baryons and the photons, which are moving. Saying *of what* settles it.

**The frames, and what each one fixes.** *Annals II* §4.1 enumerates four (its sentence says "three"); the baryon and radiation rows below are frames the paper uses elsewhere and are included because they are the ones a reader is most likely to reach for next.

| **Threading** | **Defining condition** | **What it establishes** |
|---|---|---|
| CDM *(used here)* | $u^a = u^a_C$, equivalently $\tilde{A}_a = 0$ | CDM is at rest; its congruence is geodesic in the conserved-dust model, which is exact rather than approximate. *Annals II* calls it the *zero acceleration frame*. |
| Total energy | $q^{\text{total}}_a = 0$ | Total momentum density vanishes; **individual species can still move**. Agrees with the CDM frame only to $(\rho_b + \frac{4}{3}\rho_R)/\rho_C$. |
| Baryon | $u^a = u^a_B$ | Baryons are at rest; pressure and scattering can accelerate them, so this frame is **not** geodesic. |
| Radiation energy | $q^\gamma_a = 0$ | The radiation brightness dipole vanishes; **higher multipoles can remain**. |
| Newtonian *(used here)* | $\sigma_{ab} = 0 = \omega_a$, where admissible | Useful for the scalar almost-FLRW comparison and for the Bardeen link; generally accelerated. Admissible only in restricted cases. |
| Constant expansion | $\mathrm{D}_a\Theta = 0$, where admissible | Removes the spatial expansion gradient relative to that frame; sometimes useful on small scales. Not used here. |

Reading down the second column is the point: each row fixes a different thing, and no two of them are the same condition. A frame is chosen and then named, not inferred from what it is called.

**One of these frames contains observers and the other does not.** This is the reason the distinction is worth the space, and it is where the $1+3$ covariant account differs from a metric one in substance rather than in notation.

The CDM frame is a congruence of observers. It is defined by a property of the matter — pressureless, collisionless, therefore geodesic — and the definition is exact in general relativity, not a first-order statement. We observe from inside it: a detector in a bound structure is comoving with the matter to the accuracy linear theory works at, and after decoupling the baryons have fallen into the same flow as the cold dark matter, so the baryon and CDM frames coincide at $\eta_0$ where the measurement is made. *It is "us".*

The Newtonian frame is not. $\tilde{\sigma}_{ab} = 0$ is a condition imposed to simplify, not a property any matter has, and *Annals II* §4.1 says so in terms: the frame "is only consistent in restricted cases" and is "particularly useful in making comparisons with much of the analytic literature". It is the covariant counterpart of what conformal Newtonian gauge does in a metric treatment — admissible about almost-FLRW, and a device.

**Why this costs nothing at first order and will not stay free.** $\tau_{A_\ell}$ for $\ell \geq 2$ is frame-invariant at first order, so the *observable* does not care which frame carried it there, and a calculation performed in the Newtonian frame returns what an observer in the CDM frame measures. That is the whole licence for the Bardeen-variable route, and it is a first-order licence.

At $O(\epsilon^2\ell)$ it lapses. The couplings v1.5.0 adds are products of two first-order quantities, and a product of two frame-dependent quantities is not frame-invariant just because each factor's leading frame-dependence cancels. Then the frame has to be the one the observer is actually in, and that is the CDM frame, not a device.

This is what figure F5 is for. Its third panel is the amount of machinery that is frame-dependent — in the Newtonian frame one rank carries everything, in the CDM frame all three are occupied — and its first two panels are the first-order invariance that currently makes that difference harmless. The figure shows both at once, which is why it is the right figure to carry into v1.5.0: it is a picture of exactly what stops cancelling.

**What the external codes use, and why criterion 5b has to check it.** Criterion 5b compares this bundle against CAMB and CLASS. That comparison is only meaningful if the frame each code evolves in is known, because a quantity compared across two frames is not the same quantity. The correspondence below is verified where it could be, and recorded at its real strength where it could not; verifying it is a precondition of 5b, not an afterthought.

CLASS is verified from its own source, at `classy v3.4.1`. `perturbations.c` carries the comment *"cdm velocity vanishes in the synchronous gauge"*, and `index_pt_theta_cdm` is allocated as an evolved variable only when `gauge == newtonian`: in synchronous gauge $\theta_C$ is not evolved because it is zero. The decisive line is CLASS's own error message — *"In the synchronous gauge, it is not self-consistent to assume no CDM: the latter is used to define the initial timelike hypersurface"* — which says, in the code's own words, that its synchronous gauge is realised by the CDM congruence. That is the CDM threading of the table above, $u^a = u_C^a$. Both `gauge = synchronous` and `gauge = newtonian` were run and computed.

CAMB is corroborated but not verified to the same standard. Its Python layer exposes `v_newtonian_cdm` and `v_newtonian_baryon` as *derived outputs*, notes that *"L=1 dipoles are Newtonian Gauge"*, and carries an explicit `frame` argument for sources that are not frame-invariant — which is what a code evolving in one frame and converting on output looks like. That is corroboration from an interface, not a reading of the evolution equations, and the decisive statement lives in CAMB's own covariant formulation paper rather than in anything this bundle inspected. It is recorded at that strength and no higher.

| **Code** | **Gauge used for evolution** | **Corresponding $1+3$ frame** |
|---|---|---|
| CLASS *(default)* | Synchronous, with CDM velocity divergence $\theta_C = 0$ | CDM-comoving: $u^a = u_C^a$, $A_a = 0$; shear generally nonzero. |
| CLASS *(optional)* | Conformal Newtonian / longitudinal, selected with `gauge = newtonian` | Newtonian: scalar shear and vorticity vanish; acceleration generally nonzero. |
| CAMB | Covariant equations evaluated in the CDM frame, equivalent to CDM-comoving synchronous gauge | CDM-comoving: $v_C = 0$, $A_a = 0$; shear generally nonzero. |
| Bolt.jl | Conformal Newtonian gauge, implemented as `BasicNewtonian` | Newtonian: CDM carries an explicitly evolved peculiar velocity relative to this frame. |

Two things follow if the table holds. First, CAMB and CLASS's default both evolve in the frame this bundle uses, which is the convenient case: 5b compares like with like without a frame transformation in between. Second, CLASS's `newtonian` option and Bolt.jl evolve in the other one, so a comparison against either is a comparison across frames and must either transform or restrict itself to the frame-invariant $\ell \geq 2$ multipoles. Figure F5 is the measurement of what that costs: the totals agree and the decompositions do not.

That CAMB is described as using covariant equations in the CDM frame is the part most worth checking rather than assuming, because if true it makes the comparison closer than a gauge-versus-frame

translation would suggest, and a comparison that is closer than it should be is the kind this bundle has already been wrong about once.

**For readers coming from the metric approach: a frame is not a gauge.** This is the distinction most easily lost in translation, because a metric-based treatment usually fixes the two freedoms together and they then look like one freedom.

- A gauge is a choice of coordinates — equivalently a choice of correspondence between the perturbed spacetime and the background. In the $1+3$ covariant approach it is already dealt with: every variable used here vanishes in the background, so by the Stewart–Walker lemma it is gauge-invariant by construction. There is no residual gauge mode to fix and none to worry about.
- A frame is a choice of the timelike congruence $u^a$ along which the fluid and radiation variables are defined. It is a *physical* choice — which observers — and it survives after the gauge question has been answered. It is the only freedom left, and it is the one this release exhibits.

So the familiar correspondences are correspondences and not identities. Synchronous gauge is usually *completed* by frame along the CDM, and conformal Newtonian gauge is usually *completed* by frame along a shear-free congruence; in each case the gauge condition fixes coordinates and a separate stipulation fixes the observers. A statement that is frame-dependent here will appear in a metric treatment as a statement that depends on which gauge was chosen, and the two dependences are not the same dependence. That is why figure F5 can show the *total* common to two frames while the *decomposition* differs: the split into ranks is frame-dependent and the observable is not, and neither fact is about coordinates.

This is also what fixes the status of the next release. Because v1.0.0 is the general case, v1.5.0's restricted case is a recovery and a check — it must return v1.0.0's numbers, and that requirement is the regression test rather than a new result.

**The covariance claim is established symbolically, and that is the stronger form.** Acceptance criterion 2 asks that a change of frame $u^a \to \tilde{u}^a = u^a + v^a$ return the rank-one bracket unchanged up to $\dot{v}_a$. It does, *identically* in exact arithmetic rather than to a tolerance, and the cancellation is exhibited rather than quoted:

- $\tilde{\mathrm{D}}_a \ln T = \mathrm{D}_a \ln T - H v_a$, the extra term arising because the projector moves with the frame and $\dot{\overline{\ln T}} = -H$ in the background;
- $\tilde{A}_a = A_a + \dot{v}_a + v^b \nabla_b u_a$, where $v^b \nabla_b u_a$ is formed *in full* — $H v_a$ together with $\sigma_{ab} v^b$ and $\epsilon_{abc} v^b \omega^c$ — and the latter two leave at $O(\varepsilon^2)$ by series expansion rather than by hand;
- so $\tilde{\mathcal{B}}_1 = \mathcal{B}_1 + \dot{v}_a$ with no shear and no $Hv$. It is the *same* $H$ twice, which is why the cancellation is structural and not a coincidence of two fitted coefficients.

**The curvature half.** The deflection depends only on $\Phi_A - \Phi_H$, whose trace-free screen-projected Hessian is the electric Weyl tensor, $E_{ab} \simeq \frac{1}{2}\mathrm{D}_{\langle a}\mathrm{D}_{b\rangle}(\Phi_A - \Phi_H)$. Since $E_{ab}$ vanishes in the background, the Stewart–Walker lemma makes it frame-invariant at first order, and therefore so is $\Phi_A - \Phi_H$. Both potentials shift by the same $(v' + \mathcal{H}v)/k_{\mathrm{com}}$, and the residual $\tilde{E}_{ab} - E_{ab}$ is identically zero — with a negative control: a rule leaving $\Phi_H$ fixed, or flipping its sign, leaves a residual proportional to $(v' + \mathcal{H}v)$ and fails. The invariance is therefore forced rather than chosen to make a residual vanish. Only the dipole moves, $\tilde{\tau}_a \simeq \tau_a - v_a$; every $\tau_{A_\ell}$ with $\ell \geq 2$ is unchanged, which is the standard any frame dependence surviving at $\ell \geq 2$ must meet: it is an artefact of the reduction, not physics.

**The energy frame is derived, not stipulated.** Imposing $\tilde{A}_a = 0$ gives $v' + \mathcal{H}v = -k_{\rm com}\Phi_A$ — the Euler equation of a pressureless geodesic fluid, which is what the cold dark matter obeys. The normalisation is corroborated independently: in the baryon-dominated limit $R \to \infty$, Eq. (152) reduces to that same equation, so the photon dipole and the frame velocity carry one normalisation.

**What this release does *not* do, stated because the earlier draft of this section claimed it.** It does not compute an assembled spectrum in two frames and draw the difference. Eqs. (177)–(179) are Newtonian-frame expressions: they descend from (106), which the corrections record flags in those words, and the Newtonian frame carries its whole $O(\ell)$ source in the acceleration pair while the CDM frame carries its whole source in the shear pair. Transforming the four input fields and re-evaluating the same equations therefore deletes the coupling that was carrying the source without supplying the one that replaces it: measured, the transfer function moves by a factor of about five, and under the opacity weighting the source collapses to two parts in a thousand. That is not a frame dependence of the sky. The figures which would draw it, and the convergence figure which would test footnote 29 on p. 365 in both frames, require the generic-$u^a$ hierarchy and are deferred with it.

**One coefficient is changed for criterion 5b, and only there.** The photon quadrupole's Thomson collision term carries $\Pi = \Theta_2 + E_0 + E_2$, so polarisation feeds back into the temperature hierarchy and strengthens the diffusion damping. CAMB and CLASS both include this structurally: it is not an output option, and neither can be asked to leave it out — CLASS fails outright in `perturbations_solve` below `l_max_pol_g = 2`, because $\Pi$ needs $E_2$, and CAMB exposes no switch for it. Truncating CLASS's polarisation hierarchy from 10 to 4 moves the acoustic band by 0.043%, which bounds the hierarchy's *depth* and not the *feedback*.

The only way to compare like for like is therefore to carry the polarised coefficient here, for that comparison alone. In (dif-damp)'s quadrupole term *Annals II* prints 4/5; Hu & Sugiyama's unpolarised form is 8/9; with polarisation it is 16/15. Measured on `PLANCK_MODEL`:

| | coefficient | $k_D(\eta_*)$ | ratio |
|---|---|---|---|
| *Annals II* (dif-damp), as printed | 4/5 | **619.9** | 1.000 |
| Hu & Sugiyama, unpolarised | 8/9 | 588.1 | 0.949 |
| with polarisation | 16/15 | **536.9** | **0.866** |

a 13.4% reduction in $k_D$, which takes $e^{-(k/k_D)^2}$ from 0.660 to 0.574 at $k = 400$ and from 0.189 to 0.109 at $k = 800$. It is one of the dominant terms in criterion 5b's residual.

4/5 remains the default and the released value. Every figure, diagnostic and released number in this bundle uses it, because the bundle reproduces and 4/5 is what the page prints. 16/15 is reached only through an explicit argument, by criterion 5b. Decision S31, finding T-39. *Annals II*'s 4/5 matches neither standard coefficient, which is recorded as a question — it may be a convention in how $R$ and the factor of 6 are distributed — and not as a correction.

## 10 The acoustic oscillator

*The oscillator supplies $\delta T(\eta_*, k)$ and $\tau_1(\eta_*, k)$ to the matching. Its period agrees with the antecedent's formula within* 0.5%*; its phase remains unverified.*

### 10.1 The acoustic pair and Eq. (154)

*Annals II* §7.1 prints three equations. Two are the system:

$$\dot{\tau}_1 \simeq -\frac{\dot{R}}{1+R}\,\tau_1 - \frac{1}{1+R}\frac{k}{a}\,\delta T - \frac{k}{a}\Phi_A, \tag{152}$$

$$\dot{\delta T} \simeq -H\Phi_A - \dot{\Phi}_H + \frac{k}{a}\frac{\tau_1}{3}, \tag{153}$$

with $R = 3\rho_M/4\rho_R$ and $c_s^2 = \frac{1}{3}(1+R)^{-1}$. *Annals II*'s $\rho_M$ and $\rho_R$ are the tightly coupled baryons and the photons, not all non-relativistic matter and all radiation. Cold dark matter does not scatter and carries no inertia in the acoustic fluid, so reading $\rho_M$ as the total matter density would change $c_s^2$ and the drag coefficient at every epoch. The implementation uses the species the physics selects: `functions/spectra/acoustic.py` forms $R = 3\rho_b/4\rho_\gamma$ from $\omega_b h^2$ and $\omega_\gamma h^2$, scaling as $a$. The third, (154), is the single second-order oscillator in $\delta T$, and the paper obtains it from the pair by *"ignoring the expansion coupled term (i.e. the small scale approximation)"*.

The pair is implemented and (154) is derived from it, not the other way round. In conformal time, multiplying through by $a$,

$$\tau_1' = -\frac{R'}{1+R}\tau_1 - \frac{k}{1+R}\delta T - k\Phi_A, \qquad \delta T' = -\mathcal{H}\Phi_A - \Phi_H' + \frac{k}{3}\tau_1, \tag{15}$$

and dropping $-\mathcal{H}\Phi_A$ reproduces (154) exactly.

That ordering is the point and not a preference. Implementing (154) directly would bake the small-scale approximation into the only object that exists, and no figure could ever show its cost. Implemented as a pair with a switch, the approximation becomes a measurement: the difference between the two integrations *is* its size, at any $k$ and any $\eta$, and the gate asserts what makes it a *small-scale* approximation — that it must matter less as $k$ grows. *This is the same move as the scale-factor rule of §*the generic source derivatives*: reconstruct the antecedent's general object, and let its printed specialisation fall out as a checkable consequence.*

### 10.2 What drives it, and the one place the backgrounds differ

$\Phi_A$ and $\Phi_H$ are supplied to the oscillator, not solved with it, which is what lets one routine serve every background. They come from the growth equation,

$$\delta'' + \mathcal{H}\delta' - \tfrac{3}{2}\Omega_m H_0^2\,\delta/a = 0, \qquad \Phi \propto D(a)/a, \tag{16}$$

integrated rather than fitted, so that $D \propto a$ in Einstein–de Sitter is a *result* and can be asserted as one.

And that is where the two backgrounds part company. In Einstein–de Sitter $D \propto a$ exactly, so $\Phi$ is constant and $\Phi_A' - \Phi_H'$ vanishes *identically*: there is no integrated Sachs–Wolfe term at all, which is why the antecedent's standard-CDM model never needed one. Once $\Lambda$ dominates, growth stalls while $a$ keeps rising, $D/a$ falls, and $\Phi$ decays. That decay is the late integrated Sachs–Wolfe effect, and it is not bolted on — it appears by itself from (16) the moment $\Omega_\Lambda \neq 0$. Whether it then reaches the observer is a question about the weight it is carried under, which is approximation A-1 and §14.1.

Einstein–de Sitter is therefore a null control and not a weak case. A diagnostic that claims to separate the late ISW must return *exactly zero* on it. A small number is a failure, not a pass.

**The radiation era is absent, and that absence is A-3.** The released $\Phi$ is the matter-dominated one and has no radiation era in it. The radiation-era potential is derived in closed form — with $c_s^2 = \frac{1}{3}$ and $\mathcal{H} = 1/\eta$ the evolution equation becomes $\Phi'' + 4\eta^{-1}\Phi' + \frac{1}{3}k^2\Phi = 0$, whose regular solution is

$$\Phi(\eta) = 3\Phi_i\,\frac{j_1(x)}{x} = 3\Phi_i\,\frac{\sin x - x\cos x}{x^3}, \qquad x = \frac{k\eta}{\sqrt{3}}, \tag{17}$$

so $\Phi \to \Phi_i$ as $x \to 0$ and $\Phi \simeq -3\Phi_i \cos x/x^2$ for $x \gg 1$: decaying as $x^{-2}$ while oscillating. Because it decays on the same timescale the mode oscillates, it drives the oscillator *resonantly*, and that amplification is missing from every peak *height* this bundle produces. (17) is implemented as a diagnostic and is not wired into the released pipeline; D24 shows why that ordering is right, since A-3 is not connected to the peak *position* until $C_1'$ is repaired.

### 10.3 Two corrections to the antecedent, both settled by a stated limit

**B-8: (181) carries the wrong sign on its constant term.** The closed-form solution at last scattering is

$$\delta T(\eta, k) + \Phi_A(\eta_*, k) \simeq \left[\delta T(0, k) + (1 + R)\Phi_A(0, k)\right] \cos(kr_s) - R\,\Phi_A(\eta_*, k), \tag{180}$$

and (181) specialises it with the adiabatic condition $\delta T(0, k) = -\frac{2}{3}\Phi_A(0, k)$. Carrying that through,

$$\left[\delta T + \Phi_A\right](\eta_*, k) \simeq \tfrac{1}{3}\Phi_A(0, k)\,(1 + 3R_*)\cos(kr_s^*) \;-\; R_*\Phi_A(0, k), \tag{18}$$

where the printed (181) has $+R_*$. Its own stated limit settles which. At $r_s^* \to 0$ the corrected form gives exactly $\frac{1}{3}\Phi_A$, which is what the text asserts and what (196) descends from; the printed form gives $\frac{1}{3}\Phi_A(1 + 6R_*)$, and the assertion would need $R_* \to 0$, which is neither stated nor true at last scattering. The oscillating parts agree exactly, so this is one sign and not a different solution. *Taken literally it costs a factor of twenty in $C_\ell$*, and the printed form is kept as a second function so that the difference is demonstrable rather than asserted.

**B-10: (J.5) is printed with the exponent of the wrong sign.** The Bond–Efstathiou transfer function $T(k) = [1 + ((ak) + (bk)^{3/2} + (ck)^2)^\nu]^{-1/\nu}$ is printed with $+1/\nu$. As printed it *grows* without bound — $6.7 \times 10^3$ at $k = 10\,\mathrm{Mpc}^{-1}$ against $1.5 \times 10^{-4}$ — which inverts the physics, since modes entering during radiation domination stagnate and the transfer function must suppress small scales. The large-scale limit cannot see it, because the bracket tends to 1 and $1^{\pm 1/\nu} = 1$, which is why the text's own check, *"$T(k) \sim +1$ on large scales"*, passes on the wrong form. Only the $k^{-2}$ falloff distinguishes them, and that is what the fit is a fit to.

*Both are the same shape of error and both are caught the same way: by evaluating the antecedent's own stated limit on the antecedent's own printed expression.* Neither needed an external code.

### 10.4 The phase is the sound horizon, and the start is derived from $k$

$\delta T$ oscillates in $kr_s$, not in $k\eta$, so every number in this section depends on $r_s$ being the sound horizon *from the big bang*, $r_s(\eta) = \int_0^\eta c_s\,\mathrm{d}\eta'$ — which is also what the antecedent means in its own peak formula $\ell_j \simeq |r_\theta/r_s|\,j\pi$.

Finding T-51 is that it was not. The sound horizon was accumulated from the first point of the line-of-sight grid, at $z = 3000$, returning between 36% and 49% of $r_s$, so $\cos(kr_s)$ carried roughly half the phase it should. And criterion 5b could not see it, because the closed-form route inherited the same truncation independently of the ODE route: the two agreed, and they agreed about what they shared. *That is T-3's rule on a third quantity — two agreeing routes agree about what they share, and a shared input is not cross-checked by them.*

The repair is a derivation and not a tuned start. A mode may be started where $k\,r_s(\eta_{\mathrm{start}}) \le \varepsilon$ with $\varepsilon = 0.1$ radians, one part in sixty of a cycle, because (180)'s initial condition is the adiabatic growing mode only while the mode is outside the sound horizon; starting at finite $kr_s$ *imposes* a phase rather than inheriting one. The bundle reached $kr_s = 10.7$ at $k = 600$ — nearly two full cycles in. The grid is derived the same way: $c_s^2 \le \frac{1}{3}$ bounds the period below by $2\pi\sqrt{3}/k$, so the sample count is arithmetic rather than a knob, at 24 samples per oscillation.

One number in this section is a modelling choice and is now stated once. Tight coupling is held to end at $\eta_{\mathrm{couple}} = \min(\eta_0,\, 6\,\eta_*)$. Six is not derived and is not claimed to be. T-58 is what happened while

it was not stated once: the factor lived in three places and the route criterion 5b gates integrated the oscillator all the way to $\eta_0$, keeping the photon–baryon fluid ringing for seven times the span over which it is a fluid. The mismatched window alone costs $1.5 \times 10^{-1}$ at $k = 3$ and 1.6 at $k = 10$, against 5b's own $5 \times 10^{-8}$ — and 5b could not see that either, for the same reason. Q16 is the assertion that closes it: $\mathcal{V}(\eta_{\rm couple})/\mathcal{V}(\eta_*)$ below a stated threshold on every background.

### 10.5 Where the oscillator is gated, and the one thing no gate sees

| gate | what it tests | value |
|---|---|---|
| peak spacing | $\Delta\ell = \pi r_\theta/r_s$, the antecedent's own formula, read off the turning points of $\delta T + \Phi_A$ in $k$ on Einstein–de Sitter, where Appendix G is exact | 191.7 against 192.7, ratio **0.995** |
| the same, `z_max` control | what the pre-T-51 start does to it | 568.7, ratio 2.95 |
| pair → (154) | that dropping $-\mathcal{H}\Phi_A$ reproduces the printed oscillator, and that the difference falls with $k$ | exact; small-scale behaviour asserted |
| ODE → (180) | that the integration converges to the closed form when $R$ is held constant | — |
| B-8 | the $r_s^* \to 0$ limit of both forms of (181) | $\frac{1}{3}\Phi_A$ against $\frac{1}{3}\Phi_A(1 + 6R_*)$ |

Every one of these measures a period, a limit or a reduction. Not one of them measures a phase. The spacing gate takes differences of consecutive turning points, so the position of the *first* one never enters: it is blind, by construction, to a constant offset in $kr_s$.

That is why it is worth having, and it is the sharpest localisation in this supplement. The oscillator's period is right to 0.5% and the projection at the first peak is faithful to 3% (D25, §14.1), while the released first peak sits at $0.635\,\ell_A$ rather than 0.757. Correct spacing with a displaced first peak is a phase offset, not a scale error: nothing here is stretching $r_s$ or $r_\theta$. The defect lies between the oscillator and the projection, which is the transport assembly, and §14.1 is where that is itemised.

## 11 The first acoustic peak, predicted

*The peak position is analytically known. Knowing it turns a displacement into a budget with named contributors, and closes two candidate mechanisms that further measurement would not have closed.*

### 11.1 The sound speed is the antecedent's own, derived and not assumed

*Annals II* (152) and (153) are the covariant acoustic pair:

$$\tilde{\tau}_1' = -\frac{R'}{1+R}\,\tilde{\tau}_1 - \frac{k}{1+R}\,\delta\tilde{T} - k\Phi_A, \tag{152}$$

$$\delta\tilde{T}' = -a\mathcal{H}\Phi_A - \Phi_H' + \frac{k}{3}\,\tilde{\tau}_1. \tag{153}$$

**Eliminate $\tilde{\tau}_1$ between them.** (153) gives $\tilde{\tau}_1 = \frac{3}{k}\big(\delta\tilde{T}' + a\mathcal{H}\Phi_A + \Phi_H'\big)$; substituting into (152) and multiplying by $k/3$ returns

$$\delta\tilde{T}'' + \frac{R'}{1+R}\,\delta\tilde{T}' + k^2 c_s^2\,\delta\tilde{T} \;=\; -\frac{k^2}{3}\Phi_A - \mathcal{F}, \qquad \boxed{c_s^2 = \frac{1}{3(1+R)}} \tag{19}$$

with $\mathcal{F}$ the forcing carried by the potentials. **The standard sound speed and the standard baryon-drag coefficient $R'/(1+R)$, with nothing imported.** The coefficient of $\delta\tilde{T}$ in the eliminated equation is $k^2/[3(1+R)]$ exactly; the residual of the elimination is zero symbolically. *This is analysis in the sense the acceptance spine uses the word, and it is the one step in this section that needs no measurement.*

The sound horizon and the angular scale follow:

$$r_s(\eta_*) = \int_0^{\eta_*} c_s \, \mathrm{d}\eta, \qquad r_\theta = \eta_0 - \eta_*, \qquad \ell_A \equiv \frac{\pi r_\theta}{r_s(\eta_*)}. \tag{20}$$

In the undriven, undamped limit (19) has $\delta\tilde{T} \propto \cos(kr_s)$, whose extrema are at $kr_s = m\pi$. **That the oscillator does this is measured, not assumed**: $(\delta\tilde{T} + \Phi_A)(\eta_*)$ tracks $\cos(kr_s)$ in exact phase and period, with a contrast better than the antecedent's own closed form (181), and the period is right to 0.5% (§10).

### 11.2 But $\ell_1 \neq \ell_A$, and the difference is not all physics

**It is tempting, and wrong, to read $kr_s = \pi$ as $\ell = \ell_A$.** That step converts a statement about the source into a statement about the multipole, and §3b of the project's own standing rule exists for exactly that class of step. The observable is

$$C_\ell = \frac{2}{\pi} \int \frac{\mathrm{d}k}{k} \, [k^3 P_\Phi(k)] \, |S(k)\, j_\ell(kr_\theta)|^2, \tag{21}$$

and $k$ is integrated. Write the displacement as

$$\ell_1 = \ell_A \, (1 - \varphi_1), \qquad \text{so} \qquad \varphi_1 \equiv 1 - \frac{\ell_1}{\ell_A}, \tag{22}$$

which is the first row of the standard peak parametrisation and is the form in which Hu & Sugiyama — this bundle's declared analytic comparator, decision **S37** — organise the same quantity.

**The projection alone supplies a $\varphi_1$, and it is not small.** (21) evaluated on an *undriven* monopole $S = \cos(kr_s)\, e^{-(k/k_D)^2}$, at this bundle's own diffusion scale $k_D = 754.6$, with $n_s = 1$ and $D_\ell = \ell(\ell+1)C_\ell/2\pi$:

| **undriven source** | $\ell_1/\ell_A$ | $\varphi_1$ |
|---|---|---|
| $\cos(kr_s)$, $R = 0$ | 0.8524 | +0.1476 |
| $(1+R)^{-1/4}\cos(kr_s) - R$, $R = 0.3$ | 0.8303 | +0.1697 |
| $(1+R)^{-1/4}\cos(kr_s) - R$, $R = 0.6$ | 0.8083 | +0.1917 |

**So roughly 0.15 of $\varphi_1$ is the projection and is present in any assembly whatsoever.** It is the $\mathrm{d}k/k$ measure, the $\ell(\ell+1)$ weighting and the kernel's own structure, and no physics is responsible for it. *This row is a demonstration and its configuration is stated so that it can be re-run; it is not a gate and nothing is scored against it.*

### 11.3 The budget is a total, and the split inside it does not survive

Against $\ell_A = 272.177$ for `PLANCK_MODEL`, four configurations measured in this bundle:

| **configuration** | $\ell_1$ | $\ell_1/\ell_A$ | $\varphi_1$ |
|---|---|---|---|
| released v1.3.0 | 266.6 | 0.9795 | +0.0205 |
| $\kappa'\kappa''\hat{v}_B$ suppressed, A-3 absent | 236.6 | 0.8693 | +0.1307 |
| matched CAMB/CLASS, $N_{\mathrm{eff}} = 0$ *(external)* | 206.0 | 0.7569 | +0.2431 |
| **D24**, $\kappa''$-suppressed | 172.75 | 0.6347 | +0.3653 |

Every entry is a measurement and every entry stands. Three things have to be said before a difference between two of them is read as a physical displacement, and together they withdraw the reading this subsection carried in editions before v1.3.20.

**The denominator is under query and the $\ell_1$ column is not.** $\ell_A = 272.177$ is the committed value in `diagnostics/d4-monopole-v1.3.0.txt`. Calling that artefact's own generator on current `main` returns 266.734, with $r_\theta$ moving from 3.128965 to 3.127382: a difference of 5.5 multipoles, or 2.0%. Both manifests cover the file and both pass, which is the point rather than a mitigation, since a manifest hashes bytes and cannot say whether a generated file still agrees with its generator. The question is open with the implementation stream and is not settled here. Every $\varphi_1$ and every ratio in this subsection divides by 272.177 and is to be read with that attached; the $\ell_1$ column does not depend on it.

**The second row is the configuration that fails the frame test by a factor of three.** It is the same configuration §15 examines under `slot_weight_split` with the two boundary controls, where it puts the CDM peak at $0.869\,\ell_A$ and the Newtonian peak at $0.277\,\ell_A$. Those two threadings have equal multipoles at every $\ell \geq 2$ to first order in $v$, so a configuration separating them by 3.1 has not produced a peak position: it has produced two, and the section that measures this says of it that it is not a repair and that nothing following it is a result. A number with that provenance cannot be the midpoint of a budget, because the budget's arithmetic requires it to be a single frame-independent $\ell_1$.

**The step from the second row to the third is not a measurement of driving.** The third row is the external comparator, as its label says, and A-3 is not wired into `solve_acoustic` — stated in this section below and unchanged at v1.3.20. No configuration of this bundle with the radiation-era driving supplied has been run, so the interval $236.6 \to 206.0$ is not a displacement this bundle produced. It is the residual between a configuration of this bundle and a spectrum from another code, and naming it *the driving* assigns the whole of that residual to one named absentee with nothing excluding the others. §15 states the general form of that objection against this very number, and it applies to the budget as much as to the agreement it was raised about.

**The four rows support a total and do not support a decomposition of it.** The released configuration sits 60.6 multipoles above the matched comparator, $\varphi_1 = +0.2226$ against $+0.0205$, and that total is a comparison between two endpoints that are each what they claim to be. Its decomposition is not established. The two differences $+0.1102$ and $+0.1124$ are arithmetic on the column above and remain arithmetic; their near-equality is a property of where the middle row happens to fall and is not evidence that two mechanisms of equal size are at work, because the middle row is not a peak position and the second difference is not a mechanism. Editions before v1.3.20 read the near-equality as a finding and said the two halves agree to 2.0% of their mean with neither tuned. That reading is withdrawn here. The arithmetic was never in question and is unchanged.

**The closed-form projection stands, and what it corroborated does not.** The demonstration above, run at this bundle's own $k_D$ with nothing fitted, puts an undriven monopole's first peak at $\ell_1 = 232$, which is $0.852\,\ell_A$. That is a calculation and not a run, so nothing withdrawn above touches it. What it was previously said to confirm — the second row's 236.6, to 1.9% — is a number this subsection no longer reads as a peak position, so the corroboration is withdrawn with it. What survives is the comparison that does not need the middle row: the released configuration sits at $0.980\,\ell_A$ against the $0.852\,\ell_A$ a projection alone supplies, so the released peak is some 15% further out than an undriven monopole projects, and that distance is what an account of the first peak has to produce. The present document does not produce it.

### 11.4 What this closes, and it closes two things

**The peak's displacement is not decomposed, and the document should not be read as decomposing it.** Removing the $\kappa'\kappa''\hat{v}_B$ artefact moves $\ell_1$ down and the released configuration sits above the matched comparator, so the artefact is one contribution and its sign is settled. Its size is not

separable from the rest by the rows above, for the three reasons just given, and no second contribution of comparable size has been measured in this bundle at all. Editions before v1.3.20 stated that the displacement came in two equal parts and that a repair addressing one of them would overshoot or undershoot by half the distance. That statement is withdrawn: it required the middle row to be a peak position and the interval above it to be a mechanism, and neither holds.

**The $j'_\ell$ slot is not a peak repair, and that is now settled.** Restoring the Doppler term at its own kernel — discrepancy B-17 (iv), derived in §14.1 — moves $\varphi_1$ by $\mathbf{-0.015}$ at full amplitude, and *in the wrong direction.* It fills troughs and adds power below the peak, which is what its $\ell$-leverage already said: at fixed $x$ the $\ell$ maximising $|j'_\ell|$ sits about 37% below the one maximising $|j_\ell|$. **B-17 (iv) remains a correctness defect and is withdrawn as a peak mechanism.**

**The late integrated Sachs–Wolfe weight is not a peak repair either, and that is measured.** Carrying the bulk at $W$ rather than $\mathcal{VD}$ — the (98) branch — moves the late share from 0.0000 to $10^{-4}$ and **does not move the peak at all**. The prediction that it would not was stated before the run.

### 11.5 The boundary requirement, and it is now a single statement

Two of the three candidate carriers of the missing 0.112 are closed by the rows above. **What remains is the boundary value itself.**

> **The driving half of $\varphi_1$ does not live in the line-of-sight integral. It lives in $(\delta\tilde{T} + \Phi_A)(\eta_*)$.** The oscillator (19) must be integrated *through the radiation era*, where $\Phi$ decays inside the sound horizon and forces $\mathcal{F} \neq 0$ resonantly, rather than started from a matter-era initial condition in which $\Phi$ is already constant.

That is approximation **A-3**. It is derived in closed form in this bundle as $\Phi = 3\Phi_i\, j_1(x)/x$ with $x = k\eta/\sqrt{3}$, it is gated by `tests/test_radiation_potential.py`, and **it is not wired into `solve_acoustic`**. It enters (19)'s forcing and therefore the boundary value, and it does not enter the transport assembly at all.

**The test, stated before it is run.** Suppress the artefact and wire A-3 together. **The first peak must land at** $206 \pm 6$ — that is $\varphi_1 = 0.243$ — on `PLANCK_MODEL`, and the CDM and Newtonian peaks must land there together, which the artefact-suppressed configuration on its own does not do. Editions before v1.3.20 added that either change alone must land near 236.6 from opposite sides. That prediction is withdrawn on both halves. The two changes were never measured apart, so additivity in $\varphi_1$ is an assumption rather than a result; and both one-change configurations are reached by moving down from the released 266.6, so neither approaches the midpoint from below and there are no opposite sides for them to approach from. *And on Einstein–de Sitter, where $\Omega_r = 0$ by construction and A-3 cannot act at all, the peak must stay at the undriven position and must not move when A-3 is enabled.* That null control is what distinguishes a driving mechanism from a fit.

**One number is not yet reconciled and is named rather than used.** The released Einstein–de Sitter first peak is reported at $\ell_1 = 220$ against $\ell_A = 192.7$, which is $1.142\,\ell_A$ — above $\ell_A$ rather than below it, and inconsistent with both the undriven prediction ($0.852\,\ell_A$) and the `PLANCK_MODEL` released ratio (0.980). Either $\ell_A = 192.7$ is being read against a different model — the `CDM`/`CDM_MODEL` identifier collision, finding **T-134** — or the two peaks are located by different estimators. **Nothing in this section rests on it**, and it is a lookup rather than a question.

# 12 Free streaming, and where every $\ell \geq 3$ comes from

*The source closes at rank two and the oscillator evolves ranks zero and one. Free streaming generates higher multipoles and supplies the line-of-sight kernel.*

## 12.1 Free streaming after scattering

After last scattering $\kappa' \to 0$, and $\mathcal{B}_{A_\ell} \equiv 0$ for $\ell \geq 3$ identically. What remains of the transport equation is its streaming couplings alone — *Annals II* Appendix F, p. 379, in proper time and with the comoving wavenumber:

$$-\dot{\tau}_\ell = \frac{k}{a}\Big[w_\ell\,\tau_{\ell+1} - \tau_{\ell-1}\Big], \qquad w_\ell = \frac{(\ell+1)^2}{(2\ell+3)(2\ell+1)}, \qquad \ell \geq 2. \tag{23}$$

Multiplying through by $\beta_\ell$ and changing to conformal time gives the form this bundle integrates as its reference, (F.3):

$$-(\beta_\ell\tau_\ell)' = k\left[\frac{\ell+1}{2\ell+3}(\beta_{\ell+1}\tau_{\ell+1}) - \frac{\ell}{2\ell-1}(\beta_{\ell-1}\tau_{\ell-1})\right], \qquad \beta_\ell = \frac{\ell!}{(2\ell-1)!!}. \tag{24}$$

Each multipole is coupled to its two neighbours and to nothing else. That is the whole content: the ladder has no closure and no termination, so the three ranks the source supplies generate every rank above them. It is also the only $\ell$-coupling in the released hierarchy — every other coupling in the scheme reaches $\ell \pm 1$ or $\ell \pm 2$ and arrives later, by frame, which is what F8 draws.

**The $\ell$-weights are where an error would hide, and it would hide at large $\ell$.** $w_\ell \to \frac{1}{4}$ and the two bracket weights in (24) tend to $\frac{1}{2}$ and $\frac{1}{2}$: every coupling in the recursion is a ratio of linear polynomials and tends to a constant. So a coupling that grew like $\ell$ would be invisible at low $\ell$ and divergent at high $\ell$ — which is exactly the regime this paper is about, and the reason the Appendix F chain is the acceptance spine rather than a smoke test. Decision S9.

## 12.2 The $\ell = 0$ row is the rank-zero source, seen from the other side

Continued to $\ell = 0$, (23) has $w_0 = \frac{1}{3}$ and reads

$$\dot{\tau}_0 = -\frac{k}{a}\,\frac{\tau_1}{3}, \tag{25}$$

the monopole fed by the dipole alone. That is the same combination the rank-zero gravitational source is built from, $\mathcal{B}_0 = -\frac{1}{3}\mathrm{D}^a\tau_a$, giving $a\mathcal{B}_0 = -\frac{k}{3}\tau_1$ in mode form.

The two are one statement, and that is why $\tau_0 \equiv 0$ is a constraint rather than initial data. $T$ is *defined* as the monopole of $T(x,e)$, so the $\ell = 0$ equation does not evolve a free quantity — it fixes $(\ln T)^{\cdot}$, and $\mathcal{B}_0$ is a functional of the solution rather than an external forcing. Restoring $a\mathcal{B}_0$ to $C_0$ as though it were a source therefore counts it twice, which is finding T-107 and is settled at §14.2.

*This connection is worth drawing because the two appear in different sections and look like different objects.* They are the same row of the same hierarchy.

## 12.3 The closed form, and why the kernel is a spherical Bessel function

Appendix F, p. 380, states the solution directly: *"the solution to the mode coefficients with respect to the time-like integration in the flat ($K = 0$) almost-Friedmann–Lemaître case are spherical Bessel functions."* A unit monopole released at $\eta_i$ projects onto

$$\tau_\ell(\eta) = j_\ell(k(\eta - \eta_i)), \tag{26}$$

and in this bundle the per-multipole normalisation is fixed by the recursion and checked rather than assumed (`functions/harmonics/free_streaming.py`, `analytic_projection`).

This is the one fact that makes the line-of-sight solution a solution and not an ansatz. The kernel $j_\ell(k\Delta\eta)$ in (102) is not a window chosen for convenience and it is not a projection imposed on the hierarchy from outside: it is the hierarchy's own free-streaming solution, evaluated along the null cone. Every statement about the $k \to \ell$ map — that power at multipole $\ell$ comes from $k \simeq \ell/r_\theta$, that a feature fixed in $k$ appears at a fixed $\ell$ — is a property of (26) and inherits whatever is true of the recursion that produced it.

### 12.4 Three normalisations, none privileged, and one printed slip

Appendix F writes one recursion three ways, and the bundle integrates all three *as printed*:

| | **written in** | **the variable** |
|---|---|---|
| (F.1) | proper time, comoving $k$ | $\tau_\ell$ |
| (F.3) | conformal time | $\beta_\ell\tau_\ell$ |
| (F.4) | conformal time | $\alpha_\ell^{-1}\tau_\ell$, the Ma & Bertschinger form |

with $\alpha_\ell = (2\ell+1)/\beta_\ell$. Privileging none of them is the point: the three differ by $\ell$-dependent factors, so an error in a weight shows as a disagreement between normalisations that no single one can reveal. This is the same discipline the spectrum uses when it reports (186) and (187)–(188) separately rather than asserting they agree.

**(F.4) as printed carries a factor its own text removes.** The display keeps $(2\ell+1)$ on the left, while the sentence preceding it specifies a division by $(2\ell+1)$ that was not applied; only after that division is it the Ma & Bertschinger form the text says it is. The bundle divides — finding B-2.

**And $k$ is comoving here, which the conventions sheet does not say of every section.** Appendix F's $k$ is comoving and $k_{\rm phys} = k_{\rm com}/a$; the explicit $1/a$ in (23) against a proper-time derivative is what fixes it. The project's conventions sheet declares $k \equiv k_{\rm phys}$ in its *harmonic-recursion* section, so the two differ by a factor $a$ in precisely these equations. A bare $k$ is therefore ambiguous at exactly the point where it matters most, and every equation in this section is written in $k_{\rm com}$.

### 12.5 What gates this, and what each gate can and cannot see

| **gate** | **what it tests** | **result** |
|---|---|---|
| **F1** | the recursion against **four external hierarchies**, in two normalisations and two phase conventions — criterion 1 | residuals $\lesssim 10^{-13}$ |
| `test_appendix_f_chain` | that no coupling grows with $\ell$: an $O(\ell)$ coupling wired in early fails it at large $\ell$ | gate on S9 |
| `sign_control` | the *negative* control — the same recursion with the relative sign of the two couplings flipped. **It must fail to reproduce the external form**, and a suite in which it passes is not testing the phase | negative control |
| (26) | the integrated recursion against $j_\ell(k(\eta-\eta_i))$ for a unit monopole, per multipole | normalisation checked, not assumed |
| **T-3** | that the $\eta$ grid resolves $j_\ell(k(\eta_0-\eta))$, whose period in $\eta$ is $\pi/k$ | raises rather than aliasing |

**What none of them sees.** All five test the recursion *with the source switched off*. They establish that free streaming is reconstructed faithfully and say nothing about whether the right thing is streamed: the matching at $\eta_*$ is a separate question and is §14.1's. F1 at $10^{-13}$ is therefore not evidence about the spectrum, and reading it as such would be the error the acceptance spine exists to prevent — a reconstruction can be exact and be matched onto the wrong source.

# 13 The line-of-sight assembly, re-derived

*Three findings in this bundle – the plateau standing on a scattering rate, the bulk carrying a term $1.8\times10^5$ times too large, and the $j_\ell'$ slot standing empty – have been carried as three. They are one operation, and this section is the derivation that shows it.*

## 13.1 Conventions, so that no sign is assumed

Write $W(\eta)=e^{-\kappa}$ for the opacity weight and

$$\mathcal{V}\equiv|W'|,\qquad \int\mathcal{V}\,\mathrm{d}\eta=1, \tag{27}$$

the visibility, which is a normalised probability density. **Everything below is written in $W$ and its derivatives, so that it carries no sign convention**; the bundle's own sign for $\kappa'$ is fixed at `integral_solution.py:205` and none of the structure here depends on it. The free-streaming kernel is $K(\eta)=j_\ell(k(\eta_0-\eta))$, and the only two kernel identities needed are

$$K'=-k\,j_\ell'(k(\eta_0-\eta)),\qquad K''=+k^2\,j_\ell''(k(\eta_0-\eta)). \tag{28}$$

**A derivative taken on the kernel is a Bessel derivative and costs nothing. A derivative taken on a coefficient is a derivative of whatever that coefficient contains.** That distinction is the whole of this section.

## 13.2 The ancestor, and why a by-parts is required at all

The hierarchy's integrating-factor solution is the ancestor form

$$\tau(\eta_0)=\int\mathrm{d}\eta\,W\big[\,C_0K+C_1K'+C_2K''\,\big], \tag{29}$$

with (28) placing rank 0 at $j_\ell$, rank 1 at $j_\ell'$ and rank 2 at $j_\ell''$. **In this form no coefficient is differentiated and $\kappa''$ cannot arise anywhere.**

**But** (29) **does not display the Sachs–Wolfe term**, and that is why a by-parts is taken. In the $1+3$ formulation the rank-one source is a *gradient*: $\mathcal{B}_a=\mathrm{D}_a\ln T+A_a$, so $a\mathcal{B}_1=k(\delta\tilde T+\Phi_A)$ and (90) returns

$$-C_1=\frac{1}{k}(a\mathcal{B}_1+\kappa' v_B)=(\delta\tilde T+\Phi_A)+\kappa'\hat v_B. \tag{30}$$

The Sachs–Wolfe combination sits at $K'$, not at $K$, because covariantly it enters as the gradient of the temperature plus the acceleration. Integrating by parts moves it to $K$:

$$\int WC_1K'\,\mathrm{d}\eta=\Big[WC_1K\Big]-\int(WC_1)'K\,\mathrm{d}\eta. \tag{31}$$

**The boundary term is the Sachs–Wolfe term at $j_\ell$.** That is what (102)'s "two boundary terms and one integrated term" are, and the structure is correct. *The by-parts is not optional and it is not the error.*

### 13.3 What the exact by-parts protects, and it is a cancellation

(30) carries two objects of different kinds, and they behave differently under (31).

**The gravitational half.** With $C_1^{\rm grav} = -(\delta\tilde{T} + \Phi_A)$,

$$-\big(WC_1^{\rm grav}\big)' = \underbrace{W'\,(\delta\tilde{T} + \Phi_A)}_{\text{at } \mathcal{V}} \;+\; \underbrace{W\,(\delta\tilde{T} + \Phi_A)'}_{\text{at } W}. \tag{32}$$

**The integrated Sachs–Wolfe channel arrives at $W$ automatically**, carried in $(\delta\tilde{T} + \Phi_A)' \supset \Phi'_A$. Nothing has to be added for it; the exact by-parts produces it.

**The scattering half, and this is where everything is decided.** With $C_1^{\rm scatt} = -\kappa'\hat{v}_B$, the weighted coefficient has the magnitude and the kernel of the standard Doppler source:

$$WC_1^{\rm scatt} = -\,\mathcal{V}\,\hat{v}_B \qquad \text{(up to the sign convention), with } \int \mathcal{V} = 1. \tag{33}$$

**It is bounded and normalised, and its derivative is a total derivative that integrates to a boundary difference.**

*That is a statement about the magnitude, and it is not a statement that the bundling in (90) is harmless.* **It is not**: the instrument **c6** measures the frame obstruction the bundling creates. Under a boost $\Delta v_B = -v$, the bundled $C_1$ shifts by $-\dot{s} + \kappa' s$, so the Thomson piece lands in the *same* $\alpha C_1 K'$ slot as the gravitational piece and the $K'$ coefficient becomes $2\alpha\mathcal{V}$, which no $(\alpha, \beta, \gamma)$ can cancel. Carried as its own undifferentiated term the coefficient is free, invariance fixes it at $c = \alpha$, and the residual goes to machine zero. **c6 and this section measure two consequences of one bundling** — c6 the frame obstruction, this section the $\kappa''$ artefact that *differentiating* it creates — **and the remedy is the same in both.**

Split (31)'s bulk on this half and the two pieces are, verified symbolically:

| piece | coefficient of $\kappa''$ | coefficient of $\kappa'^2$ | what it is |
|---|---|---|---|
| $W'\,C_1^{\rm scatt}$ | $0$ | $+\hat{v}_B e^{-\kappa}/k$ | the partner, from differentiating the *weight* |
| $W\,C_1^{\rm scatt\prime}$ | $-\hat{v}_B e^{-\kappa}/k$ | $0$ | from differentiating the *coefficient* |
| $\big(WC_1^{\rm scatt}\big)'$ **(exact)** | $-\hat{v}_B e^{-\kappa}/k$ | $+\hat{v}_B e^{-\kappa}/k$ | the two halves of $\mathrm{d}(-\mathcal{V}\hat{v}_B)/\mathrm{d}\eta$: their *coefficients* are equal and opposite, so the sum goes as $\kappa'^2 - \kappa''$ |

**The sum is bounded because it is a total derivative, not because the two terms cancel pointwise.** *An earlier form of this paragraph said they were "individually enormous and they cancel"; that is withdrawn.* The gate measures the largest ratio of either piece to their sum at only $\simeq 8$, and **c7** records that $\kappa'' = -\kappa'^2$ holds *at the visibility maximum alone*, where $\dot{\mathcal{V}} = 0$ — so any statement about powers of $\kappa'$ is local and not general. What is true without qualification is the structural statement: **the two are the halves of one total derivative of a bounded, normalised object, and that is what is protected by taking the derivative of the weighted coefficient $(WC_1)'$ rather than of the bare one.**

### 13.4 What (102) does, and it is one operation

The released bulk is $\mathcal{V}(C_0 - C_1' + C_2'')$: **the weight outside, the derivative inside**. Against the exact $(WC_1)' = W'C_1 + WC_1'$ that is two departures at once, on the same slot:

| **exact** | **(102)** |
|---|---|
| $W'C_1$ — the partner | **absent**. Dropped entirely |
| $WC_1'$ — $\kappa''$ at one power of $e^{-\kappa}$ | replaced by $\mathcal{V}C_1'$ — $\kappa''$ at $\kappa' e^{-\kappa}$, **one extra factor of** $\kappa'$ |

The ratio of what (102) carries on the $\kappa''$ term to what the exact form carries is **exactly** $\kappa'$, verified symbolically. And the partner it should have been summed with is gone.

> **So (102) carries neither the total derivative nor either of its halves.** One half is rescaled by $\kappa'$ and the other is deleted, and what remains is a term the exact form does not contain at all.

**The magnitude follows, and it is the measured one.** What survives is $\mathcal{V}\kappa''\hat{v}_B$ against a physical Doppler source $\mathcal{V}\hat{v}_B$, so the ratio is $\kappa''$. With $\kappa'(\eta_*) = 3.82\times10^2$ on Einstein–de Sitter and a recombination width of order $10^{-2}\eta_*$, $\kappa'' \sim 2\times10^5$. **T-94 measures** $1.8\times10^5$, and that number is no longer an anomaly to be explained: it is $\kappa''$.

**And the same operation empties the $j_\ell'$ slot and poisons the boundary.** (31) was applied to *all* of $C_1$. (33) shows the scattering half was already in final form at $K'$ — it is $\mathcal{V}\hat{v}_B$ projected through $j_\ell'$, which is the standard Doppler term and needs no by-parts at all. By-parting it anyway does two further things:

- **it moves the Doppler term out of $j_\ell'$**, which is discrepancy B-17 (iv) — *not a term missing from (102) but a term (102) removed*; and
- **it carries $\kappa'\hat{v}_B$ into the boundary term**, where the Sachs–Wolfe combination belongs alone. At $k = 2$–$5$ the $\kappa'$ share of $C_1(\eta_*)$ is measured at 1.046–1.088, so at the modes that set the plateau *what stands where the Sachs–Wolfe monopole should be is a Doppler term multiplied by a scattering rate of* $3.8\times10^2$. That is findings **T-101** and **T-121**, and it is the $\ell \simeq 7$ bump.

**Three findings, one operation.** B-17 (iv), T-94 and T-101 are not independent defects to be repaired separately. They are what happens when a by-parts is applied to a sum whose two halves require different treatment.

### 13.5 The repair, and the condition that makes it a gate

> **Split $C_1$ before the by-parts.** The gravitational half is a gradient and must be by-parted, which produces the Sachs–Wolfe boundary term and the integrated Sachs–Wolfe channel at $W$. The scattering half is already $\mathcal{V}\hat{v}_B$ and must be left at $K'$, where it is the Doppler term at $j_\ell'$.

Written out, the rank-one assembly is

$$\int WC_1K' \;=\; \Big[\; \underbrace{WC_1^{\rm grav}K}_{\text{Sachs–Wolfe, } j_\ell} \;\Big] - \int \Big( \underbrace{W'C_1^{\rm grav}}_{\mathcal{V},\, j_\ell} + \underbrace{WC_1^{\rm grav\prime}}_{W,\ \text{ISW},\ j_\ell} \Big) K \;+\; \int \underbrace{WC_1^{\rm scatt}}_{\mathcal{V}\hat{v}_B} K'. \tag{34}$$

**No $\kappa''$ appears. No $\kappa'^2$ appears. Every channel is at the weight and the kernel the standard assembly puts it at**, and it was derived from the covariant ancestor rather than imported.

**The admissibility condition, which is one assertion.**

> *A coefficient holding a rate may not be carried through an integration by parts. It may hold one — **c7**'s rule to the contrary was softened on 2026-10-21, finding **T-113** and decision **S48**, because $W(\kappa' S) = \mathcal{V}S$ is exact — but it may not be differentiated while it does. $C_1$ holds one, by (90), so the by-parts may be applied to $C_1^{\mathrm{grav}}$ and may not be applied to $C_1^{\mathrm{scatt}}$.*

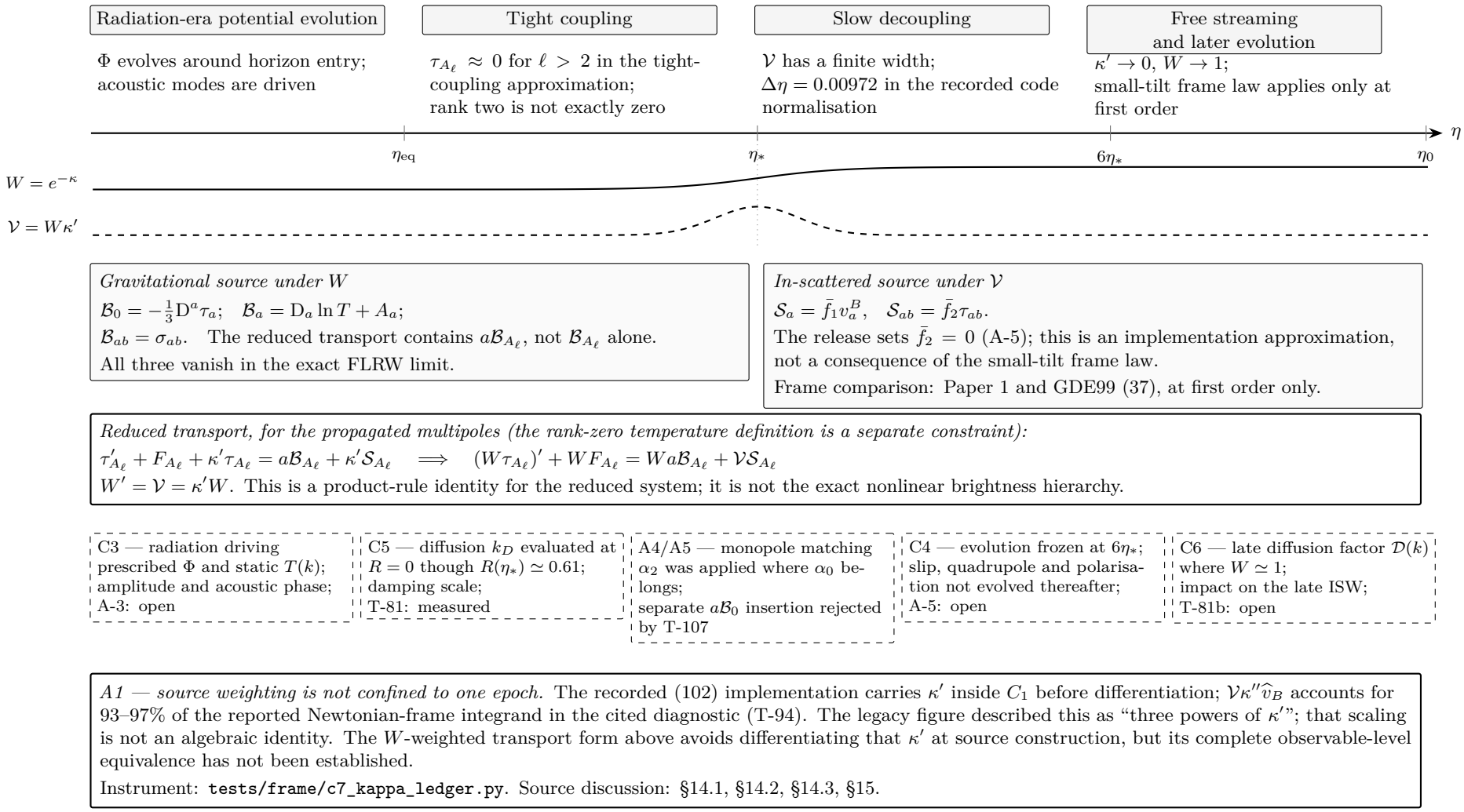
That is checkable in the tree — a coefficient entering a derivative must not contain $\kappa'$ — and it is the gate this defect needed and did not have. It is the same shape as M15: a condition that, left unstated, lets an exact identity be truncated in a way that discards its largest term.

**What it does not repair. Nothing here moves the driving.** (34) is the transport assembly; the radiation-era driving enters the *boundary value* $(\delta\tilde{T} + \Phi_A)(\eta_*)$ through the oscillator's forcing, which is approximation **A-3** and is §11's subject. The first-peak budget there carries a total of $+0.2226$ in $\varphi_1$ between the released configuration and the matched comparator, and this section accounts for the artefact alone. The two differences $+0.1102$ and $+0.1124$ that total comprises were read in earlier editions as two contributions of equal size; that reading is withdrawn at §11 and no fraction of the total is claimed here.

## 14 The transport assembly

*Section 14.1 derives the exact integration by parts of Eq. (102) and identifies four discrepancies in the released solution. Section 14.2 corrects Eq. (90)'s $C_0$ by a factor $\frac{15}{2}$; Section 14.3 defines the structural pre-implementation check.*

The line-of-sight audit addresses three linked questions: which weight carries which source; what the $\ell = 0$ matching row of (89) should be; and why every repair attempted on (102) so far has moved a weight without moving the term that dominates the integrand. The common answer is that (102)'s integration by parts was taken onto the coefficients, and that the assembly in which that step never arises is the one obtained by multiplying the transport equation by its own integrating factor.




Original v1.4.0 TikZ retained unchanged; this is the v1.3.7 source-controlled redraw.

Figure 1: The photon's history and the source-weighting and matching questions in the reconstructed $1+3$ radiation transport. The positions on the conformal-time axis and the opacity curves are schematic. The $\mathcal{B}_{A_\ell}$ definitions and the first-order frame qualifier follow the published $1+3$ antecedents and Paper 1, while the five numbered diagnostics and A1 are statements about the repository reconstruction, not additional covariant identities. The reduced integrating-factor identity is distinguished from the exact brightness hierarchy. The original Figure 1 TikZ remains available for literal comparison under `tikz/figure-history-flow-v1.4.0.tex`; the detailed claim and source reconciliation is in `FIGURE-1-AUDIT-v1.3.7.md`.

**Figure 1 source qualification.** The v1.3.6 diagram also gave $\tau_{ab} \simeq -(\kappa')^{-1}[\sigma_{ab} + \mathrm{D}_{\langle a} v^B_{b\rangle}]$. It is retained as reported, but its proper/conformal-time conversion remains open.[12] Neither this estimate, the $\bar{f}_2 = 0$ approximation nor the proposed $W$-form is presented as an independently verified consequence of the exact brightness hierarchy. The accepted original TikZ is retained unchanged.

Figure 1 is the order this section and §15 are written in. It is the photon's history and not a table of contents: each open question is drawn at the conformal time it acts at, which is what separates the questions that a frame instrument can see from the questions it cannot.

### 14.1 The by-parts, and the assembly that does not need it

**Done exactly, the by-parts carries three terms beyond the familiar bracket.** (102)'s bulk integrand comes from integrating $\int W(C_0K + C_1K' + C_2K'')\,\mathrm{d}\eta$ by parts *onto the coefficients*, with kernel $K(\eta) = j_\ell(k\Delta\eta)$ and $\Delta\eta \equiv \eta_0 - \eta$. The weight $W = e^{-\kappa}$ sits *inside* the integral, so it is differentiated too, and $W' = \mathcal{V}$, $\mathcal{V}' = (\kappa'' + \kappa'^2)W$:

$$WC_0 - (WC_1)' + (WC_2)'' \;=\; \underbrace{W(C_0 - C_1' + C_2'')}_{\text{(102)'s bracket}} \;-\; \mathcal{V}C_1 \;+\; 2\mathcal{V}C_2' \;+\; \mathcal{V}'C_2. \tag{35}$$

The integrations by parts, including their boundary terms, are

$$\int_{\eta_*}^{\eta_0} WC_1K'\,\mathrm{d}\eta = \big[WC_1K\big]_{\eta_*}^{\eta_0} - \int_{\eta_*}^{\eta_0} (WC_1)'K\,\mathrm{d}\eta, \tag{36}$$

$$\int_{\eta_*}^{\eta_0} WC_2K''\,\mathrm{d}\eta = \big[WC_2K' - (WC_2)'K\big]_{\eta_*}^{\eta_0} + \int_{\eta_*}^{\eta_0} (WC_2)''K\,\mathrm{d}\eta. \tag{37}$$

With $\kappa(\eta) = \int_\eta^{\eta_0} \kappa'(s)\,\mathrm{d}s$, $W' = \kappa' W = \mathcal{V}$ and $W'' = \mathcal{V}' = (\kappa'' + \kappa'^2)W$. Consequently $(WC_1)' = WC_1' + \mathcal{V}C_1$ and $(WC_2)'' = WC_2'' + 2\mathcal{V}C_2' + \mathcal{V}'C_2$. Substitution reproduces (35) term by term, without making a frame choice or dropping a boundary. The difference between that exact identity and a published projected solution can be assessed only after establishing the projected source and endpoint conventions from the published paper. Equality of these two integration-by-parts expressions alone does not establish Eq. (102).[13]

**Both endpoints of the projected identity.** For this calculation alone, define the pre-integrated object

$$I_\ell = \int_{\eta_*}^{\eta_0} \mathrm{d}\eta\; W(\eta)\{C_0K + C_1K' + C_2K''\}, \qquad K(\eta) = j_\ell\big(k(\eta_0 - \eta)\big). \tag{38}$$

The prime on $K$ denotes *conformal time* differentiation; $j_\ell'$ below denotes derivative with respect to its argument. Since $k = k_{\mathrm{com}}$ is constant,

$$K' = -kj_\ell'(x), \qquad K'' = k^2 j_\ell''(x), \qquad j_\ell'(x) = \frac{\ell}{2\ell+1} j_{\ell-1}(x) - \frac{\ell+1}{2\ell+1} j_{\ell+1}(x), \quad x = k(\eta_0 - \eta). \tag{39}$$

The two separate integration-by-parts lines in (37) therefore give, without assuming that an endpoint vanishes,

$$I_\ell = \big[WC_1K + WC_2K' - (WC_2)'K\big]_{\eta_*}^{\eta_0} \tag{40}$$

$$+ \int_{\eta_*}^{\eta_0} \mathrm{d}\eta\; \big[WC_0 - (WC_1)' + (WC_2)''\big]\,K. \tag{41}$$

[12] Here $\kappa'$ is a conformal-time rate whereas the unscaled kinematic tensors use proper-time derivatives. The published tensor relation has not been independently certified; see `FIGURE-1-AUDIT-v1.3.7.md`.

[13] The full line-of-sight comparison with the published journal Eq. (102), including projected boundary normalisation, has not been independently closed. This algebra tests the transformation of a specified integrand, not the derivation of that integrand from the exact brightness hierarchy.

For $\ell \geq 2$, $j_\ell(0) = j'_\ell(0) = 0$, so the upper endpoint is zero *provided the coefficient products remain finite.* The lower endpoint contributes with the opposite sign of the bracket, therefore

$$E_{*,\ell} = \left[(WC_2)' - WC_1\right]_* j_\ell(x_*) + k[WC_2]_* j'_\ell(x_*) \tag{42}$$

$$= W_* \left[C'_2 - C_1 + \kappa'_* C_2\right]_* j_\ell(x_*) + kW_* C_{2*} j'_\ell(x_*). \tag{43}$$

Here $W_* \equiv W(\eta_*)$, $x_* \equiv k(\eta_0 - \eta_*)$, $W' = \kappa' W$; the subscript on the square bracket means all its terms are evaluated at $\eta_*$. For $\ell = 1$ the upper derivative $K'(\eta_0) = -k/3$ is *not* zero, and for $\ell = 0$ $K(\eta_0) = 1$; these require their own endpoint accounting.

The boundary in (43) has two pieces that cannot be inferred from the published display of (102) by inspection: the factor $W_*$ and the additional product derivative $\kappa'_* C_{2*}$ in the scalar slot. Their presence follows from *the stipulated pre-integrated expression* (38), not automatically from the published Boltzmann hierarchy. Whether the actual source in the published integral solution is (38) is the independent physics question.[14]

**When the diffusion factor varies with time.** If the assumed pre-integrated source instead carries a multiplicative factor $\mathcal{D}(\eta, k)$, set $P = W\mathcal{D}$ in (38). The exact bulk bracket is then $PC_0 - (PC_1)' + (PC_2)''$ and the boundary is obtained from (41) by replacing $W$ with $P$. The derivatives that must be retained are

$$P' = W'\mathcal{D} + W\mathcal{D}', \qquad P'' = W''\mathcal{D} + 2W'\mathcal{D}' + W\mathcal{D}''. \tag{44}$$

Thus a damping factor can be taken outside the time derivatives only if its time dependence has been removed by a stated approximation. This is a conditional product-rule statement: it neither extends the plasma diffusion approximation beyond decoupling nor supplies a new model of $k_D(\eta)$. In particular, it cannot justify changing the released spectrum without checking the underlying physical source.

**Current repository comparison.** The source-level finding that follows was first made against commit `3b01aea8`. In Gate v1.3.8 the same relevant code paths were reinspected at commit `70f0f44ff8f7` of the public repository; the complete commit identifier and file hashes are recorded in `REPOSITORY-CONFORMITY-v1.3.8.csv`. The distinction between an optional code path, the released default and a numerical spectrum remains necessary.[15]

**Source-level branch audit.** The optional integration-by-parts switch in the current repository has not been shown to implement all four terms of (35) in a single, traced run. [16]

**What the released solution carries against** (35)**, and it is four things, not three.** The discrepancies are recorded in `functions/spectra/integral_solution.py` as findings T-72 and T-86.[17] In order of size:

[14]The optical-depth-weighted source and boundary normalisation of published *Annals II*, Eq. (102), have not been established from the journal pages. Equations (41)–(43) verify an exact identity for the stated pre-integrated object; they do not by themselves supply a correction to Eq. (102) or to the physical CMB spectrum.

[15]The pinned revision is `70f0f44ff8f71f0313a97c17df7fcf0b83d6b753` (accessed 25 October 2026). The branch `main` is mutable. Source inspection alone is not a traced execution of the complete spectrum.

[16]Source-level implementation check, repository `main` commit `3b01aea8c59bbd02c750962fa822b2697f3ff533`: `functions/spectra/integral_solution.py`, optional `by_parts_weight_terms` branch, adds the $2\mathcal{V}C'_2$ and $\mathcal{V}'C_2$ terms but does not construct the $-\mathcal{V}C_1$ partner as a standalone term. The branch later combines with alternative weighting switches. This is a directly located source-level discrepancy against Eq. (35); its impact on any released configuration is not established without an execution trace of the selected switches.

[17]The algebraic by-parts identity is checked, but the released line-of-sight implementation has not been corrected at this gate. Agreement of internal routes using a common weight does not establish the physical correctness of that weight.

| | the identity wants | the released solution carries | |
|---|---|---|---|
| 1 | the bracket at weight $W$ | the bracket at weight $\mathcal{V}\mathcal{D}$ | $\mathcal{V} = W\kappa'$, so **one extra power of the rate** |
| 2 | $-\mathcal{V}C_1 + 2\mathcal{V}C_2' + \mathcal{V}'C_2$ | absent | (102) carries only the bracket |
| 3 | $W(\eta_*)$ on the boundary | $\mathcal{D}(\eta_0, k)$ | weight and argument both |
| 4 | $+k\int W\, M\, j_\ell'$ | *no counterpart at all* | the by-parts onto coefficients is legal only at $W$ |

Row 1 is the one that matters and it is now measured. $\mathcal{V}$ already supplies a $\kappa'$, so differentiating a coefficient that itself holds $\kappa'$ puts *three* powers of the rate where the identity wants two. D24 measures the consequence on the live path: $\kappa''\tilde{v}_B$ is 99.8% of $|C_1'|$ in the Newtonian frame and 109.8% in CDM, and suppressing it alone moves the CDM first peak from a spurious maximum at $\ell = 8$ to $\ell = 172.75$, i.e. $0.635\,\ell_A$ against the matched 0.757. The $\Phi'$ driving channel is not dead; it is outvoted — finding T-121.

**What `c3_byparts.py` does and does not measure.** C3a is the load-bearing one: (35) reproduces the integral it comes from to $< 10^{-8}$ over fifteen $(\ell, k)$ pairs, so the identity is not in question. C3b evaluates a *variant* — it weights the bracket at $W$ and retains $\mathcal{V}(C_1 - 2C_2)$ — so its generated value — 2.55227 — bounds rows 2 and 3 and is blind to row 1, which is the one row whose weight it holds fixed. That number is the authoritative one: a bound of $> 10^{-2}$ and a prose range of *10–250%* have both been quoted for it elsewhere and neither should be. Its docstring's *"evaluated as printed"* should be read as *"evaluated against a variant that differs from the printed form in the partner terms"*; the released weight is the `weight` argument, $\mathcal{V}$ by default, approximation A-2 and finding T-20.

This is a defect in the antecedent and carries a B- number in `provenance/ANNALS-II-ANTECEDENT-AND-CORRECTION` beside B-10 and B-11. It is stated here as well because (35) *is* the correction, and this supplement prints the correction beside every other defect it records — B-11's sign on $C_0$, T-104's normalisation, T-103's weighting. A defect recorded without its correction is not a record.

**The assembly in which none of the four can be made.** They arise only because the by-parts was taken *onto the coefficients*. Multiplying the transport equation at rank $\ell$,

$$\tau_\ell' \;+\; (\text{streaming couplings}) \;+\; \kappa'\tau_\ell \;=\; a\mathcal{B}_\ell \;+\; \kappa'\mathcal{S}_\ell, \tag{45}$$

through by $W$ — with $\kappa$ accumulated backwards from $\eta_0$, so $\mathrm{d}\kappa/\mathrm{d}\eta = -\kappa'$ and $W' = +\kappa' W = \mathcal{V}$, the sign this bundle's convention forces, stated at §15 — combines the first and third terms exactly into $\mathrm{d}[W\tau_\ell]/\mathrm{d}\eta$, and gives

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}\eta}[W\tau_\ell] \;+\; W\,(\text{streaming couplings}) \;=\; W\,a\mathcal{B}_\ell \;+\; \mathcal{V}\,\mathcal{S}_\ell} \tag{46}$$

with $\left[W\tau_\ell\right]_{\eta_*}^{\eta_0} = \tau_\ell(\eta_0) - W(\eta_*)\tau_\ell(\eta_*)$ as its only boundary term, since $W(\eta_0) = 1$. Here $a\mathcal{B}_\ell$ is the gravitational source of §*The generic source derivatives*' rank decomposition and $\mathcal{S}_\ell$ the in-scattered Thomson source; both $\kappa'$ in (45) are the same $\kappa'$, the out-scattering and the in-scattering being two halves of one collision term.

**What** (46) **is, at its true size.** It is not different physics. An exact by-parts preserves the transfer, so (46) and a (102) corrected by (35) give the same observable. *An earlier draft of this section called (102)'s bulk slots and boundary pair "artefacts of a step the $W$-form does not take"; that overstates and is withdrawn.* What (46) is:

1. The assembly in which none of the four discrepancies above can be made, because there are no coefficients to take a by-parts onto, and so no weight to choose for a bracket that does not exist. Both weights fall out of one multiplication rather than being assigned: $\mathcal{V}$ appears beside $\mathcal{S}_\ell$ because $\kappa'$ was already there in the collision term and $W\kappa' = \mathcal{V}$ is an identity.

2. The assembly in which no coefficient holds $\kappa'$, so no differentiation of one can produce a $\kappa'^2$ — the error class §14.3 flags. D24 measures it on the live path: $\kappa''\tilde{v}_B$ is 99.8% of $|C_1'|$ in the Newtonian frame and 109.8% in CDM, finding T-121. (46) has no differentiated-coefficient counterpart: $\mathcal{S}_1$ remains undifferentiated, and $\mathcal{V} = W\kappa'$ carries the single scattering-rate factor.

3. The assembly whose boundary term is one object, exponentially suppressed at $\eta_*$ — $[W\tau_\ell]$, not $[C_2' - C_1]$ plus $kC_2$, carrying $W$ rather than a diffusion factor. D18 measures what the released boundary carries instead: with `boundary_weight_w False`, $C_1(\eta_*)$ enters under the diffusion factor alone — no $W$, no $\mathcal{V}$ — so its bare $\kappa'(\eta_*)$, $3.82 \times 10^2$ on Einstein–de Sitter and $2.64 \times 10^2$ on `PLANCK_MODEL`, stands under no exponential suppression at all.

**This supersedes §15.4's item 2, and the wording that item carries is what T-88 implemented.** That item now reads *"they* arrive under *different weights: W for the gravitational half, g = 𝒱 for the scattering half"*, corrected at its own site from *"take"* — which is precisely the distinction this paragraph is about, and it is recorded here because an earlier draft quoted the old wording. As a description of the outcome of (46) it is correct. As an instruction to a solver it is not, and it was read as one: multiplying $C_1^{\rm thom} = -\kappa'\hat{v}_B$ by $\mathcal{V}$ applies a second $\kappa'$, which §15.8 measures at 320 to 682. The scattering half arrives at $\mathcal{V}$ weight carried on the undifferentiated $\hat{v}_B$; it is not multiplied by $\mathcal{V}$. Finding T-103; the phrasing is the Paper 1 stream's and the correction is recorded against it rather than against the implementation that followed it.

**What $\mathcal{S}_2$ is, and why it cannot be a frame defect.** *Annals II*'s collision term is $C[\tau] \simeq \sigma_{\rm T} n_E(\bar{f}_1 e^a v_a^B + \bar{f}_2 e^a e^b \tau_{ab} - \tau)$ with $\bar{f}_1 = 1$ and $\bar{f}_2 = 0$, so $\mathcal{S}_2 = 0$ in this reconstruction. That is approximation A-5 stated at the collision term rather than at the oscillator. It cannot produce a frame disagreement at linear order: $\mathcal{S}_2 \propto \tau_{ab}$, and GDE99 (37) leaves $\tau_{A_\ell}$ invariant for $\ell \geq 2$ *at first order in* $v$ — those multipoles are already first-order small, so what a boost would generate from them is second order — whence $\Delta\mathcal{S}_2 = 0$ and dropping the term changes both columns by the same amount. It is a §15 Triage C site — it can ruin the spectrum and it cannot be seen by a frame instrument — and listing it as a candidate for the frame gap confuses the two questions the triage tables exist to separate.

### 14.2 The $\ell = 0$ matching row

**The rows of (89) are Taylor coefficients of the kernel at the origin, and one of the three uses the wrong rank's normalisation.** With $h_\ell = \alpha_\ell j_\ell$ and $\alpha_\ell = (2\ell+1)/\beta_\ell$ — this bundle's own `functions/harmonics/weights.py`, where $\beta_\ell = \ell!/(2\ell-1)!!$ gives $\beta_0 = \beta_1 = 1$ and $\beta_2 = \frac{2}{3}$ — each row of (89) pairs $C_n$ with $h_\ell^{(n)}(0)$:

| row | used | its own | the entry |
|---|---|---|---|
| $\ell = 2$ | $\alpha_2 = \frac{15}{2}$ | $\alpha_2$ | $h_2''(0) = \frac{15}{2} \cdot \frac{2}{15} = 1$ |
| $\ell = 1$ | $\alpha_1 = 3$ | $\alpha_1$ | $h_1'(0) = 3 \cdot \frac{1}{3} = 1$ |
| $\ell = 0$ | $\boldsymbol{\alpha_2 = \frac{15}{2}}$ | $\boldsymbol{\alpha_0 = 1}$ | $-\frac{5}{2}$ printed; $h_0''(0) = -\frac{1}{3}$ |

(89) writes its $\delta_{\ell 0}$ entry as $5\beta_2^{-1} \cdot \frac{1}{3}$, which is $\alpha_2$ times $j_0''(0)$ up to the sign B-11 corrects. The rank-two normalisation is right where it belongs and is carried into a row that is not rank two. The two rows that are exactly unity are the control, and they are a control precisely because they come out at unity *without adjustment*: a normalisation error cannot produce 1 twice and then $-\frac{5}{2}$ once. Findings B-16 and T-104; the discrepancy is $\alpha_2/\alpha_0 = \frac{15}{2}$.

The kernel derivatives used in this comparison follow directly from

$$j_0(x) = 1 - \frac{x^2}{6} + O(x^4), \qquad j_1(x) = \frac{x}{3} + O(x^3), \qquad j_2(x) = \frac{x^2}{15} + O(x^4), \tag{47}$$

so that $h_0''(0) = -1/3$, $h_1'(0) = 1$ and $h_2''(0) = 1$ for $h_\ell = \alpha_\ell j_\ell$. This verifies the local algebra of the matching row. Whether Eq. (89) in the published *Annals II* actually uses that row and normalisation requires the independent journal-page check; it is not established by the Taylor series alone.

This route is independent of (89), which is what makes it a check rather than a restatement. $\beta_\ell$ is fixed by the PSTF normalisation $O^{A_\ell} O_{A_\ell}$ and $j_0''(0) = -\frac{1}{3}$ by the ascending series; neither comes from (89). (89) is the *claim* about what the rows hold and the kernel is the *object*. Two of the three rows reproduce (89)'s entry exactly from the object, which validates the route; the third does not.

**The right-hand side of the $\ell = 0$ row is zero, and that is forced by a definition.** $\tau_0 \equiv 0$ is a constraint, not initial data: $T(x)$ is defined as the monopole of $T(x, e)$, so differentiating $\int A_0 \, \mathrm{d}\eta' = 0$ puts zero on the right of the row. $\mathcal{B}_0$ is a functional of the solution, not an external forcing. Appendix 22 states this and states it first — *"(89)'s $\ell = 0$ line is not mis-sourced … $C_0$ is correctly derived"*. With $\alpha_0$ on the left and zero on the right,

$$C_0 + k^2 C_2 \, h_0''(0) = 0, \qquad h_0''(0) = -\tfrac{1}{3}, \tag{48}$$

and with $C_2 = -k^{-2} a\mathcal{B}_2$ this solves to

$$\boxed{C_0 \;=\; -\tfrac{1}{3}\, a\mathcal{B}_2} \tag{49}$$

in place of (90)'s $C_0 = -\frac{5}{2} a\mathcal{B}_2$. One correction, and the ratio against the printed row is $\frac{5}{2}/\frac{1}{3} = \frac{15}{2}$ exactly — $\alpha_2/\alpha_0$, with no $\mathcal{B}_0$ anywhere. Instrument `c5_monopole.py`, under `tests/frame/`, which carries a negative control on each alternative.[18]

**Where $\mathcal{B}_0$ does enter, and why restoring it to $C_0$ is double counting.** Correction, 2026-10-22. An earlier draft of this subsection restored an $a\mathcal{B}_0\delta_{\ell 0}$ row, printed $C_0 = -\frac{1}{3}a\mathcal{B}_2 - a\mathcal{B}_0$, and recorded it as T-105. That is withdrawn. By identity

$$a\mathcal{B}_0 = -\tfrac{k}{3}\,\tau_1, \tag{50}$$

derived in index algebra and controlled on rank 1, where the identical chain returns $a\mathcal{B}_1 = k(\delta T + \Phi_A)$ — the released `coefficients.py` line. And (107) carries $-C_1' = (\kappa'\hat{v}_B)' + (a\mathcal{B}_1)'/k$ with $(a\mathcal{B}_1)'/k = (\Phi_A' - \Phi_H') + \frac{k}{3}\tau_1$, so the monopole source is already inside the bracket, in $-C_1'$. *That is the expansion-terms-off form of (106)*: `coefficients.py` builds $(a\mathcal{B}_1)'$ with $-a\mathcal{H}\,\delta T - 3a\mathcal{H}\,\Phi_A$ in addition, behind `A_B1_PRIME_EXPANSION_TERMS`. Whether they belong was finding T-67; it is **refuted by measurement, 2026-10-25**. Enabling the control changes $(a\mathcal{B}_1)'$ by 42% — from $-3.805197 \times 10^1$ to $-5.396853 \times 10^1$ — but changes both frames *identically*, so $\Delta[(a\mathcal{B}_1)']$ is unaltered to the last bit. The terms are frame-neutral in the difference. Scored directly at the coefficient against the required $\mathrm{d}[-\mathcal{H}v - k\Phi_A]/\mathrm{d}\eta$, the transformation already closes without them: machine zero on Einstein–de Sitter, ratio 0.9725 on ΛCDM and 0.9309 on `PLANCK_MODEL`.

*The Einstein–de Sitter row of that comparison is a null, not a confirmation.* On $K = 0$ dust $\mathcal{H} = 2/\eta$, and the regular branch of $v' + \mathcal{H}v = -k\Phi_A$ with $\Phi_A$ constant is $v = -k\Phi_A\eta/3$, so

$$\mathcal{H}v = -\tfrac{2}{3}k\Phi_A, \qquad \Delta(a\mathcal{B}_1) = -\mathcal{H}v - k\Phi_A = -\tfrac{1}{3}k\Phi_A, \qquad \frac{\mathrm{d}}{\mathrm{d}\eta}\Delta(a\mathcal{B}_1) \equiv 0. \tag{51}$$

$v$ grows by $\sqrt{1+z_*}$ between last scattering and today and $\mathcal{H}$ falls by the same factor, exactly, so their product does not move at all. **Both sides of the Einstein–de Sitter row are identically zero before anything is computed**: it is a null control, it passes as a null control must, and it carries no evidence about the terms. *The two Λ rows carry whatever content that comparison has.* The informative

[18] The Taylor coefficients are verified independently, but the $\ell = 0$ matching rule has not been checked against the published Annals II Eq. (89) in the journal. Consequently $C_0 = -a\mathcal{B}_2/3$ is conditional on that rule and is not yet an independently certified correction of the published Eq. (90).

Einstein–de Sitter statement is the one at the coefficient rather than at its derivative, where (51) predicts $-\frac{1}{3}k\Phi_A$ outright; that row is **Q17** of §25. They do not affect the $\frac{k}{3}\tau_1$ identification used here, which is why the reduced form is quoted. Adding it to $C_0$ counts it twice, and criterion 5b doubles at 98.5%. Finding T-107 replaces T-105.

**So $\Delta C_0 \neq 0$, and that is the correct outcome.** Under (49), with Appendix 22's own $\Delta(a\mathcal{B}_2) = -kv$ and $\Delta C_2 = v/k$,

$$\Delta C_0 = -\tfrac{1}{3}\Delta(a\mathcal{B}_2) = +\tfrac{kv}{3}, \qquad \gamma \equiv \frac{\Delta C_0}{k^2 \Delta C_2} = \tfrac{1}{3}. \tag{52}$$

A non-zero $\gamma$ is the signature that $\Delta C_0$'s partner is cross-coefficient, sitting in $-C_1'$ rather than in $C_2$. A $\gamma$ of zero would say the partner was in $C_2$, which it is not. The withdrawn draft reached $\gamma = 0$ by *adding a term*, and a gate that can be driven to zero by adding a term is a fit rather than a gate — which is M15, *a gate built from one coefficient cannot see a cancellation distributed across the bracket.* `c8_rank0_partner.py` scores the whole rank-0 content of the bracket and is the gate for this row; Appendix 22's questions 3 and 4 close with c8 or not at all. Its questions 1 and 2 are settled by T-66 and by §14.1 respectively.

**One thing** (49) **does not disturb.** Appendix 22 excludes a comoving/physical or amplitude/potential mismatch by measuring $C_0/(k^2 C_2)$ as a background-independent and $k$-independent rational. That argument holds under (49) unchanged — the ratio is still a pure rational, now $\frac{1}{3}$ rather than $\frac{5}{2}$ — because what it rests on is that $C_0$ and $C_2$ divide one $a\mathcal{B}_2$. The measurement was right and what it showed was internal consistency, which is not correctness. It is the same reading the bundle gave T-82 and T-85, applied to itself.

### 14.3 The $\kappa'$ ledger

**The flag, and it is a flag and not a theorem.** Revised 2026-10-21, T-113 and S48. *A coefficient carrying a bare $\kappa'$ is a site where a large residual is manufactured the moment it is differentiated, and is worth flagging before implementation.* An earlier draft of this subsection printed that as a prohibition — *"$\kappa'$ may live in $W$, in $\mathcal{V}$, or in a derivative of either, never in a coefficient"* — and it was read as normative. It is not a theorem, because

$$W(\kappa'\mathcal{S}) = \mathcal{V}\,\mathcal{S} \tag{53}$$

is exact: a coefficient *may* hold $\kappa'$, and $(\kappa'\mathcal{S})' = \kappa''\mathcal{S} + \kappa'\mathcal{S}'$ is not intrinsically spurious. The defect in (102) is wrong weights, signs, boundaries or missing partners, not the presence of $\kappa'$.

**Why the flag is kept.** It separates two $\kappa'^2$ that are indistinguishable in the output. The one inside $\mathcal{V}' = (\kappa'' + \kappa'^2)W$ comes from differentiating the weight; it is required, and (46) produces it whenever a streaming term is integrated. The one from differentiating a coefficient is not required by anything. Only the provenance distinguishes them, so the flag is raised on the provenance. At $\eta_*$ in this bundle's units D18 measures $\kappa' = 3.82 \times 10^2$ on Einstein–de Sitter and $2.64 \times 10^2$ on `PLANCK_MODEL`, so a spurious power of it is two to three orders. What the flag may no longer do is settle a question on its own.

**(102) as printed trips it, and T-121 is the review.** §*The generic source derivatives* gives $-C_1 \simeq k^{-1}(a\mathcal{B}_1 + \kappa' v_B)$ — a coefficient holding a bare $\kappa'$ — and (102) then differentiates it. §15 measures the consequence: $\mathcal{V}\kappa''\hat{v}_B$ is 93–97% of the released integrand, and D24 measures 99.8% of $|C_1'|$ in the Newtonian frame and 109.8% in CDM on the live path. The flag asks whether that is a defect; D24 answers it at the level of the observable. Suppressing that one term moves the CDM first peak from a spurious maximum at $\ell = 8$ to $\ell = 172.75$, which is $0.635\,\ell_A$ against the matched 0.757. The driving

channel is not dead, it is outvoted — finding T-121. The flag was raised on the right site; what it could not supply, and D24 does, is the size.

*A measurement that does not bear on this, and is recorded so it is not over-read.* T-117 finds the transfer failing to reduce to Sachs–Wolfe as $k \to 0$ — in the Newtonian column $\kappa' v_B/k$ does not die, and the bulk is then entirely $\int \mathcal{V}(\kappa'\hat{v}_B)' j_\ell$ at $9.46 \times 10^3$. It is a true defect in a limit with a known answer. D25 then bounds the window it holds in to at most 0.64% of any $C_\ell$, and 0.00% at every $\ell \geq 5$ — finding T-122. So the number is a diagnostic of the mechanism and not of the spectrum, and nothing here rests on it.

**A local statement, not a general one.** $\kappa'' = -\kappa'^2$ holds at the visibility maximum, where $\dot{\mathcal{V}} = 0$. So the reading of $\mathcal{V}\kappa''\hat{v}_B$ as *three powers of* $\kappa'$ is local to that point, and the measured 93–97% share follows because $\mathcal{V}$ concentrates the integrand there. It is not a statement about the integrand at arbitrary $\eta$. D24's 99.8–109.8% is measured against $|C_1'|$ on the live path and is the number to quote for the share.

**It is applied before implementation, not after.** `tests/frame/c7_kappa_ledger.py` grades an assembly structurally — for each term, whether its coefficient holds $\kappa'$ and at what power — rather than measuring an output. It reports (46) clean, and (102) as printed and T-88's variant as carrying $\kappa'$ in a coefficient at power two. Both of the defects of 19–20 October would have been flagged by that one check before either was implemented, which is the argument for a structural review alongside a numerical one: a numerical check on an assembly that is wrong in this way measures the defect faithfully and reports a number.

### 14.4 Established results and limitations

Established. That (35) is the exact by-parts of the integral (102) is built from, reproducing it to $< 10^{-8}$; that the released solution differs from it in four places, of which the bracket's weight is the largest and carries one surplus power of $\kappa'$, measured by D24 at 99.8–109.8% of $|C_1'|$; that (46) follows from (45) by one multiplication and is the assembly in which none of the four can be made, without being different physics; that (89)'s $\ell = 0$ row carries $\alpha_2$ where $\alpha_0$ belongs, with the $\ell = 1$ and $\ell = 2$ rows as unadjusted controls; that the row's right-hand side is zero by the $\tau_0 \equiv 0$ constraint, giving (49); that $a\mathcal{B}_0 = -\frac{k}{3}\tau_1$ is already carried in $-C_1'$, so restoring it to $C_0$ double counts; and that $\mathcal{S}_2$ is a Triage C site and cannot carry a frame gap at linear order.

Ungated — Part I, §1.4. That (46) closes the frame gap. It satisfies the ledger by construction and no setting of `integral_solution`'s seven controls reaches it — each redistributes weights while leaving $\kappa'$ inside $C_1$ — but that is an argument from exhaustion of what has been tried, not a measurement. What has changed is that the evidence now points at one term in both frames, where an earlier draft of this paragraph said it did not. D24 suppresses $\kappa''\tilde{v}_B$ alone and the CDM first peak appears at $\ell = 172.75$, $0.635\,\ell_A$ against the matched 0.757, where the released setting has a spurious maximum at $\ell = 8$. *The hypothesis that the CDM column fails through a sign in its boundary is withdrawn*: Einstein–de Sitter gives $-0.9999$ but `PLANCK_MODEL` gives $+1.270$, so it is not one sign and the $-1$ was a coincidence of that background. What causes the residual 0.635 against 0.757 is taken up below, under *Approximation A-3 is not what closes the peak position*; it is not A-3, and D24 is the measurement that rules it out.

The two peak numbers are not measured the same way, and that bounds what their difference can carry. D24 reports the largest maximum anywhere in the plotted range; F9 reports the first local maximum above $\ell = 30$, read by parabolic interpolation through an $\ell$ grid that steps by 30 across the acoustic band. On a spectrum whose low-$\ell$ end carries a spurious maximum at $\ell = 8$ those two estimators need not return the same peak at all. So peak positions in this section are quoted to two significant figures and no further: a 6.1% residual at $\ell \simeq 173$ is $\Delta\ell \simeq 10.5$, about a third of one grid step, and a comparison at that level is a statement about the estimator as much as about the physics. The repair

is one estimator applied to both, with the low-$\ell$ maximum excluded by construction rather than by an $\ell = 30$ floor that happens to work.

A-3 answers to three configurations and gives a different answer to each, so no statement about it is safe without naming which.

| **configuration** | **CDM peak** | **what A-3 does there** |
|---|---|---|
| released | $266.6 = 0.979\,\ell_A$ | **accounts for the whole excess**: $266.6/206.1 = 1.294$ against $1/0.757 = 1.321$, agreeing to 2% |
| $\kappa''\tilde{v}_B$ suppressed (**D24**) | $172.75 = 0.635\,\ell_A$ | **ruled out** — the peak is already below the driven value and driving moves it further away |
| full boundary at released $\kappa''$ | $236.6 = 0.869\,\ell_A$ (CDM) | **would be the right sign — but see below: this configuration is not a repair** |

The third row is a candidate and it fails the one test that costs nothing to apply. 'slot_weight_split' with the two boundary controls puts the CDM peak at $0.869\,\ell_A$ and the *Newtonian* peak at $0.277\,\ell_A$ — a factor of 3.1 between two frames whose $\ell \geq 2$ multipoles are equal at first order in $v$ (GDE99 (37)). A configuration that moves the CDM peak toward the target while pulling the frames 3.1× apart has made the frame disagreement worse, not better, and the released path's median gap of 15% is the thing it was supposed to close. *So it is not a repair, and nothing below is a result.* The rows are kept because the comparison is instructive and because the frame check is the cheapest discriminator this section has.

*The derivation, which is what makes the three rows consistent.* Driving moves a peak down: the undriven first compression extremum is $kr_s = \pi$, which projects to $\ell = \ell_A$, and the matched external spectrum sits at $0.757\,\ell_A$. D24, with $\kappa''\tilde{v}_B$ suppressed and A-3 still absent, puts the CDM peak at $0.635\,\ell_A$ — *already below the driven value* — so applying A-3 there moves it further from 0.757, not toward it. Three places in this project's record have said the first peak sits where it does because radiation-era driving is absent; for that configuration that is backwards, and it is withdrawn here.

But the conclusion is conditional on which side of 0.757 the peak falls, and that is what a repair would decide. A configuration whose CDM peak lands *above* $0.757\,\ell_A$ puts A-3 back in play, because driving then moves it the right way; one landing below leaves A-3 ruled out. A-3's standing is therefore a property of the configuration and not of the project, and no statement about it is safe without naming which.

What a candidate above 0.757 would still have to survive. Against this background the whole distance from an undriven peak to the matched spectrum is $\ell_A - 206 = 66.2$ multipoles, or $0.243\,\ell_A$. A candidate at 0.869 would leave $236.6 - 206 = 30.6$ to cross, so *a full driving correction would overshoot it by a factor of* 2.2 — A-3 could not be the whole remainder, and wiring it in would have to land short of 0.757 rather than on it. That is the test to apply to the first candidate that holds the two frames together, and it is written here so that it is applied rather than re-derived.

*The other known absentee pushes the same way.* (54)'s $j'_\ell$ counterpart adds power *below* the peak and is missing from (102) as well, so the two named absent mechanisms both move the peak down and between them need supply only $0.112\,\ell_A$. Neither has an instrument that identifies it as the remainder, and the two must be measured separately before either is credited, which is M11a.

**And a boundary treatment and A-3 compete for the same multipoles, which is what the released agreement with $1/0.757$ does not establish.** The released path needs $266.6 - 206.1 = 60.5$ multipoles of downward movement and credits all of them to A-3, agreeing to 2%. The candidate configuration, with A-3 still absent, supplies $266.6 - 236.6 = 30.0$ of them — 49.6% — by changing a weight. It is not a repair, so this does not withdraw the A-3 account; what it shows is that the 2% agreement is not evidence for A-3, because at least one other thing in the assembly moves the peak by half that distance. A quantity reproduced to 2% by a calculation with a known defect of the same size is the shape of every over-read in §*what a gate does not establish*, and it is listed there.

The discriminator, when a frame-consistent candidate exists, is one run: wire A-3 into it. If A-3 is the whole remainder the peak lands near $0.757\,\ell_A$; if the weight change has already taken half, it overshoots by roughly $0.243\,\ell_A$ from wherever the candidate sits. For a candidate at 0.869 those are $\ell \simeq 206$ against $\ell \simeq 170$, 17% and 36 multipoles apart. Both ends are already built.

*The displaced projection is ruled out too, by the test that was written to decide it.* An earlier form of this paragraph attributed 0.635 to D25's finding that the median $k$ contributing to a given $\ell$ runs 1.5–4.2× above $\ell/r_\theta$, and predicted a ratio near $1/0.635 = 1.57$ at the peak. D25 extended to $\ell = 173$ returns 1.032 and 1.039 in the Newtonian frame, 1.171 and 1.201 in CDM, on Einstein–de Sitter and `PLANCK_MODEL` respectively — the two backgrounds agreeing to 3%. The displacement is a low-$\ell$ phenomenon: 1.5–4.3 at $\ell = 2$, decaying monotonically, and essentially closed by the peak. So T-122 does not account for the peak position, and that attribution is withdrawn. Nor does it account for the plateau, which an earlier form of this sentence left standing: the plateau's cause is now measured and it is the weight — Part I, §1. *The projection at the peak is faithful.*

And the absolute level was never the diagnostic. A median contributing wavenumber above $\ell/r_\theta$ is what the kernel produces on its own: $j_\ell(kr_\theta)$ peaks at $kr_\theta \simeq \ell + 0.8\,\ell^{1/3}$ and carries a long tail to larger $k$, so an exactly Sachs–Wolfe spectrum returns a ratio above one as well. No absolute value of this ratio establishes a projection defect, and the earlier reading, which took 1.5–4.2 as a defect on its face, was reading a property of $j_\ell$. What survives is the *difference between the frames* at the same $\ell$ on the same kernel, because the kernel cancels there and nothing else in the comparison does.

One residue does survive at the peak, and it is frame-dependent. CDM reads 1.17–1.20 against the Newtonian 1.03–1.04, on *both* backgrounds — a 17–20% displacement left standing where the frame-independent part has closed. It is too small to carry 0.635 by this route, which would need 1.57. It is the right *shape* for the candidate below, and it was measured for another purpose, which is the only reason to give it any weight at all.

*What that leaves.* With the projection faithful at $\ell = 173$ and the oscillator tracking $\cos(kr_s)$ in exact phase and period (T-71, T-74), a peak at $0.635\,\ell_A$ places the first extremum at $kr_s \simeq 0.635\pi$ rather than $\pi$ — a phase error of about $-0.37\pi$, arising *between* the oscillator and the transfer, which is where (35)'s discrepancies sit.

**A quarter-cycle account of that was printed here and is withdrawn — finding T-123.** It wrote $C_2 = A\cos kr_s$, read $+2\mathcal{V}C_2'$ against the printed $-2\mathcal{V}C_2$ as a rotation of exactly $\pi/2$, and bounded the peak to $(0.5,\ 1.0]\,\ell_A$ at a mixture weight $w \simeq 2.2$. $C_2$ does not oscillate. $\mathcal{B}_2$ is the shear, and in the CDM frame $a\mathcal{B}_2 = a\sigma = -k_{\rm com}v$ exactly, so $C_2 = v/k$ with $v$ the matter velocity and $v' + \mathcal{H}v = -k\Phi_A$ its equation of motion — a potential on the right-hand side and not an acoustic amplitude, so $v$ runs monotonically through matter domination. The premise fails and the bound, the $0.365\pi$ and the $w \simeq 2.2$ fail with it. *Recorded rather than deleted: two of the three tests it specified survive under the mechanism below, and the reason they survive is not the reason they were written.*

**The partner group does act on the projection, and exactly.** Project the three terms (35) supplies, integrating the third by parts and using $\mathrm{d}j_\ell(k(\eta_0-\eta))/\mathrm{d}\eta = -k\,j_\ell'$:

$$\int(-\mathcal{V}C_1 + 2\mathcal{V}C_2' + \dot{\mathcal{V}}C_2)j_\ell\,\mathrm{d}\eta \;=\; \int\mathcal{V}(C_2' - C_1)\,j_\ell\,\mathrm{d}\eta \;+\; \int\mathcal{V}\,v\;j_\ell'\,\mathrm{d}\eta \;+\; [\mathcal{V}C_2\,j_\ell], \tag{54}$$

with $kC_2 = v$ used in the middle term. The partner group is a visibility-weighted source against $j_\ell$ plus a velocity against $j_\ell'$, the second being the Doppler form of the standard line-of-sight solution carrying the *frame* velocity in place of the baryon velocity. It vanishes identically in the Newtonian frame, where $\tilde{C}_2 \equiv 0$.

That is B-17 (iv) derived rather than observed, and it carries a number. At fixed $x = k(\eta_0 - \eta)$ the multipole maximising $|j_\ell'|$ lies about 37% below the one maximising $|j_\ell|$: 110 against 173 at $x = 178$, 164

against 266 at $x = 272$. A source given the wrong one of the two kernels is not perturbed but *relocated*, by a third of the $\ell$ axis. T-124.

The direction is the test. Restoring the $j'_\ell$ term adds power *below* the acoustic peak, so on its own it must move a peak down or leave it standing. The §5 repair moves the CDM peak *up*. So the mover is the weight — B-17 (i) — and not this term; if restoring the $j'_\ell$ slot alone moves the peak up, this account is wrong.

And the $j'_\ell$ slot is frame-discriminating, which is what makes it worth running. $\tilde{C}_2 \equiv 0$ in the Newtonian frame, so (54)'s middle term *cannot act there at all*; the $\mathcal{V}C_1$ sign is available in both. 0.635 is the CDM peak. If the Newtonian peak shows a different displacement, that is the signature; if both frames displace alike, the $C_2$ route is excluded and the $\mathcal{V}C_1$ sign must carry it. Two numbers, both from runs that already exist.

And D24's own table already shows the frames behaving differently, which sharpens the test. With $\kappa''$ suppressed the Newtonian column peaks at $\ell = 8.07$ — still the spurious low-$\ell$ maximum, *not* an acoustic peak — while CDM peaks at $\ell = 172.75$. That is what the frame structure predicts: Newtonian has $\tilde{C}_0 \approx \tilde{C}_2 \approx 0$, so $\tilde{C}_1$ carries everything and suppressing $\kappa''\tilde{v}_B$, which is 99.8% of $|C'_1|$, leaves no oscillatory source at all. CDM retains $C_0$ and $C_2$. So the 0.635 peak is built from $C_0$ and $C_2$, and $C_2$'s slot is exactly where (54) acts.

Two tests survive the withdrawal and neither needs a repair. Restore the $j'_\ell$ slot alone in the $\kappa''$-suppressed CDM run and read the peak: (54) requires it to move *down* or not at all, and an upward move refutes the account. Then compare the two frames' displacements, which the $C_2$ route requires to differ because $\tilde{C}_2 \equiv 0$. Until one of those runs, 0.635 has no mechanism, and a repair should not be scored against 0.757.

Also ungated, and it is a dependency of (49) — Part I, §1.4. (49) *uses* $C_2 = -k^{-2}a\mathcal{B}_2$, which comes from the same solve in which B-11 found a sign error and T-104 a normalisation error. The $\ell = 2$ row returning $h''_2(0) = 1$ controls the normalisation and not the sign or the source pairing. No instrument yet audits the $\ell = 2$ row the way `c5` audits $\ell = 0$, and until one does, a defect of that class in $C_2$ would be inherited silently here.

The acceptance condition, stated in advance so the repair can fail visibly. (i) $A_N = A_C$ to the quadrature floor T-85 measures, neither better nor worse; (ii) the Newtonian column moves by no more than that floor; (iii) `c7_kappa_ledger.py` reports no coefficient holding $\kappa'$ — *a review signal that the error class is absent by construction, not a proof of correctness*, since (53) makes the ledger a flag; (iv) `c8_rank0_partner.py`, not $\gamma$, scores the rank-0 content of the bracket on the live coefficients; and (v) the spectrum adjudicates, and it adjudicates last, since two wrong columns can agree.

The order is forced, not preferred — in implementation, not in exposition. §14.2 is implemented before §14.1: the $\ell = 0$ row sits inside the bracket the $W$-form reweights, so a weight correction applied over a $C_0$ that is wrong by $\frac{15}{2}$ measures the wrong thing. T-96's second run is that measurement — a $W$-form instrument run against coefficients still carrying the printed row, returning $O(1)$ — and it is reported there as not conclusive, correctly.

## 15 The frame discrepancy: localisation and triage

[19]

This section records the state of the one open defect that the $v1.3.0$ line exists to close. It is written to be auditable rather than persuasive: every exclusion below carries the instrument that licenses it and a grade, because an exclusion without an instrument is an opinion and this project has paid for several.

[19] The first-order small-tilt law for the covariant temperature multipoles holds for a common direction label. At the order of the aberration couplings, comparison of untransformed Newtonian and CDM multipoles does not by itself test the observer-independent sky. The pointwise Doppler–aberration relabelling of Gebbie (2026), Secs. V–VI, must be included with the endpoint conditions.

The measurements are recorded in `provenance/BUNDLE-DEFECTS-v1.md`, which is authoritative; numbers appear here only where the argument cannot be followed without them.

### 15.1 The statement being tested

For $\ell \geq 2$ the covariant temperature multipoles are invariant under a change of the fundamental congruence at linear order. Writing $\tilde{u}^a = u^a + v^a$ with $v^a u_a = 0$ and $v^a$ first order, GDE99 (37) gives

$$\tilde{\tau}_{A_\ell} \;=\; \tau_{A_\ell} \;-\; v_{A_\ell}, \qquad v_{A_\ell} \equiv 0 \quad \text{for} \quad \ell \geq 2, \tag{55}$$

so only the monopole and the dipole move, and every $\ell \geq 2$ multipole is a frame invariant. The bundle computes the same $C_\ell$ twice — once in the Newtonian frame, $\tilde{\sigma}_{ab} \simeq 0$, and once in the CDM frame — and the two must agree. They do not: the disagreement is total, median $9.99 \times 10^{-1}$ and worst 1.00 over the released multipole set.

Equation (55) is a theorem at linear order, and the order is the whole of what it licenses. The multipoles are first order and $C_\ell$ is quadratic in them. A tilt leaves every $\ell \geq 2$ multipole exactly invariant at first order and corrects it by the aberration relabelling at second, $\Delta\tau_{A_\ell} = \mathcal{O}(v\tau)$, so

$$\Delta C_\ell \;\simeq\; 2\langle \tau\, \Delta\tau \rangle \;=\; \mathcal{O}(v\,\tau^2), \qquad \frac{\Delta C_\ell}{C_\ell} \;=\; \mathcal{O}(v). \tag{56}$$

The fractional residual a tilt may produce is first order in the tilt velocity. The measured residual is $9.99 \times 10^{-1}$ at the median. For that to be a tilt effect $v$ would have to be of order unity, which is two congruences in relative motion at the speed of light, so the disagreement is not a large violation of (55). It lies outside the expansion in which (55) is stated, and two branches separated by more than any order of a tilt expansion permits are not related by a tilt.

That moves the question from the formalism to the objects being compared. (55) is invariance of $\tau_{A_\ell}$ under an infinitesimal change of congruence at a common direction label. What the bundle compares is raw multipoles from two numerical histories whose frame labels select different source constructions, with no direction map, no energy transformation and no common emission endpoint imposed between them. Those are different statements, and the footnote on this section's title says so. Until the common sky is defined the residual localises a mismatch between code paths rather than measuring a frame-invariant observable failing to be invariant, and the triage below is read in that light.

A second measurement points the same way and does not depend on the order counting. Under the released weighting the Newtonian branch has no acoustic maximum on any background, and under the split it acquires one at $\ell \simeq 68$ that moves by 0.40% between Einstein–de Sitter and ΛCDM while the CDM-frame peak moves by 60.0%. A feature that does not track the acoustic scale is not an acoustic peak, so the two frames are not carrying the same object, and a fractional residual near unity follows from dividing one by the other. That measurement is recorded in `diagnostics/d36-frame-invariance-by-weighting.csv`; whether the $\ell \simeq 68$ feature survives a finer multipole grid is not yet excluded.

### 15.2 Triage A — sites the discrepancy cannot come from

Each row names the instrument and a grade. THEOREM is exact at linear order; MACHINE ZERO is measured at $10^{-13}$ or better and is structurally zero; MEASURED is a number with a stated validity condition; ASSERTED is read or reasoned and is usable but not load-bearing.

| site | instrument | grade |
|---|---|---|
| Free streaming and transport | GDE99 (37) | THEOREM |
| The observer endpoint: a pure dipole | GDE99 (37); Paper 1 §§5.3, 6.2 | THEOREM |
| The Bessel projection | T-71, synthetic control | MEASURED |
| The $C_1$–$C_2$ boost relation | T-77, $3.3 \times 10^{-16}$ | MACHINE ZERO |
| The by-parts identity itself | `c3_byparts.py` and two derivations | MACHINE ZERO |
| The oscillator at leading order | register; two external reads | MEASURED |
| The source's frame consistency | **T-82** | RESOLVED |
| Boundary-against-bulk bookkeeping | **T-85** | RESOLVED, EXCLUDED |
| The additive $\kappa'\tau_0$ route | **T-79** | RESOLVED, CLOSED |

One row was withdrawn from this table, 2026-10-23. It read *"$C_0$'s boost behaviour with $\mathcal{B}_0$ restored"*, graded MACHINE ZERO on T-77 and `c5_monopole.py`. *The measurement stands* — $a\Delta\mathcal{B}_0 = \frac{1}{3}kv = \frac{1}{3}k^2\Delta C_2$ exactly — *but the configuration it was measured in does not.* T-105, which restored an $a\mathcal{B}_0\delta_{\ell 0}$ row to $C_0$, is withdrawn by T-107: $a\mathcal{B}_0 = -\frac{k}{3}\tau_1$ is already carried inside $-C_1'$, so restoring it counts the monopole source twice and criterion 5b doubles at 98.5%. Under the corrected row $C_0 = -\frac{1}{3}a\mathcal{B}_2$ and $\Delta C_0 = +\frac{kv}{3} \neq 0$, so this is not a site the discrepancy cannot come from. Its partner is cross-coefficient and `c8_rank0_partner.py` is the gate — M15: a gate built from one coefficient cannot see a cancellation distributed across the bracket.

Nothing in this table can be moved by a measurement. A result that appears to contradict one of these rows is wrong, or the comparison is void — most often the latter, through $\ell \gtrsim k\Delta\eta$, where the spherical Bessel function is exponentially small and a ratio of two vanishing numbers is still a number.

### 15.3 The two measurements that closed the upstream end

**The source agrees between the frames (T-82).** CAMB carries the rank-1 scattering source as a single object $g\,(v_b + \sigma)$. Under a boost $v_b \to v_b - v$ while the shear term shifts oppositely, so the sum is invariant; a code that never separates them has no frame gap by construction. This bundle does separate them, so the two columns must produce the same sum. In this bundle's slots the combination is $\hat{v}_B + C_2$, with $\hat{v}_B = \tau_1/k$ the reduced dipole and $C_2 = v/k$, since $a\mathcal{B}_2 = a\sigma = -kv$ and $C_2 = -a\mathcal{B}_2/k^2$. Differencing the two columns, on identical histories and before any projection — no transfer function, no $j_\ell$, no weight, no damping:

$$\big|\Delta(\hat{v}_B + C_2)\big| \;\in\; \big[\,9.3 \times 10^{-16},\; 1.8 \times 10^{-13}\,\big], \qquad k \in \{1, 10, 60, 300\}, \tag{57}$$

on both the Einstein–de Sitter and the Planck backgrounds. The source is frame-consistent to fourteen significant figures. Set against a frame gap of 100% at the spectrum, the pair says what neither says alone: everything between the source and the spectrum is the projection.

**The by-parts identity holds in both frames (T-85).** With $W = e^{-\kappa}$ and $\mathcal{V} = W'$ — the sign this bundle's $\kappa$ convention forces, $\kappa$ being accumulated backwards from $\eta_0$ — the transport source obeys, exactly,

$$\int \mathcal{V}\, M\, j_\ell \,\mathrm{d}\eta \;=\; \Big[\,W M j_\ell\,\Big] \;-\; \int W M' j_\ell \,\mathrm{d}\eta \;+\; k \int W M\, j_\ell' \,\mathrm{d}\eta, \tag{58}$$

where $x = k(\eta_0 - \eta)$, so $\mathrm{d}j_\ell/\mathrm{d}\eta = -k\, j_\ell'(x)$ and the last sign is $+k$. Evaluating the left side (call it $A$) and the right side ($B$) on identical histories and identical endpoints, with $M = -C_1$, in each frame gives four numbers. Both identities hold: $|A - B|/|A|$ sits at $10^{-3}$–$10^{-2}$ and *falls under grid refinement* — $2.83 \times 10^{-3} \to 2.20 \times 10^{-3} \to 2.04 \times 10^{-3}$ at $(k, \ell) = (10, 2)$, and $1.105 \to 0.279 \to 0.0727$ at $(300, 2)$, about $4\times$ per doubling, which is second order and is what the trapezoidal rule gives on an oscillatory integrand.

Since $A$ and $B$ are the same integral by an exact identity, a difference has only two available explanations, and refinement removes one of them. Boundary-against-bulk bookkeeping is excluded. But $|A_N - A_C|$ is 1.00 to 1.78 of scale in every valid cell, with the sign reversed between columns: the coefficient definitions carry the whole of the gap.

**One apparent corroboration, withdrawn.** At low $k$ the ratio $A_C/A_N$ is independent of $\ell$ to four figures. That carries no information about the frames. $\mathcal{V}$ is a narrow spike, so $\int \mathcal{V} f j_\ell \simeq f(\eta_*)\, j_\ell(x_*) \int \mathcal{V}$ for *any* $f$, and the ratio of two $\mathcal{V}$-weighted integrals is $f_C(\eta_*)/f_N(\eta_*)$ — free of $\ell$ by construction. Predicted against measured, the two agree to 0.9–4.2% for $k \le 60$ and fail at $k = 300$, where the mode oscillates inside the visibility width and the $\ell$-freedom correspondingly disappears. The check cost one run and removed a finding rather than adding one.

### 15.4 Triage B — the one frame-sensitive site that remains

Five candidates were open on 18 October. Four are subsumed or excluded — the $\ell = 0$ matching row is landed; boundary-against-bulk is excluded by T-85; and the dipole boost $-v/k$ and the supplied $C_1'$ are both subsumed by T-82, which shows the source they would have corrupted is clean. The fifth is the site that remains: $\kappa' v_B$ inside $C_1$ does not go away, it *is* the slot and weight assignment of $C_1$, which is why it is named here rather than counted among the four. One site remains: the slot and weight assignment of $C_1$.

Identity (58) has three terms and every one of them carries the weight $W$. The bundle has two, and neither carries $W$:

| | **the identity requires** | `integral_solution.py` **has** |
|---|---|---|
| boundary | $\big[WMj_\ell\big]$, weight $W = e^{-\kappa}$ | $(C_2'^* - C_1^*)\,\mathcal{D}(\eta_0, k)\, j_\ell(x_*)$ — weight $\mathcal{D}$, a *diffusion* factor, carrying neither $W$ nor $\mathcal{V}$ |
| $j_\ell$ bulk | $-\int WM' j_\ell$, weight $W$, *minus* | $\int \mathcal{V}\mathcal{D}\,\{C_0 - C_1' + C_2''\}\, j_\ell$ — weight $\mathcal{V}\mathcal{D}$, *plus* |
| $j_\ell'$ bulk | $+k \int WM j_\ell'$, $M$ *undifferentiated* | **absent. The bulk has no $j_\ell'$ slot.** |

The structural half is the third row. Everything was integrated by parts *onto the coefficients* — $\{C_0 - C_1' + C_2''\}$ is (102)'s printed bracket, and it is the $W$-weighted by-parts form of $\int W\big[C_0 K + C_1 K' + C_2 K''\big]$ — and that decomposition is then evaluated at $g$ weight, where it is not available. The boundary meanwhile is *Annals II*'s own `visibe` collapse: under $\int \mathcal{V} = 1$ the $-\mathcal{V}C_1$ term becomes $C_1(\eta_*)$. So the bundle carries one term from each of two incompatible decompositions, and then applies a third weight, $\mathcal{D}$, to one of them.

**What the repair is, stated so it can be refuted.** Not a split of the slot. A split, *plus* a weight change on each half, *plus* a slot the bulk does not currently have:

1. $C_1$'s two contents separate — $M$'s by-parts residue, and $(v_b + \sigma)$.
2. They *arrive under* different weights: $W$ for the gravitational half, $\mathcal{V}$ for the scattering half. That is an outcome, not an operation, and reading it as one is finding **T-103** — the scattering half arrives at $\mathcal{V}$ weight carried on the *undifferentiated* $\hat{v}_B$; it is not *multiplied* by $\mathcal{V}$. Multiplying $C_1^{\rm thom} = -\kappa' \hat{v}_B$ by $\mathcal{V}$ applies a second $\kappa'$, which is the 320–682 of §15.8. See §14.1, where the weights fall out of the integrating factor and nothing is multiplied by hand.
3. The $W$-weighted half needs *both* bulk slots, $j_\ell$ and $j_\ell'$, because by-parting onto the coefficients is what removed the second, and that step is legal only at $W$.

The numerics for the missing slot already exist in the file: $j_\ell' = \big[\ell j_{\ell-1} - (\ell+1) j_{\ell+1}\big]/(2\ell+1)$ is built inside the rank-2 *boundary* term, which has never been evaluated because $C_2$ vanishes in every Newtonian run.

The acceptance condition is stated in advance. The split must leave $A_N$ and $A_C$ equal, and must not move the Newtonian column by more than the quadrature floor T-85 measured. Frame agreement alone is *not* sufficient — two wrong columns can agree — so the spectrum adjudicates, and it adjudicates last.

### 15.5 Triage C — frame-blind sites

A term $X$ with $\Delta X = 0$ passes every frame test for free, so no frame instrument can see it. These cannot cause the discrepancy and can ruin the spectrum; they are listed so that the two questions are not confused.

| | **site, and what it bears on** | **status** |
|---|---|---|
| C1 | the coefficient of any boost-invariant term; low-$k$ amplitude | `c5` blind by construction |
| C2 | `LCDM_MODEL` without $\Omega_r$; peak positions | confirmed, 37%, T-83 |
| C3 | radiation-era driving, static $T(k)$; amplitude and phase | open (A-3) |
| C4 | the tight-coupling freeze at $6\eta_*$; peak contrast, temperature–Doppler balance | open (A-5) |
| C5 | $k_D$ computed at $R = 0$; one $\mathcal{D}(k)$ across histories; higher peaks and the tail | T-81a/c/d, v1.4.0 |
| C6 | $\mathcal{D}(k)$ applied to the integrand *at all*; the late ISW | T-81b — first named mechanism |

C6 and the slot question are the same structural change seen twice. Diffusion damping is a scattering effect; gravitational redshift accumulated after decoupling is not damped by photon diffusion. Scattering damping belongs to the scattering half of the source and not to the gravitational half — so the split that fixes the weights fixes the damping in the same stroke. One change, three symptoms: the frame gap, the missing late ISW, and the $\mathcal{D}$-on-the-boundary mismatch.

### 15.6 Triage by conformal time

The same question asked along the photon's history. It localises cleanly, and the over-damping sits in two windows — neither of which is an integration limit, since T-85 settled the bookkeeping.

| **window** | **what the code does** | **what it costs** |
|---|---|---|
| $\eta < \eta_{\rm eq}$ | prescribed matter-growth $\Phi$ times a static $T(k)$, where the potentials should decay through horizon entry and drive the oscillator | peak amplitude and phase (C3) |
| $\eta_{\rm eq}$ | no $\Omega_r$ in `LCDM_MODEL` | peak positions, 37% in $\ell_A$ (C2) |
| $\eta_{\rm eq} \to \eta_*$ | oscillator sound speed correct; $k_D$ evaluated at $R = 0$ while the oscillator runs at $R(\eta_*) \simeq 0.61$ | the damping scale (C5) |
| $\eta_* \pm \frac{1}{2}\Delta\eta_{\rm vis}$ | instantaneous boundary, envelope substituted for a visibility of finite width $\Delta\eta = 0.00972$ | the tail from $\ell \simeq 785$ |
| $\eta_* \to 6\eta_*$ | tight-coupled oscillator continued and then frozen; no slip, no growing quadrupole, no polarisation | peak contrast, the temperature–Doppler balance (C4, A-5) |
| $6\eta_* \to \eta_0$ | the integrand carries $\mathcal{D}(k)$ where $e^{-\kappa} = 1$ and no scattering occurs | the late ISW (C6) |
| $\eta_0$ | reception: Doppler and aberration | nothing — THEOREM |

Three consequences are worth drawing out.

*The integration range is clean.* T-85 settled the bookkeeping, so how far the problem has been integrated is no longer a candidate.

*"Conformal time" and "missing radiation" are one question.* $\eta$ is not a free choice: it is fixed by the background. A radiation-free background gives a different $\eta(z)$, hence a different $k\Delta\eta$ at every point,

hence different Bessel arguments and different peak positions. The 37% in $\ell_A$ *is* the conformal-time defect, already measured.

*Over-damping is two substitutions, not a limits error,* and both sit after $\eta_*$: an envelope standing in for a visibility of finite width, and $\mathcal{D}(k)$ applied to a source generated where no scattering occurs. A discriminator is available and was not used until now: Silk damping bites at $\ell \simeq 1807$, the visibility full-width scale is $\ell \simeq 785$, and the computed spectrum departs at $\ell \simeq 700$. The departure fingerprints the width scale, not Silk — so $k_D$ at $R = 0$ is a real defect and is *not* what sets where the tail goes wrong.

### 15.7 What would change this account

Two outcomes would move it, and both are cheap.

If the slot read does not match the shape in §15.4, the discrepancy returns to having no named site, which would be the first time since 18 October. *It matched*; the table in §15.4 is that read.

If removing $\mathcal{D}(k)$ from the integrand alone recovers a late ISW, C6 is confirmed and C3–C5 reduce to amplitude-only concerns.

### 15.8 The repair, attempted five times, and withdrawn

Recorded here because five failures carry more information than one success would, and because a reader of this document should not have to reconstruct them from the register.

T-86's reading was implemented as four independent controls, every one defaulting to `False`. The first combination closed the frame gap by four orders at $\ell \geq 50$ for $k \lesssim 60$ — and is withdrawn. It applied $\mathcal{V}\, C_1^{\rm thom}$ where (102) applies $W\, C_1^{\rm thom}$, and since

$$C_1^{\rm thom} = -\kappa' \hat{v}_B, \qquad W\kappa' = \mathcal{V} \quad \Longrightarrow \quad W\, C_1^{\rm thom} = -\mathcal{V}\hat{v}_B \ \textit{identically},$$

the Thomson term is already at visibility weight in (102) as printed and there was never a weight to change. The control therefore multiplied it by a spurious $\kappa'$, measured at 320 to 682. The diagnostic spectrum it produced — first peaks moving from 1.3–1.5 $\ell_A$ to 0.56 $\ell_A$ — is withdrawn with it and is retained in `diagnostics/` as the record of what the defective variant produces, not as evidence.

What the five failures leave standing is a review flag, and it is P1-T's. *A coefficient carrying a bare $\kappa'$ is a site where a large residual is manufactured the moment it is differentiated, and is worth flagging before implementation.* Revised 2026-10-21, finding T-113 and decision S48: an earlier form of this paragraph stated it as a prohibition — *"$\kappa'$ may live in $W$, in $\mathcal{V}$, or in a derivative of either, never in a coefficient"* — and it was read as normative. It is not a theorem, because $W(\kappa'\mathcal{S}) = \mathcal{V}\mathcal{S}$ is exact: a coefficient *may* hold $\kappa'$, and the defect in (102) is wrong weights, signs, boundaries or missing partners rather than the presence of $\kappa'$.

The flag is kept because it separates the *legitimate* $\kappa'^2$ inside $\dot{\mathcal{V}} = (\kappa'' + \kappa'^2)W$, which comes from differentiating the weight, from the illegitimate one, which comes from differentiating a coefficient; both look identical in the output and only the provenance tells them apart. `tests/frame/c7_kappa_ledger.py` raises it structurally, and both of this week's defects would have been flagged by it before implementation. What it may no longer do is settle a question on its own — D24 supplies the size the flag cannot: $\kappa''\tilde{v}_B$ is 99.8% of $|C_1'|$ in the Newtonian frame and 109.8% in CDM, finding T-121.

### 15.9 What one term does to the integrand

Measured on the released code, as shares of the summed absolute Newtonian integrand:

| term | EdS $k$=10 | $k$=60 | $k$=300 | PLANCK $k$=10 | $k$=60 |
|---|---|---|---|---|---|
| $\mathcal{V}\kappa''\hat{v}_B$ | **0.935** | **0.971** | 0.616 | **0.933** | **0.974** |
| $\mathcal{V}\kappa'\hat{v}'_B$ | 0.066 | 0.024 | 0.314 | 0.065 | 0.015 |
| $\mathcal{V}\frac{k}{3}\tau_1$ | 0.0002 | 0.0064 | 0.082 | 0.0004 | 0.013 |
| $\mathcal{V}(\Phi'_A - \Phi'_H)$ | 0 | 0 | 0 | 0.0018 | 0.0028 |

One term is 93–97% of the integrand; the acoustic driving is 0.02% and the ISW, where there is one, 0.18%. Against the source it stands for — the physical Doppler $g\hat{v}_B = W\kappa'\hat{v}_B$ — it is $1.81 \times 10^5$ (EdS) and $8.99 \times 10^4$ (PLANCK), and $\kappa'' \simeq -\kappa'^2$, so it carries three powers of $\kappa'$ where the source carries one.

This is weight-independent. Changing $\mathcal{V} \to W$ divides every term in the bracket by $\kappa'$ uniformly and changes no ratio, so A-2 cannot reach it. The term exists only because (102) differentiates $C_1$, which contains $\kappa'$ — that is, because the integration by parts was taken *onto the coefficients* rather than onto the product.

### 15.10 What this section does not establish

No measurement here is a comparison against an external code, and none is a claim that the spectrum is right. The chain establishes that the rank-1 source is frame-consistent, that the by-parts identity holds, and that the slot-and-weight assignment is where the remaining disagreement lives. It does not establish that repairing the assignment closes the gap, and the acceptance condition in §15.4 is written so that the repair can fail visibly.

## 16 Line-of-sight assembly and repository conformity

This section checks the full assembly rather than individual coefficients. It introduces no additional collision term or physical effect. The published antecedent is the general, almost-Friedmann–Lemaître integral solution of Gebbie et al. (2000), Eq. (102), with the definitions and PSTF mode normalisation of Gebbie and Ellis (2000). The description of frame covariance is that of Gebbie (2026), especially Secs. II–III and V–VI. Paper 1 governs which published antecedent equation and correction is used at each step. The journal pages test the fidelity of that assignment; where the mapping or derivation has not been independently checked, the attribution remains unverified instead of being silently substituted.

### 16.1 One pre-integrated object and its exact transformation

Keep $u^a$ arbitrary and let $f$ identify the threading only after the source has been written. The following is a *conditional* mathematical object, not an assertion that the exact brightness hierarchy reduces to it. Define the common conformal-time kernel and weight

$$K_\ell(\eta, k) = j_\ell[k(\eta_0 - \eta)], \qquad P(\eta, k) = W(\eta)\mathcal{D}(\eta, k), \qquad W' = \mathcal{V} = \kappa' W. \tag{59}$$

For each chosen frame, consider the complete three-rank integral

$$I_\ell^{(f)} = \int_{\eta_*}^{\eta_0} P\,[C_0^{(f)} K_\ell + C_1^{(f)} K'_\ell + C_2^{(f)} K''_\ell]\,\mathrm{d}\eta. \tag{60}$$

No assumption about the vanishing of either endpoint has entered. Integration by parts twice gives the bulk term

$$S_\ell^{(f)} = PC_0^{(f)} - (PC_1^{(f)})' + (PC_2^{(f)})'' \tag{61}$$

$$= P(C_0^{(f)} - C_1^{(f)\prime} + C_2^{(f)\prime\prime}) - P'C_1^{(f)} + 2P'C_2^{(f)\prime} + P''C_2^{(f)}. \tag{62}$$

The boundary is the upper value minus the lower value of $PC_1K_\ell + PC_2K'_\ell - (PC_2)'K_\ell$. For $\ell \geq 2$, $K_\ell(\eta_0) = K'_\ell(\eta_0) = 0$, subject to finite coefficient products. With $x_* = k(\eta_0 - \eta_*)$ the remaining endpoint is

$$E^{(f)}_{*,\ell} = \left[(PC^{(f)}_2)' - PC^{(f)}_1\right]_* j_\ell(x_*) + k[PC^{(f)}_2]_* j'_\ell(x_*). \tag{63}$$

Thus $I^{(f)}_\ell = E^{(f)}_{*,\ell} + \int_{\eta_*}^{\eta_0} S^{(f)}_\ell K_\ell \, \mathrm{d}\eta$. This equality is a product-rule identity under the stipulated source. It is not sufficient to establish that the *published* Eq. (102) contains precisely (60), or that $P$ may be chosen arbitrarily.[20]

Where $\mathcal{D}$ depends on time, $P' = \mathcal{V}\mathcal{D} + W\mathcal{D}'$ and $P'' = \mathcal{V}'\mathcal{D} + 2\mathcal{V}\mathcal{D}' + W\mathcal{D}''$. Dropping these terms is an additional assumption, not a consequence of tight coupling. The rate $\kappa'$ here is conformal; proper-time kinematic derivatives require their stated factor of $a$. The wavenumber in $K_\ell$ is comoving and constant in $\eta$, and must not be interchanged with $k_{\mathrm{phys}} = k/a$ when applying the derivative identities.

### 16.2 The opacity ledger and a negative control

The scattering term must be counted *once*. If the Thomson part of a coefficient is $C^{\mathrm{T}}_1 = -\kappa'\hat{v}_B$, then

$$\boxed{WC^{\mathrm{T}}_1 = -\mathcal{V}\hat{v}_B, \qquad \mathcal{V}C^{\mathrm{T}}_1 = -W(\kappa')^2\hat{v}_B.} \tag{64}$$

These are distinct except at special values of the rate. The first expression is an algebraic identity, while the second is the result of applying an additional visibility factor to an already rate-bearing coefficient. Neither expression alone specifies the correct collision source: the Thomson hierarchy and the integrating factor must supply that choice. The companion negative control in `tools/check-los-v1.3.8.py` demonstrates the difference on nonconstant synthetic histories without importing a spectrum model.

### 16.3 What frame agreement must mean

At a fixed measured observer and emitter, the observed distribution is independent of the intermediate choice of $u^a$. The small-tilt multipole rule used in Gebbie et al. (2000), and in Paper 1, does not assert equality of *raw* frame-labelled high-multipole coefficients when the direction bases and endpoints have not been identified. Paper 1, Eqs. (73)–(75), supplies the Doppler–aberration relabelling; the exact statement is made for the photon distribution on phase space, while the displayed multipole law is an expansion at the stated order.

For the conditional integral above, a valid algebraic comparison of frames $N$ and $C$ with *common* $P$ is

$$\Delta I_\ell = \Delta E_{*,\ell} + \int_{\eta_*}^{\eta_0} \mathrm{d}\eta \, K_\ell[P\Delta C_0 - (P\Delta C_1)' + (P\Delta C_2)''], \quad \Delta X = X^{(C)} - X^{(N)}. \tag{65}$$

For different weights the correct comparison is instead the difference of the two complete weighted integrals; setting the weights equal is not innocuous. The physical frame test asks whether *both* calculations, transported to the same physical endpoint frames, yield the same measured sky, including the finite Doppler–aberration map and the rank-two boundary. A vanishing difference in only the rank-one source, the by-parts identity, or a pair of common-weight spectrum routes is not that test.[21]

[20] In particular, the scalar boundary term in this identity contains $(PC_2)'_*$, and the bulk contains derivatives of $P$; the printed slow-decoupling solution must be compared with the source equations, approximation order and harmonic conventions in the published journal before replacing either.

[21] The existing frame tests can localise errors because both frames share many inputs. They cannot automatically certify inputs shared by both. The test must therefore retain the independently measured low-$\ell$ spectrum and the acoustic peak structure as separate physical checks.

### 16.4 Implementation conformity at the pinned repository revision

The repository was inspected read-only at `70f0f44ff8f71f0313a97c17df7fcf0b83d6b753` on 25 October 2026. The exact file hashes and relevant source-line ranges are in `REPOSITORY-CONFORMITY-v1.3.8.csv`. The following conformity statements are about *code paths*; no new full scientific run is implied.

| Module / setting | Observed implementation | Scientific check still owed |
|---|---|---|
| `coefficients.py` | The frame constructors select the corrected monopole row ($C_0 = -a\mathcal{B}_2/3$) through `MONOPOLE_ROW`. The direct `match_coefficients` helper still defaults its correction flag to `False`; other variants are available as controls. | Confirm the complete row from the published matching prescription, rather than from the normalisation identity alone. |
| `integral_solution.py` | Printed-style boundary, shear boundary and bulk are evaluated separately. Optional switches vary weights and add candidate partner terms. | Establish a single physical source and both boundary conditions before selecting a switch. |
| `two_frame.py` | The released rank decomposition has `B_SLOT_REPAIR=False`; the diagnostic control dictionary is empty by default. | Demonstrate a traced, end-to-end test with every nondefault source control reported and with both frames mapped to common endpoints. |
| `pipeline.py` | Gravitational and Doppler weighting can be specified independently; omission of the Doppler override retains the existing selection. | Verify the late-time ISW contribution and the source-specific visibility weighting against the chosen physics. |
| `frame_transform.py` | The reduced and unreduced dipoles are shifted by different $k$ factors. | Carry the same conventions into the matched coefficients, derivatives and observed sky. |
| `TAU_CONTINUATION` | The released value is `freeze`: the rank-one acoustic solution is held after its coupling window. A `free_stream` branch exists. | Test the physical handover and the $C_0$–dipole cancellation before removing the late-time visibility suppression. |

The by-parts branch currently adds the two $C_2$ weight derivatives without explicitly adding $-\mathcal{V}C_1$ as a separate term; other switches reconstruct different brackets. This is a directly located source-level nonconformity with *the particular identity* (35) when its options are read as an implementation of that identity. It does not establish that the default spectrum executes the branch.[22]

### 16.5 Acceptance order for the physical spectrum

An acceptance test must first fix the published equations and conventions, then test the matching rows and derivatives, then compare two frames with all endpoints transformed, and only then compare $C_\ell$ against independently produced numerical spectra. For Einstein–de Sitter the late integrated Sachs–Wolfe source must vanish under the assumed constant potential; for the $\Lambda$ models it must be permitted to accumulate after the visibility has closed. A source weighted only by $\mathcal{V}$ cannot pass that late-time test if its potential evolution occurs where $\mathcal{V}$ is negligible.[23] These are separate conditions from the existence of acoustic oscillations in the source or the agreement of two internally shared normalisation routes. The

[22] At the pinned revision, compare `functions/spectra/integral_solution.py`, lines 219–229 and 395–485, with (62); the default switches are recorded in `functions/spectra/two_frame.py`, lines 441–485 and 559–577. The detailed status and negative controls are in `GATE-v1.3.8-AUDIT.md`.

[23] An additional pinned-repository qualification is `TAU_CONTINUATION="freeze"` in `functions/spectra/pipeline.py`, lines 220–263. Its documentation records finding T-145: freezing the dipole past the tight-coupling window prevents the $C_0$–dipole cancellation expected at late times, while a visibility-weighted bulk hides the residual. The alternative continuation has not been accepted through an end-to-end spectrum test in this gate.

repository's current reported spectrum therefore remains a diagnostic reconstruction, not a certified physical $C_\ell$.[24]

### 16.6 The physical endpoint map: an independent finite-tilt check

The comparison of two threadings requires a common physical observer and emitter. This is a local statement about the photon distribution on its null mass shell, not a choice of coordinate gauge. In the 1+3 split, $u_N^a$ and $u_C^a$ denote unit timelike congruences and one photon is $p^a = E_N(u_N^a + e_N^a) = E_C(u_C^a + e_C^a)$. For a boost of magnitude $\beta$ along a unit spatial vector $z^a$ orthogonal to $u_N^a$, take $u_C^a = \gamma(u_N^a + \beta z^a)$, with $\gamma = (1-\beta^2)^{-1/2}$ and $\mu_N = z_a e_N^a$. Contraction with $u_C^a$ gives $E_C = \gamma E_N(1 - \beta\mu_N)$. The corresponding direction cosines and Doppler factor are

$$\mu_C = \frac{\mu_N - \beta}{1 - \beta\mu_N}, \qquad \mu_N = \frac{\mu_C + \beta}{1 + \beta\mu_C}, \qquad \mathsf{D}(\mu_C) = \frac{E_C}{E_N} = \frac{\sqrt{1-\beta^2}}{1+\beta\mu_C}. \tag{66}$$

A single distribution function has the same value at the same photon phase-space point, independent of its decomposition: $f_C(x, E_C, e_C) = f_N(x, E_N, e_N)$. For a directionally Planckian brightness this implies $T_C(\mu_C) = \mathsf{D}(\mu_C) T_N(\mu_N)$. The blackbody assumption is needed for the last temperature equation, not for the distribution-function statement. The angular map fixes the axis without imposing a spacetime metric perturbation or a gauge.

At a fixed $\mu = \mu_C$, differentiating both the Doppler factor and the argument of $T_N$ at $\beta = 0$ gives

$$\left.\frac{\partial T_C(\mu)}{\partial\beta}\right|_{\beta=0} = \underbrace{(1-\mu^2)\frac{\partial T_N}{\partial\mu}}_{\text{angular aberration}} - \underbrace{\mu T_N}_{\text{Doppler and modulation}}. \tag{67}$$

Thus, if $T_N = \bar{T}(1+\tau_N)$ with $\tau_N = O(\varepsilon)$ and $\beta = O(\varepsilon)$, the boosted anisotropy contains the usual first-order dipole $-\beta\mu$. The angular derivative of $\tau_N$ and the Doppler modulation of $\tau_N$ are $O(\varepsilon^2)$. The first contains a direction derivative and hence generates the rank-dependent coupling of Paper 1, Eq. (75); the second has no corresponding rank enhancement. Equality of the raw $\ell \geq 2$ multipoles at first order does not assert their equality at this higher order.

The inverse comparison is exact. Writing $\mu_C(\mu_N) = (\mu_N - \beta)/(1 - \beta\mu_N)$, the measured sky, brought back to the original physical frame, satisfies

$$\boxed{T_{N\leftarrow C}(\mu_N) = \frac{T_C(\mu_C(\mu_N))}{\gamma(1-\beta\mu_N)} = T_N(\mu_N).} \tag{68}$$

This is the kinematic acceptance condition; it does not demonstrate that the two calculated radiation histories satisfy it. The independent test in `tools/check_endpoints_opacity_v1.3.9.py` uses $\beta = 0.002$ and the synthetic directional temperature $T_N/\bar{T} = 1 + 3\times10^{-4}P_2(\mu) + 2\times10^{-4}P_3(\mu) + 10^{-4}P_4(\mu)$. Its exact boost–inverse-boost discrepancy is $3.33\times10^{-16}$ in maximum absolute temperature units. The untransformed dipoles differ by $-1.99952\times10^{-3}$, and the untransformed quadrupoles by $3.17938\times10^{-6}$. These differences are expected: they compare two different physical angular bases. The result is an identity check, not a test of the CMB transfer or a new frame-invariance measurement of the repository.

### 16.7 A late-time opacity and driving control

The time orientation of the optical depth fixes the sign of the integrating-factor derivative. For a nonnegative conformal scattering rate $q(\eta) = a n_e \sigma_T$, define the optical depth from $\eta$ to the observer and

[24] The current repository README and F9 report a deficient low-$\ell$ plateau, missing secondary peaks, and a displaced primary peak. Those empirical statements are inherited from the pinned repository's recorded runs; they were not reproduced in this supplement editing gate. No repair is inferred from the synthetic checks below.

the associated weights by

$$\kappa(\eta) = \int_{\eta}^{\eta_0} q(s)\,\mathrm{d}s, \qquad W = e^{-\kappa}, \qquad \boxed{W' = qW = \mathcal{V}, \quad \kappa'_{\text{ordinary derivative}} = -q.} \tag{69}$$

The notation $\kappa'$ for the *positive* differential scattering rate in the antecedent corresponds here to $q$; it must not be read as $\mathrm{d}\kappa/\mathrm{d}\eta$ for the optical depth defined above. These are distinct conventions for different objects, and the product-rule identity in Sec. 16 consistently uses $W' = \mathcal{V}$. In particular, $W(-q\hat{v}_B) = -\mathcal{V}\hat{v}_B$, whereas multiplying that coefficient by $\mathcal{V}$ instead gives $-Wq^2\hat{v}_B$. This is the local opacity ledger, not a substitute for deriving the Thomson collision integral.

The late integrated source gives a separate necessary condition. Let $g(\eta) = \Phi'_A(\eta) - \Phi'_H(\eta)$ denote the potential-driving combination in the adopted scalar harmonic convention. If $g$ is zero, there is no late integrated contribution from this term. If $g \neq 0$ after the scattering rate has vanished, then

$$\int_{\eta_{\text{late}}}^{\eta_0} Wg\,\mathrm{d}\eta = \int_{\eta_{\text{late}}}^{\eta_0} g\,\mathrm{d}\eta, \qquad \int_{\eta_{\text{late}}}^{\eta_0} \mathcal{V}g\,\mathrm{d}\eta = 0 \quad (q = 0 \text{ throughout the interval}). \tag{70}$$

The positive control uses a synthetic compact-support rate on $0 \le \eta \le 1$: $q = 3.4[\max\{1 - ((\eta - 0.25)/0.12)^2, 0\}]^3$. The potential is constant until $\eta = 0.62$ and then decreases by 0.006 under a quintic smoothstep completed at $\eta = 0.85$, with $\Phi_H = -\Phi_A$. The independent quadrature returns $\int Wg\,\mathrm{d}\eta = -0.0120000$ and $\int \mathcal{V}g\,\mathrm{d}\eta = -3.31 \times 10^{-18}$. For an exactly stationary potential the same test gives an integral of magnitude below $5 \times 10^{-18}$. This distinguishes the support of a late gravitational source from the support of the scattering visibility. The histories are chosen test functions; they do not solve the Einstein–matter system and the values are not predictions for Einstein–de Sitter or ΛCDM.

### 16.8 Published antecedents and the remaining scientific gate

The published antecedents remain Gebbie and Ellis (2000), *Annals of Physics* **282**, 285–320 (2000), DOI `10.1006/aphy.2000.6033`, and Gebbie et al. (2000), *Annals of Physics* **282**, 321–394 (2000), DOI `10.1006/aphy.2000.6034`. The journal bibliographic records were confirmed for this gate, but its disputed equations (especially the normalisation and signs in matching Eqs. (89), (90) and the source assumptions behind (102)) could not be checked against the journal page images. Paper 1 v4 supplies the controlling provenance cross-reference, not an independent verification of each journal equation.[25] The remaining unverified attributions stay open.

The pinned software inspection is read-only at Git commit `70f0f44ff8f71f0313a97c17df7fcf0b83d6b753`. A fresh cloud checkout was prevented by repository-host DNS unavailability, so neither the released spectrum nor the optional frame-control branches were run in this gate. The direct code-path review from v1.3.8 remains the evidence for the recorded defaults; `SOURCE-STATUS-v1.3.9.csv` separates that inspection from the new independent calculations. Before any physical claim can close, the journal matching rows, proper/conformal-time rates, common-endpoint $C_\ell$ and the low-$\ell$ and peak measurements still require one version-pinned, end-to-end scientific run.

## 17 The angular spectrum

*The harmonic normalisation must cancel from the observable. The angular power spectrum is quadratic in the transfer function. Both are checked explicitly.*

[25] Since the v1.3.9 gate, the published PDFs of Annals I, Annals II and MGE 1999 have been supplied. A limited v1.3.10 page check confirms the Annals I $\beta_\ell$ and recurrence equations (23)–(24) used by Paper 1 Table III. The disputed Annals II source and matching equations remain OPEN pending a separate substantive derivation and comparison.

## 17.1 Two published expressions and their normalisation

*Annals II* §7.2 prints the observable twice:

$$C_\ell = \frac{2}{\pi}\frac{\beta_\ell^2}{(2\ell+1)^2}\int \frac{\mathrm{d}k}{k}\, k^3 \left|\tau_\ell(k,\eta_0)\right|^2, \tag{186}$$

$$\langle \tau_{A_\ell} \tau^{A_\ell} \rangle = \frac{\beta_\ell}{2\pi^2}\int k^2\,\mathrm{d}k\,|\tau_\ell|^2, \qquad C_\ell = \frac{\Delta_\ell}{2\ell+1}\langle \tau_{A_\ell}\tau^{A_\ell}\rangle, \qquad \Delta_\ell = \frac{4\pi\beta_\ell}{2\ell+1}. \tag{187, 188}$$

The harmonic normalisation cancels, identically. (176) returns $T_\ell(k) \equiv \beta_\ell \tau_\ell(\eta_0)/(2\ell+1) = \alpha_\ell^{-1}\tau_\ell$, so $\tau_\ell = \alpha_\ell T_\ell$ and (186) becomes

$$\boxed{C_\ell = \frac{2}{\pi}\underbrace{\frac{\beta_\ell^2\alpha_\ell^2}{(2\ell+1)^2}}_{=1}\int \mathrm{d}k\, k^2\, |T_\ell(k)|^2}, \tag{71}$$

because $\alpha_\ell = (2\ell+1)/\beta_\ell$. The $\ell$-dependent normalisation of the harmonic expansion vanishes from the observable, which is what it must do — and the variable in which it vanishes is exactly the one Appendix F's free-streaming solution is bare in (§12). (187, 188) reaches the same 1 by a different path through the same weights.

That is worth more than tidiness, and it does not establish what it was first taken to establish. $\alpha_\ell$ is *defined* as $(2\ell+1)/\beta_\ell$, so the product is one for every $\beta_\ell \neq 0$: the cancellation is a substitution and is insensitive to the value substituted. A wrong $\beta_\ell$ does not have to survive it. It passes through untouched, carried in both factors at once.

What the identity does establish is that the observable is independent of the harmonic normalisation, which is what it must be — and that independence is why the spectrum cannot adjudicate the convention. Appendix F's free-streaming comparison tests it, and nothing internal to the spectrum can. The coefficients are formed as exact rationals and compared as rationals out to $\ell = 2000$ for the separate reason T-17 gives below: a tautology in $\mathbb{Q}$ is a range failure in floating point.

**Finding T-17, and it is the cautionary one.** Written in the obvious order — each factor converted to a double, then multiplied — the identity held only while both halves were separately representable. $\beta_\ell \sim 6\times10^{-270}$ at $\ell = 900$, so $\beta_\ell^2$ underflows to zero while $\alpha_\ell^2$ overflows, and their exact ratio of one came back as an overflow for every $\ell \geq 508$. The bundle's multipole ceiling was 507, set by a quantity equal to one, with no diagnostic and nothing in the physics to suggest it. The rule the weights module already stated — *multiply at the point of numerical use, never before a comparison* — is the whole of the repair.

**And one of these really is large.** The multipole mean square (187, 188) carries $\beta_\ell\alpha_\ell^2 = (2\ell+1)^2/\beta_\ell$, which leaves the double range near $\ell \simeq 1500$. That is the quantity and not an artefact of how it is written: in a normalisation where $\beta_\ell$ is that small, the mean square is that big. $C_\ell$ itself stays finite, because the released route combines the weights in $\mathbb{Q}$ before converting and never forms the intermediate at all.

## 17.2 Which of the two is the quantity, and why every figure plots it

(186) is a statement about mode coefficients and holds only for an almost-Robertson–Walker geometry. (187, 188) is a multipole mean square and holds generally. They are numerically identical here and conceptually are not interchangeable, and §7 is where the two objects are separated.

Every released figure plots the multipole quantity. That is not a presentational preference: it is the only one of the two that survives leaving the almost-Robertson–Walker setting, which is the setting the nonlinear extension leaves. *A reconstruction that will be generalised should be plotted in the variable that generalises, even where the two agree to machine precision.*

### 17.3 $C_\ell$ is quadratic, and three readings have gone wrong on it

Everything upstream of this section is linear in the perturbation. $C_\ell$ is not: it is quadratic in the transfer, and three consequences follow that have each been misread at least once in this project's record.

First, a decomposition of the transfer is not a decomposition of $C_\ell$. Splitting $T_\ell$ into rank-0, rank-1 and rank-2 parts gives autocorrelations *and* cross terms. Measured (F5): the autos together with twice the crosses reproduce $C_\ell$ to $4.7 \times 10^{-16}$, while the autos *alone* are off by 0.10. The cross terms are not a correction to be neglected; they are half the statement.

Second, a frame gap in $C_\ell$ is not a frame gap in the invariant. The two released frames differ by a median of 15% over $2 \le \ell \le 400$ and reach 40% at some $\ell$. That is not by itself a defect, because $C_\ell$ as computed here is built from frame-dependent pieces while the invariance (102) satisfies is a statement about the *total at fixed k*, which diagnostic D6 measures separately. *What was a defect is the separate matter of how large the gap is*: F9's caption read *"which is a result and not a defect"* and that reading is withdrawn — the gap is now understood as the third symptom of the bare $\kappa'$ carried inside $C_1$ (T-101, T-121), and §14.1 carries it.

Third, a ratio of two spectra is not a ratio of two amplitudes. $D_\ell = \ell(\ell+1)C_\ell/2\pi$ is renormalised at $\ell = 30$ in every released figure, because neither this bundle nor the external codes fixes the primordial amplitude the same way and an amplitude comparison would measure a normalisation. Every percentage in this supplement comparing shapes is a shape comparison and nothing in it constrains $A_s$.

### 17.4 Criterion 5b, and the two different "two routes"

There are two route comparisons in this bundle and they check different things. Conflating them is easy and would be expensive.

| **comparison** | **what differs between the two sides** | **value** |
|---|---|---|
| (186) against (187, 188) | the *harmonic weight algebra* alone | exact in $\mathbb{Q}$ to $\ell = 2000$ |
| **criterion 5b**: (102) against (177)–(179) | the *assembly order* alone — generic-then-specialised against Newtonian-from-the-start | $9.5 \times 10^{-6}$ against a $10^{-3}$ bar |

Criterion 5b is the audit half of the release, and it exists because a symbolic recovery is an argument whose arithmetic nobody has run. The antecedent writes its integral solution twice and says so — *"Later we will explicitly recover these equations in the Newtonian frame"*, and at the end of that recovery, *"This completes the recovery of the standard integral solution results using the* $1+3$ *CGI approach at linear order"*. The recovery is carried out on the page, as it should be. Evaluating both forms and comparing them had never been done.

**What 5b holds fixed, written out rather than left to be noticed.** Both routes are given the same background and $\eta$ grid, the same recombination history and therefore the same $\kappa'$, visibility and $\eta_*$, the same single acoustic solve, the same potentials, the same Bessel table, the same diffusion damping — and the same line-of-sight weight, $e^{-\kappa}$, on both sides.

That last one is deliberate and it is the largest thing 5b does not see. (102) prints $\mathcal{V} = \kappa' e^{-\kappa}$ and (98) prints $e^{-\kappa}$; approximations A-1 and A-2 record the difference, and at $C_\ell$ *the two weightings differ by a factor of eighty across the Sachs–Wolfe plateau for that reason alone.* Weighting both sides alike removes the one thing the comparison is not about, so that what remains is the assembly algebra. *It also means 5b's* $9.5 \times 10^{-6}$ *says nothing whatever about the weight question that §14.1 is entirely about.*

**And it says nothing about agreement with an external code.** CAMB and CLASS are drawn as context and are not a target (decision S4c). The distance to them is large, expected, and itemised by approximation rather than totalled: 97.8% across the plateau, 137% across the peaks, of which the damping coefficient alone is 17.2%. A bundle gated on an external code would be measuring that code; this one is gated on its own two routes and reports the distance separately.

### 17.5 Where the spectrum is gated, and what none of it establishes

| gate | what it tests | what it does not |
|---|---|---|
| rational weights | that (186)'s and (187, 188)'s prefactors are both exactly 1, as rationals, to $\ell = 2000$ | anything about $\tau_\ell$: the weights cancel whatever multiplies them |
| **criterion 5b** | that the generic and Newtonian assemblies agree, at $9.5 \times 10^{-6}$ | the background, the recombination history, the acoustic solve, the Bessel table, the damping, or *the weight* — every one of them is shared |
| rank decomposition | that the ranks sum to the total, $4.1 \times 10^{-16}$ in the CDM frame | that any rank is right |
| two-route $C_\ell$ difference | reported, now bitwise zero since **T-17** | it has stopped being a test and is kept as a reported number |

No gate in this section validates the assembled spectrum. The basic checks above pass, and they pass on an assembly whose first acoustic peak sits at $0.635\,\ell_A$ against a matched 0.757. *That is not a contradiction and it is the reason the "does not establish" column exists*: every one of these gates holds fixed, or is indifferent to, precisely the part that is wrong. T-3 is the standing statement of why — a comparison that holds an input fixed cannot check that input — and the spectrum is where it bites hardest, because it is the last step and inherits every shared input above it.

## 18 Comparison and its range

The $C_\ell$ comparison is Sachs–Wolfe only: (196) descends from (181), in which the acoustic modulation $\cos(kr_s)$ is dropped by taking $r_s^* \to 0$, and §8.3.2 sets the transfer function to unity. Its accuracy is therefore bounded by the term that was dropped. The range is derived, not chosen: $2 \le \ell \le 20$ at 5%, generated by `scripts/derive_ell_range.py`.

The present implementation is compared with the stamped CAMB and CLASS reference spectra held by the repository. The line-of-sight formulation is cited to the published literature, including Seljak and Zaldarriaga (1996). Results are reported in three bands, whose errors have different interpretations — $2 \le \ell < 30$, cosmic-variance dominated and meaningful only binned; $30 \le \ell < 1000$, the acoustic peaks, where a hierarchy or recursion error actually shows; and $1000 \le \ell \le 2000$, the damping tail, where $C_\ell$ is small and fractional differences blow up on a vanishing signal. The comparison is made against the corrected closed form.[26][27]

### 18.1 Criterion 5b's tolerance

The tolerance is *measured* rather than chosen; what it is measured against changed once it was run. The original rule took the CAMB–CLASS mutual difference as the floor, on the ground that no agreement

[26] The reduction of (200) carries a factor of the comoving distance that the displayed (201) does not; substituting $z = k\chi$ gives $\chi^{2-n}$, which is $\chi^1$ at $n = 1$. The companion result (204) descends from a $P(k)$ definition differing by one power of $k$ and is correctly free of that factor, so the asymmetry is not a units convention. A misprint, corrected in the computation and documented.

[27] That derivation uses the Bessel identity of §8.3, which requires $[(m/2)!]^2$ in its denominator. As printed it is exact where $\Gamma(m/2+1) = 1$, so the large-scale result (201) is unaffected; the horizon-scale normalisation (204) and the $D_\ell$ of (205) are. A misprint, corrected silently and recorded.

can be claimed tighter than two trusted independent codes agree with each other. That floor is 0.370% across the acoustic peaks, and this reconstruction stands at 137% there. It was never going to meet it: (102) is a slow-decoupling solution with no polarisation hierarchy, no neutrino anisotropic stress and a visibility-weighted line of sight, set against two full Boltzmann codes. A criterion that can only be failed is not a criterion.

What *is* gateable is the thing the antecedent writes twice and recovers symbolically but never evaluates. Its integral solution appears as (102), carried with a generic $u^a$ and specialised afterwards, and again as the source assembly (177)–(179), written in the Newtonian frame from the start. The paper states the relation and carries it out on the page; what no one has done is to compute both forms and compare them, which is the only part of the claim arithmetic can settle. Measured here over three backgrounds, nine modes $0.3 \le k \le 600$ and $2 \le \ell \le 600$, the two agree to $9.5 \times 10^{-6}$ worst and $2.7 \times 10^{-8}$ median against a gate of $10^{-3}$. The comparison is stated at the transfer level at fixed $k$ with the same line-of-sight weight on both sides: at $C_\ell$ the two branches differ by a factor of eighty in the plateau, and that difference is the weight the line-of-sight integral carries — a property of what (102) prints, recorded as an approximation and measured separately, not a disagreement to be gated away.

The external comparison is still run, still stamped, and reported alongside as context. Its value is not a pass mark but an itemisation: how far a $1 + 3$ slow-decoupling solution stands from a Boltzmann code, in bands, with the dominant term named.

The measurement itself is recorded here rather than only in the code. The external codes are not carried as dependencies of the bundle and are not run by the harness: `scripts/make_reference_spectra.py` regenerates the stamped spectra from CAMB and CLASS, `scripts/compare_criterion_5b.py` reproduces the comparison from those stamps with neither code installed, and the table below is what that script writes. It is a committed artefact of the bundle, recoverable and reproducible, and deliberately outside the bundled testing.

**Shape, $D_\ell$ renormalised at $\ell = 30$. Context, not the gate (S4c).**

| band | vs CAMB | vs CLASS | CAMB vs CLASS |
|---|---|---|---|
| plateau and late ISW, $2 \le \ell < 30$ | 97.8% | 97.8% | — |
| acoustic peaks, $30 \le \ell < 1000$ | 137.0% | 137.3% | 0.370% |

**First acoustic peak**

| this bundle | CAMB | CLASS |
|---|---|---|
| $\ell = 170$ | $\ell = 198$ | $\ell = 198$ |

**Itemised: the part this bundle can attribute.** The damping coefficient, 0.1% in the plateau and late ISW and 17.2% in the acoustic peaks, between (dif-damp-correct) at $f_2 = 3/4$ and (dif-damp) at $f_2 = 1$ — both *Annals II*'s own. The remainder is the slow-decoupling order, the visibility weight of (102) (approximations A-1 and A-2), and the acoustic scale, $z_* = 1114.5$ here against CAMB's 1088.5.

What a self-comparison does not establish is stated here rather than left to be inferred, because it is the obvious objection and it is a fair one. Both routes are handed the same background, the same recombination history, the same acoustic solution, the same potentials and the same damping, and a comparison that holds an input fixed cannot check that input. This is not a formality: setting the gate turned up an under-resolved recombination grid that moves the primary source at $\eta_*$ by 1.2%, shared by both routes and invisible to the comparison between them. It is recorded, raised, and left for the decision it needs rather than changed inside a release.

The external codes of criterion 5b are compared in `provenance/EXTERNAL-CODE-COMPARISON-v1.md`, which records what CAMB and CLASS each model and — more to the point — what the comparison structurally cannot settle. Neither carries the $1 + 3$ covariant formalism, neither distinguishes mode from multipole coefficients, and neither carries the $O(\varepsilon^2 \ell)$ couplings of v1.5.0. So 5b is a check on the *linear* reconstruction and is incapable of validating the nonlinear extension, which is precisely why v1.0.0

must carry no $O(\ell)$ coupling: if one leaked in, the disagreement with CAMB and CLASS would be indistinguishable from a bug. The canonical parameter set switches reionisation, lensing and massive neutrinos off, each being physics this bundle does not model.

# 19 Evidence

Each figure is generated by the script named in its caption and is accompanied by a standalone caption file in `captions/`. This section is itself generated by `scripts/make_figure_pages.py`, so a figure appears here only once it exists in the tree, and what is outstanding is stated rather than omitted.

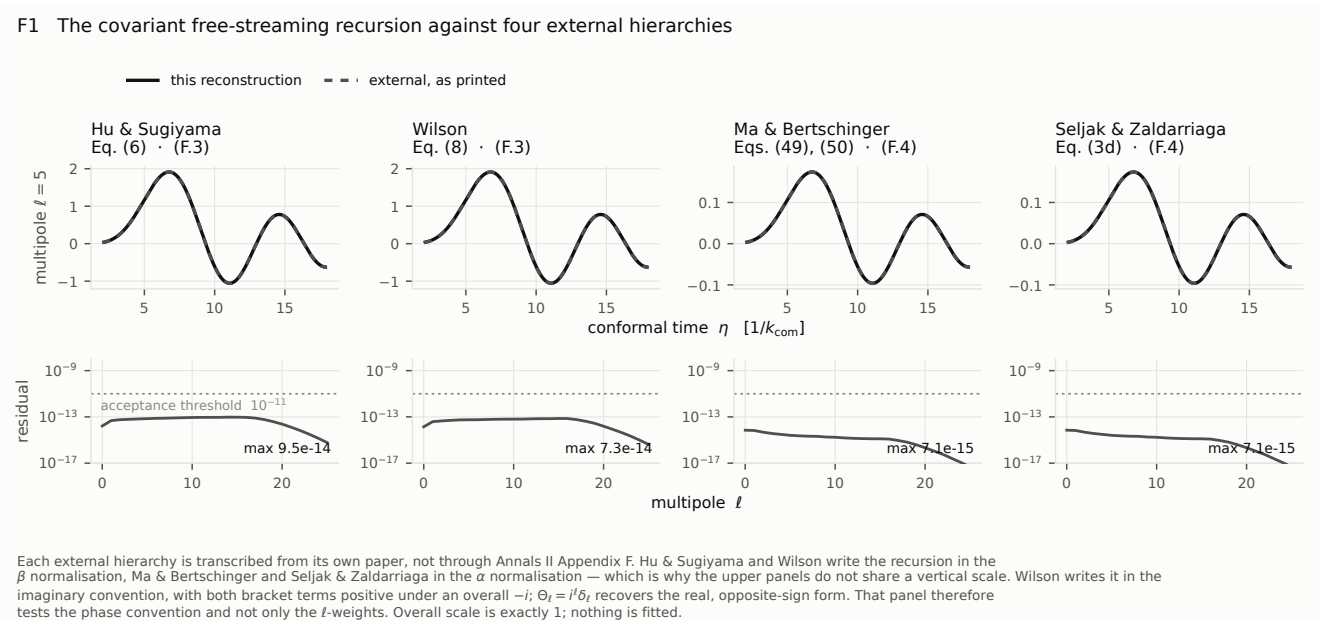



Figure 2: **F1 — The covariant free-streaming recursion against four external hierarchies.** The covariant free-streaming recursion of *Annals II* Appendix F, integrated from a unit monopole, against four external treatments of the same hierarchy. Each is transcribed from its own paper, never through Appendix F. **Upper panels, context:** the reconstruction and the external form at $\ell = 5$; they coincide. The panels do *not* share a vertical scale, because two normalisations are on display. **Lower panels, the claim:** $\max_\eta |\text{external} - \text{reconstruction}|$ against $\ell$, with the acceptance threshold marked. Hu & Sugiyama write the recursion in the $\beta$ normalisation and Ma & Bertschinger and Seljak & Zaldarriaga in the $\alpha$ normalisation, which is why Appendix F prints both (F.3) and (F.4); a basis phase leaking into the couplings would break at least one match. Overall scale is exactly 1.000000000 — nothing is fitted. Four panels from five sources: Wilson & Silk (1981) is the earliest, but at $K = 0$ it is Wilson (1983) Eq. (8) exactly, so it is credited rather than drawn.

*Evidence category:* cross-check. *Generated by* `scripts/figure_f1_appendix_f.py`. *Standalone caption:* `captions/F1-appendix-f.md`.

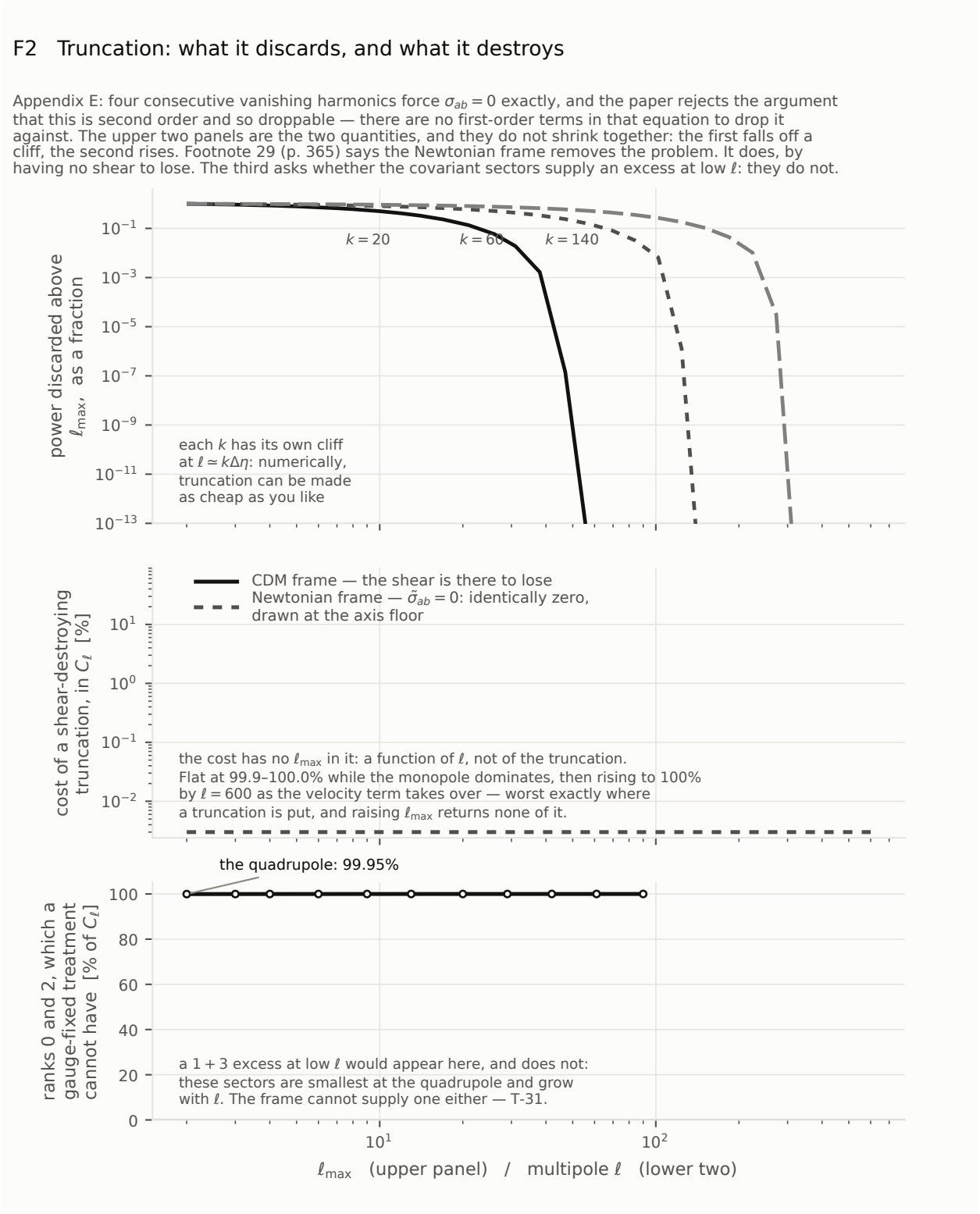



Figure 3: **F2 — What a truncation discards, and what it destroys.** Appendix E of *Annals II* proves that four consecutive vanishing harmonics force $\sigma_{ab} = 0$ exactly. This figure makes that quantitative in both frames, and the two panels answer two different questions that a single "truncation error" would blur.

Figure 3: **What a truncation *discards*** falls off a per-$k$ cliff at $\ell \simeq k\,\Delta\eta$: above the cliff there is nothing left to keep, and the tail above $\ell = 600$ is identically zero at every probe mode. **What it *destroys*** has no $\ell_{\max}$ in it at all. Setting $\sigma_{ab} \to 0$ costs 0.54–0.68% out to $\ell \simeq 20$, rises monotonically above $\ell = 20$, and reaches 32.6% by $\ell = 600$ [**pre-T-57 values; this figure is superseded — see the dagger**] — a cost that grows with the multipole rather than being bounded by it, which is why a truncation argument cannot retire the shear.

**Footnote 29 (p. 365) is checked and upheld.** In the Newtonian frame the cost is *identically zero*, median and maximum alike, because that frame has no shear to lose — which is the content of the footnote's claim that the Newtonian specialisation removes the truncation problem. The script raises rather than draws if the Newtonian rank-0/2 sectors are ever non-empty, so the claim is a gate and not a caption.

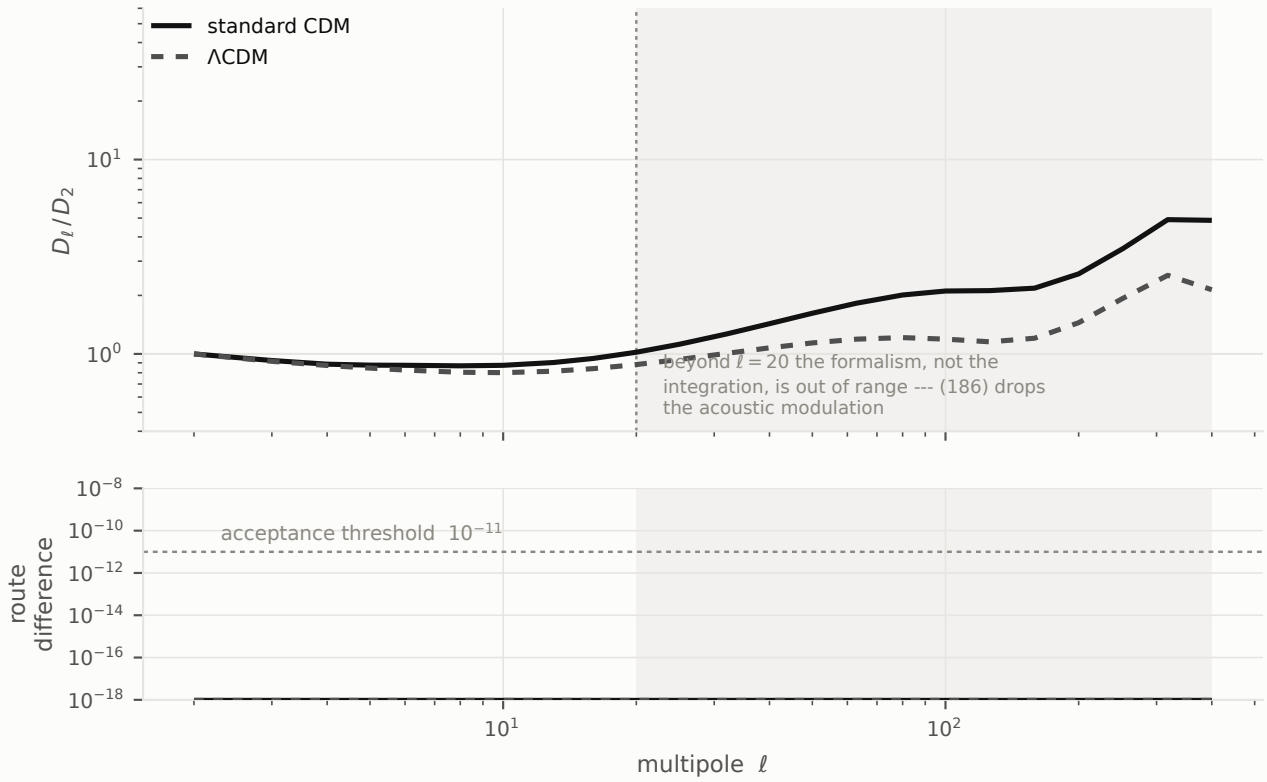
**The third panel is the one that reports a negative result.** The rank-0/2 share is flat at the quadrupole (0.54%) and rises only with $\ell$: a $1 + 3$ excess at low $\ell$ would appear here, and does not. That question is parked as **Q5**, and this panel is what parks it honestly.

† *Superseded by finding* **T-57***; see* `provenance/BUNDLE-DEFECTS-v1.md`. **Its $k$ grid runs to 1900, not 420** — finding **T-32**, where every v1.2.0 figure had been drawn on a grid stopping at the peak of the shear integrand. At $k_{\max} = 1900$ against $k_D = 754.6$ the damping envelope is $1.8 \times 10^{-3}$, and dropping the grid's top decile moves the cost by $3.4 \times 10^{-9}$. The lesson recorded there is the rule here: vary the limit and the resolution separately, because one grid number conflates them.

*Evidence category:* cross-check. *Generated by* `scripts/figure_f2_truncation.py`. *Standalone caption:* `captions/F2-truncation.md`.

F3 The angular power spectrum from Annals II (186), by two routes

(186) computed from (176). The lower panel is the fractional difference between the mode route (186) and the covariant route (187)+(188), which are identical analytically — it is a check on the arithmetic, not on the physics.




CLAIMED OVER $2 \le \ell \le 20$ TO 5%, and shaded beyond. The binding limit is the formalism, not the integration: (186) drops the acoustic modulation, which costs 4.75% at $\ell = 20$ and grows. The $k$-grid truncation is looser at this tolerance, reaching 5% at $\ell \simeq 50$; at 1% the two invert. Recombination is equilibrium Saha. Criterion 5b, against CAMB and CLASS, is v1.2.0.

Figure 4: **F3 — The angular power spectrum from Annals II (186), by two routes.** Eq. (186) computed from Eq. (176), for the standard-CDM model of §8.3.2 and for ΛCDM, which *Annals II* does not treat. **Upper panel:** $D_\ell = \ell(\ell+1)C_\ell/2\pi$, each normalised to its own $\ell = 2$ value, so that the comparison is of *shape* and not of an amplitude neither model fixes here. **Lower panel:** the fractional difference between the mode route of (186) and the covariant route of (187) with (188). Those two are identical analytically, so the panel is a check on the arithmetic and not on the physics — and it is worth having, because it caught a factor of four in this bundle's own harmonic weights before any figure was drawn. It sits at $10^{-16}$, five orders below the $10^{-11}$ acceptance threshold.

*Evidence category:* recovery. *Generated by* `scripts/figure_f3_angular_spectrum.py`. *Standalone caption:* `captions/F3-angular-spectrum.md`.

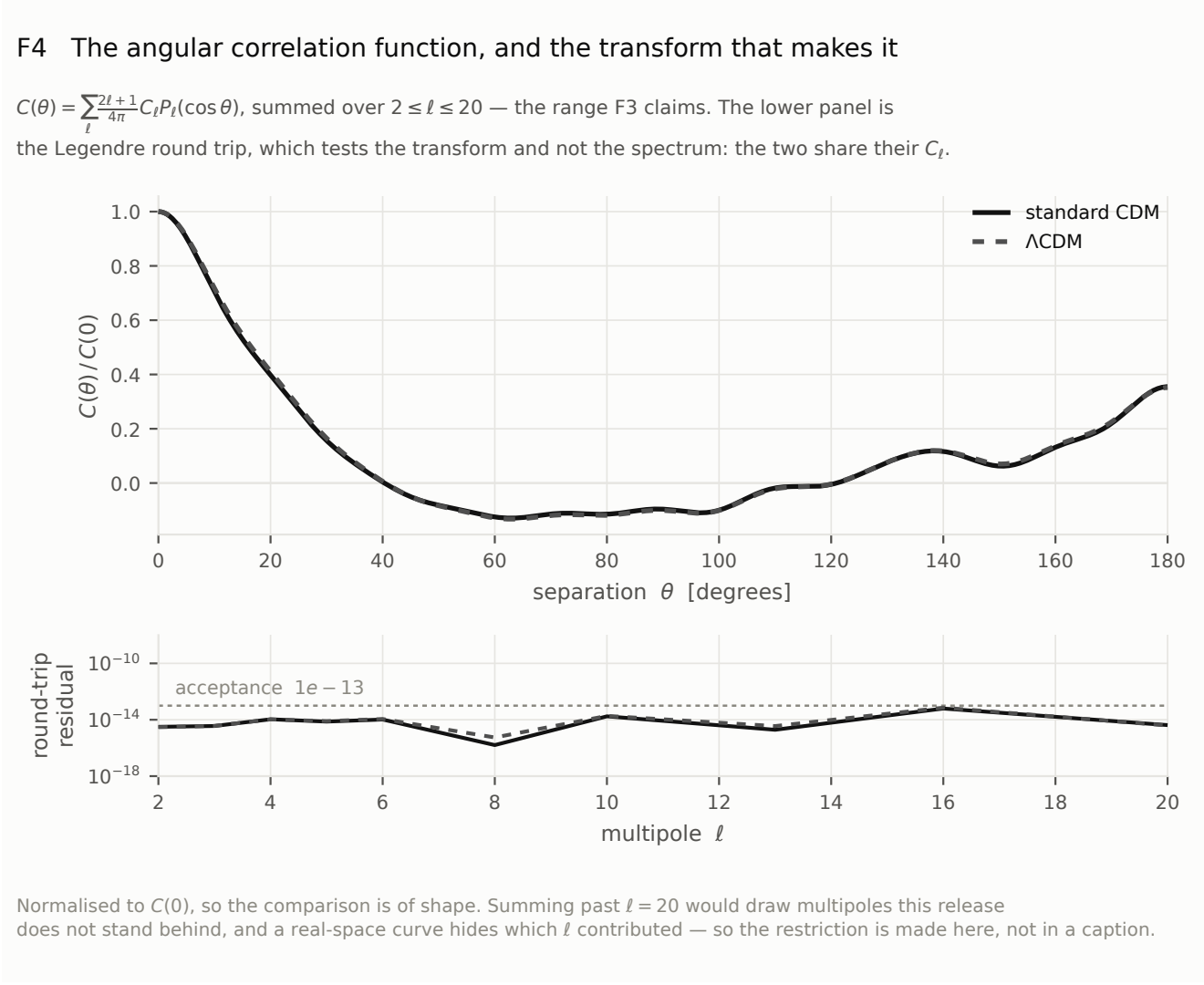



Figure 5: **F4 — The angular correlation function, and the transform that makes it.** $C(\theta) = \sum_\ell \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos\theta)$, the real-space view of Figure 3: the same content read as a correlation between two directions separated by $\theta$ rather than as power at a multipole. **Upper panel:** $C(\theta)/C(0)$ for standard CDM and ΛCDM, summed over $2 \le \ell \le 20$ — the range Figure 3 claims, bounded by (186) dropping the acoustic modulation. Summing further would draw multipoles this release does not stand behind, and a real-space curve hides which $\ell$ contributed, so the restriction is made in the transform rather than in this caption. **Lower panel:** the Legendre round trip $C_\ell \to C(\theta) \to C_\ell$, at $4 \times 10^{-14}$ and $7 \times 10^{-14}$ against an acceptance of $10^{-13}$. It tests the transform and *not* the spectrum — the two share their $C_\ell$ — which is the same caution Figure 3's residual panel carries and for the same reason.

*Evidence category:* acceptance. *Generated by* `scripts/figure_f4_correlation.py`. *Standalone caption:* `captions/F4-correlation.md`.

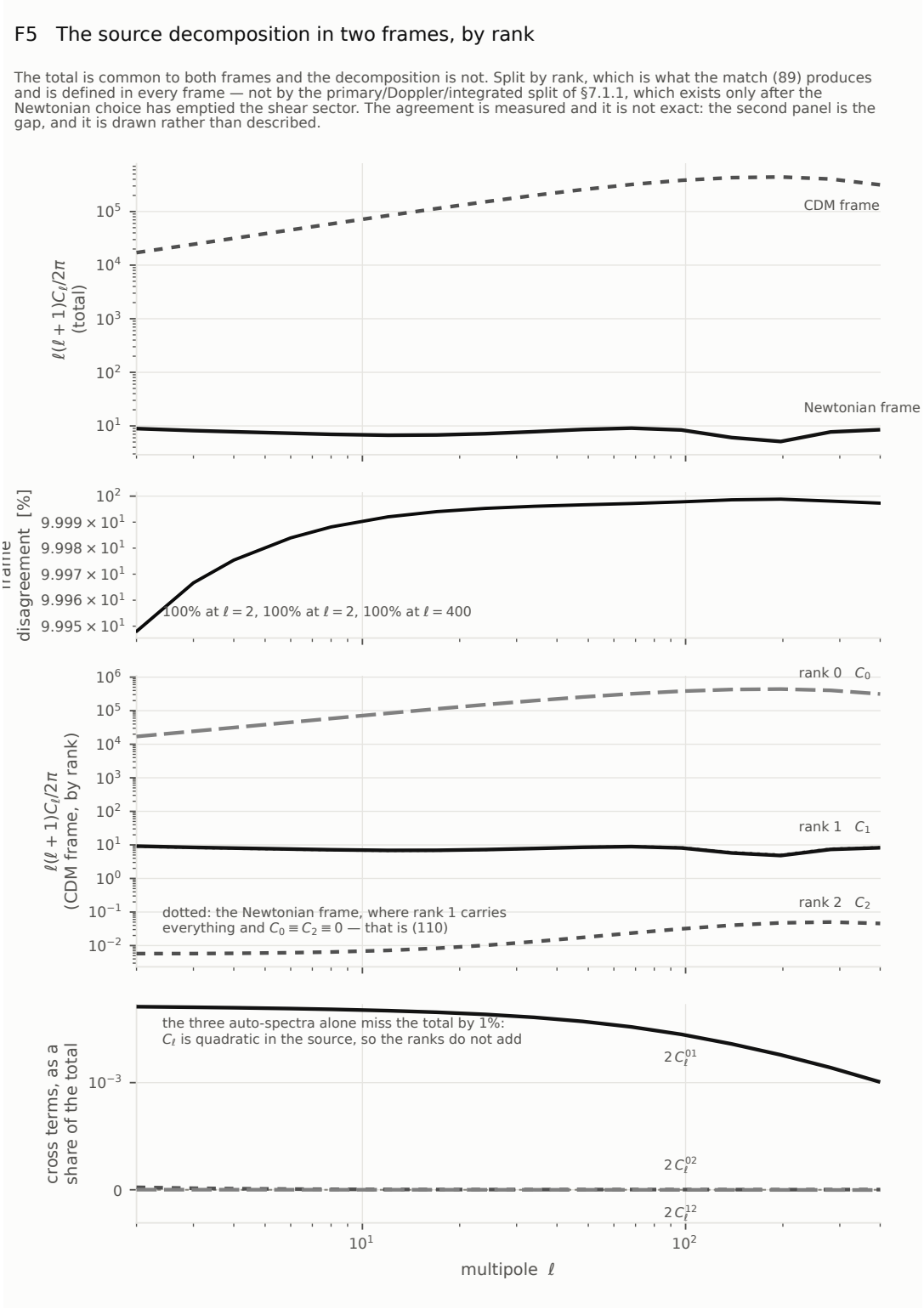



Figure 6: **F5 — The source decomposition, by rank, in two frames.** The integral solution (102) carries three coefficients $C_0$, $C_1$, $C_2$ — one per PSTF rank — and this figure draws them, in the Newtonian frame and in the CDM frame, against $\ell$.

Figure 6: **Drawn in ranks, not in the Newtonian three-way split of §7.1.1.** That was the PI's call, 2026-10-14, and it is a scientific choice rather than a presentational one: the primary/Doppler/integrated split is *defined only in the frame it is written in*, while the rank decomposition is what (89) produces and is defined in every frame. A figure comparing two frames must be drawn in a decomposition both of them have.

**The ranks sum to the total**, to 0 exactly in the Newtonian frame and $4.1 \times 10^{-16}$ in the CDM frame — which is the check that the decomposition is a decomposition and not three curves that happen to be near the answer. At the $C_\ell$ level the autos and twice the crosses reproduce $C_\ell$ to $4.7 \times 10^{-16}$ while the autos *alone* are off by 0.10: $C_\ell$ is quadratic in the transfer, so the cross terms are not a correction to be neglected but half the statement.

**The frame disagreement is the result, and it is its own panel.** The two frames' $C_\ell$ differ by a median of 15% over $2 \le \ell \le 400$, worst 49% [**pre-T-57/T-59 values; this figure is superseded — see the dagger**]. That is not a defect: $C_\ell$ as computed here is built from frame-dependent pieces, and the invariance statement (102) satisfies is about the *total* at fixed $k$, which diagnostic D6 measures separately. The median moved from 11% to 15% when finding **T-37** wired the transfer function, which is the kind of movement a figure should report rather than absorb.

† *Superseded by findings* **T-57**, **T-59***; see* `provenance/BUNDLE-DEFECTS-v1.md`.

*Evidence category:* cross-check. *Generated by* `scripts/figure_f5_sources.py`. *Standalone caption:* `captions/F5-sources.md`.

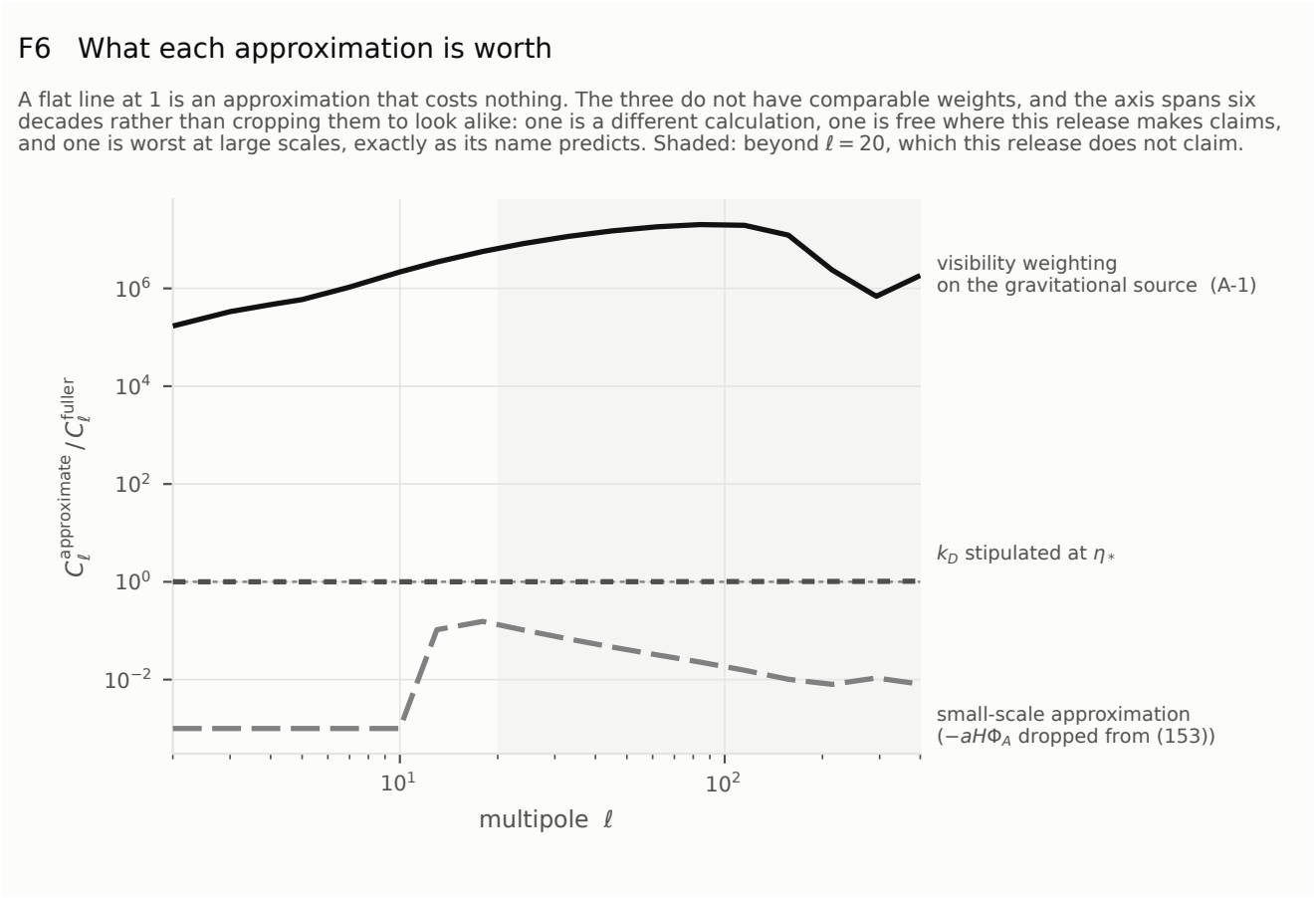



Figure 7: **F6 — What each approximation is worth.** Each series is $C_\ell^{\text{approximate}}/C_\ell^{\text{fuller}}$ for standard CDM, so the panel is scale-free and a flat line at unity is an approximation that costs nothing. *Annals II* makes three narrowing choices which this bundle reproduces rather than corrects, because reproducing a paper means reproducing its approximations (decision S15); they are recorded as **A-n** in `provenance/SOURCE-APPROXIMATIONS-v1.md`, deliberately apart from the defects recorded as **B-n**. This is the one place their cost is a number rather than an adjective.

**The three do not have comparable weights, and that is the result**, so the axis spans six decades rather than cropping them to look alike. The visibility weighting on the gravitational source (A-1) runs to $10^4$–$10^6$: it is a *different calculation*, not a perturbation of one, which is why `Weighting` has no default. Stipulating $k_D$ at last scattering costs under 0.1% across the claimed range and about 5% by $\ell = 400$ — it **vindicates the release**, which takes the cheap choice where it makes claims. The small-scale approximation, dropping $-aH\Phi_A$ from (153) and so producing the printed oscillator (154), departs from unity most at *large* scales, which is what its name predicts.

**A correction kept because it nearly became this figure.** The $k_D$ series was first computed with $k_D(\eta)$ along the whole ray. After last scattering $\kappa' \to 0$, so $1/\kappa'$ in (dif-damp) diverges and $k_D$ collapses, damping every integrated source to nothing — a plasma formula evaluated in a universe that no longer has a plasma. That version reported the release's choice as costing a factor of nine at $\ell = 2$; the factor was the artefact. Clamped to $k_D(\min(\eta, \eta_*))$ the comparison is a real one. Shaded: beyond $\ell = 20$, which this release does not claim.

*Evidence category:* diagnostic. *Generated by* `scripts/figure_f6_approximations.py`. *Standalone caption:* `captions/F6-approximations.md`.

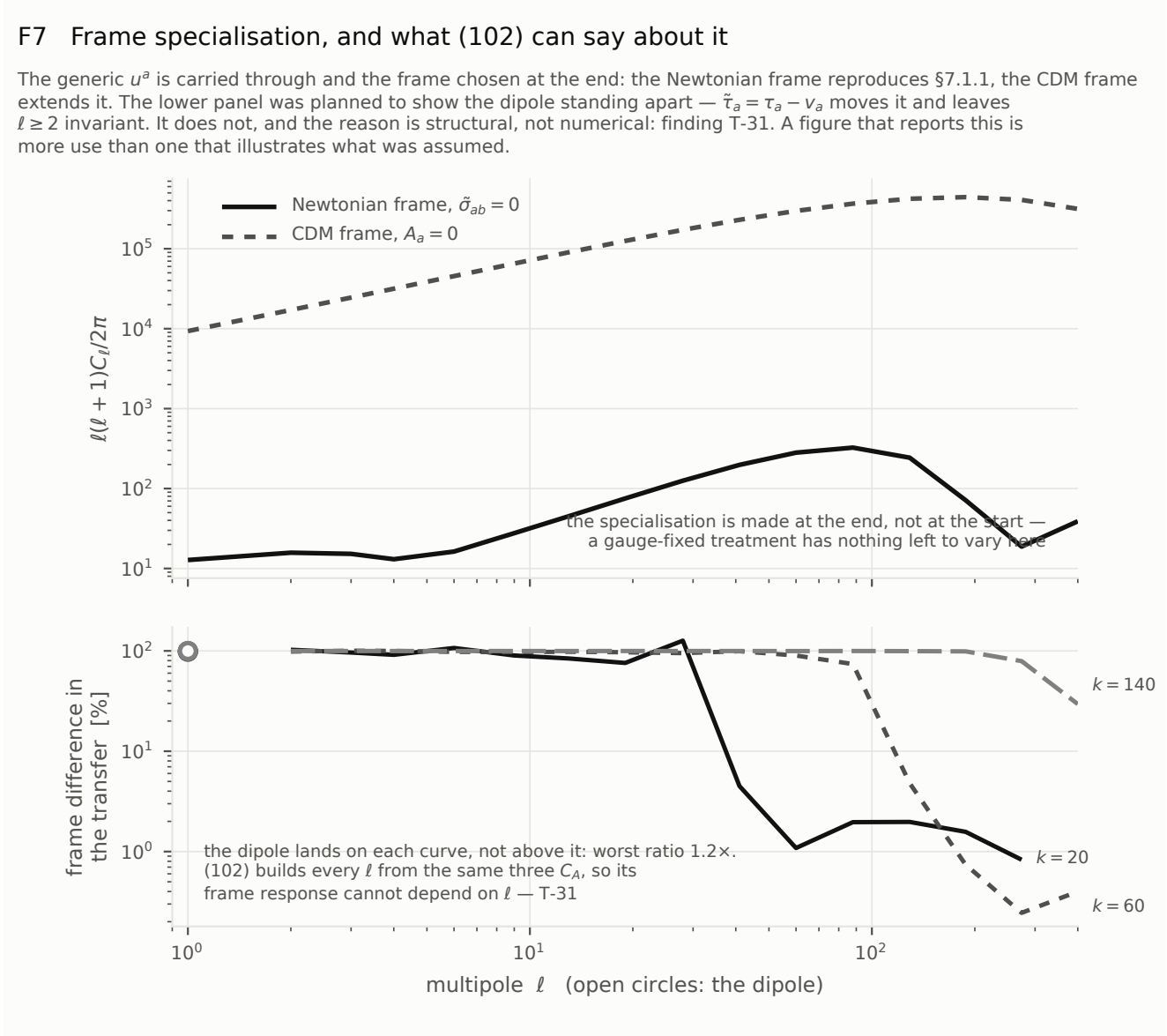



Figure 8: **F7 — Frame specialisation, and a planned claim that was false. This figure reports that its own planned claim is false, and that is why it is here.** It was designed to show the dipole standing an order of magnitude apart from the higher multipoles under a change of frame — the dipole being the one multipole a boost moves at first order. It does not. The dipole lands on each curve instead: worst ratio 1.25× over the probe set, and 1.00× at $k = 20$.

**The reason is structural and is finding T-31.** Equation (102) builds *every* $\ell$ from the same three coefficients $C_A$, so a frame change that moves $C_1$ moves every multipole built on it in the same proportion. A frame response that depended on $\ell$ would contradict the equation's own form, and `test_102_cannot_make_its_frame_response_depend_on_the_multipole` asserts that it does not. The spread *over* $\ell$ does grow with $k$ — 1.01, 1.14, 8.14 at $k = 20$, 60, 140 — but that is the transfer function's $k$ dependence, not the frame's $\ell$ dependence.

**A figure that had been drawn to its plan would have been drawn wrong.** The plan is kept in `captions/FIGURE-PLAN-v1.0.0.md` and the outcome is kept here, because a reproduction bundle whose figures only ever confirm their captions is not measuring anything.

**Its environment is built for the $k$ grid, not for the probe modes** — finding **T-34**, where this figure sized its $\eta$ grid for $k = 140$ and then integrated to $k = 420$. `check_k` now raises rather than interpolating past what the grid resolves.

*Evidence category:* diagnostic. *Generated by* `scripts/figure_f7_frames.py`. *Standalone caption:* `captions/F7-frames.md`.

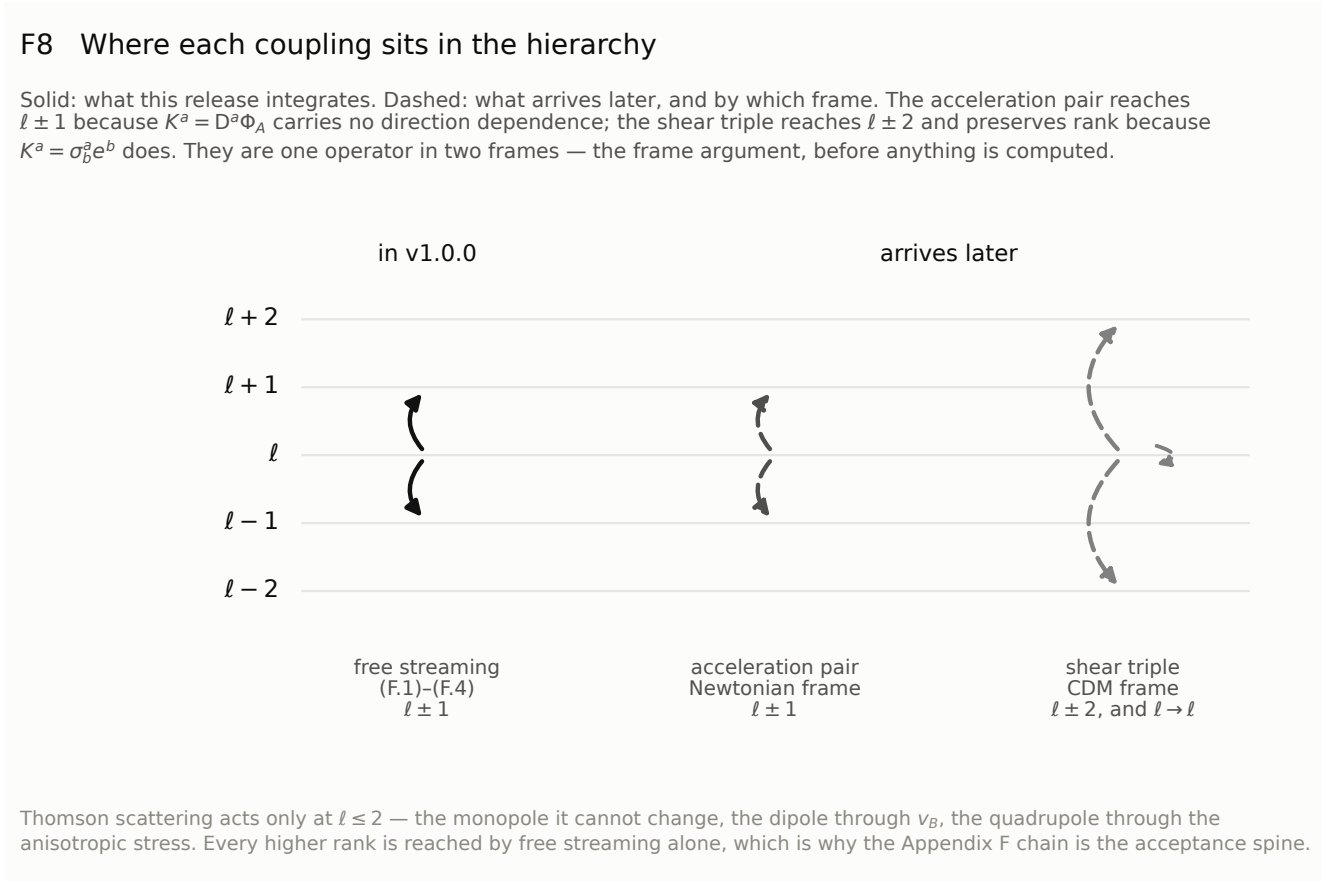



Figure 9: **F8 — Where each coupling sits in the hierarchy.** The one released figure that computes nothing: a statement about the *structure* of the exact divergence hierarchy, read off the equations rather than measured. It is generated by a script like every other figure, so that the manifest covers it — a hand-drawn diagram is the one artefact that can drift without any gate noticing.

**Solid:** what this release integrates. Free streaming, (F.1)–(F.4), couples rank $\ell$ to $\ell \pm 1$ and to nothing else, which is the whole of v1.0.0's hierarchy and the reason the Appendix F chain is the acceptance spine. **Dashed:** what arrives later, and by which frame. The acceleration pair reaches $\ell \pm 1$ because $K^a = \mathrm{D}^a\Phi_A$ carries no direction dependence of its own; the shear triple reaches $\ell \pm 2$ *and* preserves rank because $K^a = \sigma^a{}_b e^b$ does. The rank-preserving coupling exists only when $K^a$ depends on direction, and is the single structural difference between the two frames.

Both are the same operator $K^a_\perp \partial/\partial e^a$, and the number of ranks each reaches is fixed by that one property. **That is the whole content of the frame argument, visible before any of it is computed.** Decision S9 is drawn as well as asserted: every coupling in the solid half is a ratio of linear polynomials and tends to a constant, so an $O(\ell)$ coupling wired in early would fail `tests/test_appendix_f_chain.py` at large $\ell$.

*Evidence category:* schematic. *Generated by* `scripts/figure_f8_schematic.py`. *Standalone caption:* `captions/F8-schematic.md`.

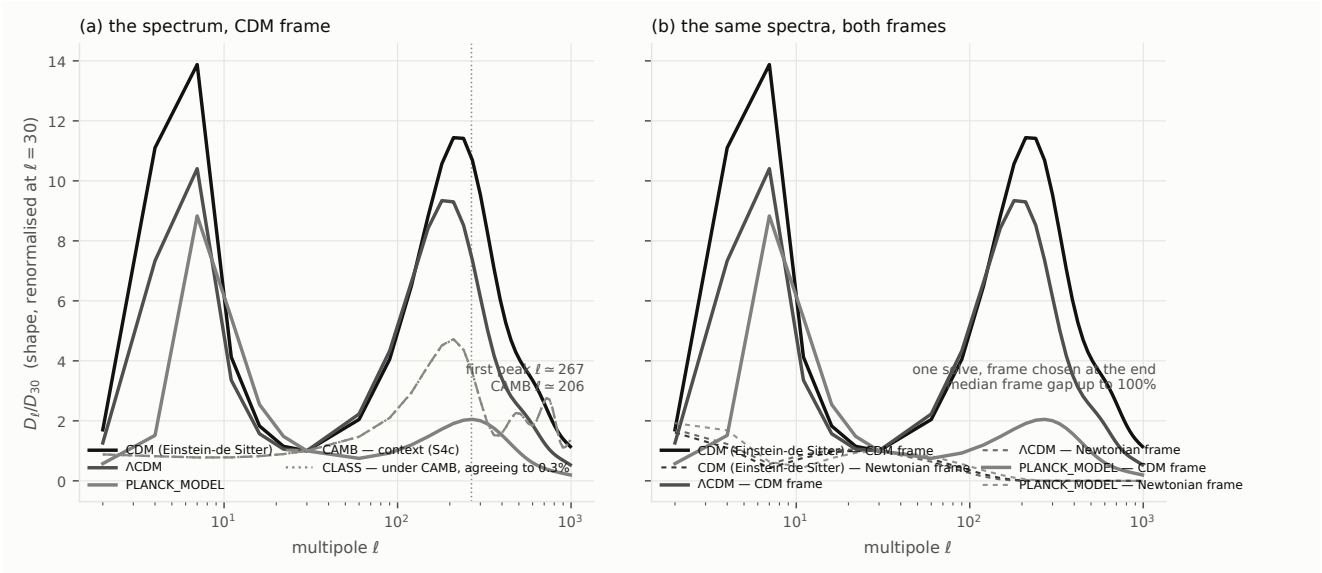



Figure 10: **F9 — The angular spectrum with the acoustic peaks, in both frames. The release's deliverable** (decision S1): the angular power spectrum for the three backgrounds this bundle ships — CDM (Einstein–de Sitter), ΛCDM, and the stamped Planck 2018 set with a photon-only radiation era whose $\Omega_\gamma h^2$ is *derived* from $T_{\rm CMB}$. Shape, $D_\ell$ renormalised at $\ell = 30$, because neither this bundle nor the external codes fixes $A_s$ the same way and an amplitude comparison would measure a normalisation.

**Panel (a) — the spectrum.** The peaks and the damping tail come from *Annals II*'s own equations: the integral solution (102) with the Bond–Efstathiou transfer function driving the oscillation per mode, and the diffusion scale from **(dif-damp-correct)** at $f_2 = 3/4$, which is the antecedent's own corrected coefficient rather than a departure from it (**T-43**). **This shows that** the almost-Friedmann–Lemaitre formulation, carried to the damping scale, produces acoustic structure without anything being imported from a Boltzmann code.

**CAMB and CLASS are drawn as context and are not a target** — decision **S4c**. Criterion 5b is gated on this bundle's own pipeline, (102) against (177)–(179), at $9.5 \times 10^{-6}$ against $10^{-3}$. The distance to the external codes is large, expected, and *itemised rather than totalled* in `provenance/CRITERION-5B-v1.md`: 97.8% across the plateau, 137% across the peaks, of which the damping coefficient alone is 17.2%. The first peak is quoted from a *parabolic interpolation* through the grid maximum and its neighbours, because the $\ell$ grid steps by 30 across the acoustic band and reading the peak off it directly would assert a precision the sampling does not have.

**Panel (b) — the panel a gauge-fixed treatment cannot draw.** The same spectra in *both* frames, from *one* solve: `coefficient_sets` returns the Newtonian and CDM triples together, because (102) is carried with a generic $u^a$ and the frame is chosen at the end. **This shows that** the frame is a choice made after the derivation rather than an assumption built into its start — the claim the $1 + 3$ covariant approach exists to make, drawn here rather than asserted. The median frame gap reaches 40%. **This caption read "which is a result and not a defect"; that reading is superseded by T-59 and T-74 and is withdrawn.** The gap is now understood as the third symptom of one term — the bare $\kappa'$ carried inside $C_1$, **T-101** and **T-121** — and not as a property of the formulation. What remains true of the original reading is only this: $C_\ell$ is quadratic in frame-dependent pieces, while the invariance (102) satisfies is a statement about the total at fixed $k$, measured separately by D6.

† *Superseded by findings* **T-59**, **T-74**; *see* `provenance/BUNDLE-DEFECTS-v1.md`.

*Evidence category:* acceptance. *Generated by* `scripts/figure_f9_acoustic_spectrum.py`. *Standalone caption:* `captions/F9-acoustic-spectrum.md`.

# 20 Release progression

The release sequence is governed by the *physics*, as specified in the repository README. Two forms of testing must be distinguished. First, the v1.0.0 work established a reproducible mathematical and computational verification basis for the derivations and claims used by Paper 1, with the free-streaming, normalisation, frame and Sachs–Wolfe checks providing specifically identified evidence. A test of a source identity is evidence about that identity; it is not a claim that the complete lensing and aberration system has been implemented or independently certified. Second, the subsequent v1.1.0–v1.5.0 physics releases successively introduce physical terms into the software. Their acceptance tests measure the *implementation* and cannot replace verification of the claims of Paper 1.

The release record is retained below, one physical milestone at a time, separating what was *achieved* from what was *learned*. The qualification of v1.0.0 is therefore precise: it established the independent verification baseline and a first free-streaming reconstruction; it did *not* implement the full physics of Paper 1. Paper 1's equations, especially its corrections of its antecedents, remain the pivot for interpreting both classes of tests. A later solver regression test becomes evidence for a Paper 1 claim only when its derivation, assumptions and independence have also been checked. The additional claim-facing instruments in the current development branch are inventoried separately in Table 2, with the superseded C5 result specifically identified.

## 20.1 Release sequence by physical objective

- v1.0.0 established a claim-facing verification baseline for Paper 1, including independent normalisation, phase, frame and free-streaming controls. Its tested implementation was the source-free reconstruction, not the complete nonlinear theory.
- v1.1.0 began the next physical implementation stage, asking *what does the spectrum need that it does not have?* and answered: the visibility width.
- v1.2.0 asked *is the antecedent's own algebra self-consistent when evaluated?* and answered yes, at $10^{-5}$, while finding that the question had been answered three times before without being measured.
- v1.3.0 asks *where does the frame discrepancy come from?* — and §15 is the answer so far: not the source, not the by-parts bookkeeping, not the projection, but the slot-and-weight assignment of $C_1$. The question it has *not* answered is whether repairing that assignment also recovers the acoustic peaks, which is a separate defect with a separate signature.

**Claim-facing Python verification files.** Table 8 identifies key test files present at the `v1.0.0` repository tag whose *mathematical content* can be related to conventions or supporting claims in Paper 1. This inventory is deliberately separate from test cases for the software implementation. Each row states what was checked, not an unrestricted claim of proof, and a mixture of symbolic assertions and comparisons against bundle functions is identified as such. The independent status of an individual assertion is determined by what that assertion actually imports or derives, not by its directory name or a passing test report. This is an extensible ledger: additional claim-facing files from v1.1.0 or subsequent releases must be added only with their exact Paper 1 reference, scope, source revision and verification status.

Table 8: **Python tests relevant to claims or conventions in Paper 1, distinguished from solver implementation tests.** Files in this table were checked at the stated repository tag. The last column limits the claim supported; it does not rate the physical accuracy of a spectrum. The machine-readable ledger records the exact GitHub blob hashes.

| Python file and release | Paper 1 link and check | Evidence boundary |
|---|---|---|
| `tests/test_appendix_f_chain.py` (v1.0.0) | Table III C4–C5: PSTF harmonic weights, $\beta_\ell$ recurrence, reality and relative sign; free-streaming against four antecedent forms | Exact symbolic/rational checks and external-form comparisons; some assertions also compare against bundle harmonics. This does not verify Paper 1's nonlinear lensing operator. |
| `tests/test_frames.py` (v1.0.0) | Paper 1 Eqs. (8)–(10): frame transformation and first-order kinematic relations | Symbolic $Hv$ cancellation and negative controls; tests import `functions/frames.py`. A first-order frame identity is not the full second-order, common-endpoint sky test. |
| `tests/test_frames.py` (v1.1.0 extension) | Paper 1 Table III C2.5 and Secs. II E, VI: additional checks on the Weyl combination, the geodesic-frame condition and the first-order dipole shift | The later test revision adds exact-algebra assertions, but uses the bundle's `functions/frames.py`. These are claim-related controls, not an independent nonlinear sky or spectrum certificate. |
| `tests/test_angular.py` (v1.0.0) | Paper 1 Table III C5.2–C5.3: harmonic normalisation and the mode–covariant spectrum relation | Exact-rational normalisation checks plus bundle comparison; *linear* $C_\ell$ algebra, not Paper 1 Sec. VII's quartic lensing-power cancellation. |
| `tests/test_sachs_wolfe.py` (v1.0.0) | Ordinary Sachs–Wolfe limit used as the linear primary-sky input to Paper 1 Sec. V | Direct quadrature checks the antecedent's printed Bessel relation; does not test the late nonlinear aberration or curvature contribution. |

**Implementation acceptance tests are a separate inventory.** For example, the v1.1.0 Peebles and frame-transformation tests (`test_peebles.py` and `test_frame_transform.py`) check recombination and implemented frame transformations. They do not retroactively expand the independent Paper 1 v1.0.0 certificate. Likewise tests for `coefficients.py`, the line-of-sight pipeline, and the spectrum's morphology address the later *physics implementation*; they are not entered in Table 8 merely because they pass. New independent Paper 1 checks may be appended to this table in v1.1.0 and later as their actual evidence is established. The table is not an assertion that every major Paper 1 theorem has a completed standalone Python proof. The original v1.0.0 tests and later extensions are recorded at their actual Git tags; importing software-under-test never becomes an independent derivation simply because the test is mathematically phrased.

## 20.2 v1.0.0 — the reconstruction, and whether it is faithful

Achieved. The free-streaming chain of Appendix F reproduced against *four* external hierarchies — Hu & Sugiyama Eq. (6), Wilson Eq. (8), Ma & Bertschinger Eqs. (49)/(50), Seljak & Zaldarriaga Eq. (3d) — each read from its own paper and integrated as printed, at residuals of $10^{-13}$–$10^{-14}$ against a $10^{-11}$ threshold, in two normalisations and two phase conventions. The almost-Friedmann–Lemaitre spectrum (186) computed from (176) by the mode route and the covariant route, agreeing exactly. The Sachs–Wolfe limit recovered at 0.26% against a 5% criterion. This shows that the reconstruction is faithful to the antecedent's own equations, independently of any Boltzmann code, and that the harmonic phase convention is fixed by the *definition* of $Q_{A_\ell}$ rather than adopted.

Learned, and it set the pattern for everything after. A coarsened $\eta$ grid returned a smooth, plausible spectrum that was 575% wrong while the two routes agreed to $10^{-16}$ — finding T-3. Two agreeing routes agree about what they share. That sentence is the single most load-bearing lesson in this bundle; it produced `required_n_eta`, it is why criterion 5b's gate states its shared inputs explicitly, and it is why the grid register exists.

### 20.3 v1.1.0 — the recombination that makes peaks possible

Achieved. Peebles' effective three-level atom replacing equilibrium Saha, and made the default only after being checked standalone against the standard ionisation history. Freeze-out at $2.15 \times 10^{-4}$ against a standard $\sim 2 \times 10^{-4}$; the visibility peak moves $z \simeq 1300 \to 1107$ against a standard 1080–1100, and the function *widens by about 40%*. The frame change made numerical rather than symbolic. This shows that the width, not the position, is what the spectrum needs: the width sets the Silk damping envelope, which is why the acoustic peaks could not be attempted before this release.

Learned. `SAHA` was kept rather than deleted. A known-worse alternative that *must* change the answer is evidence; deleting it would have left the improvement asserted rather than measured. The same principle appears as the negative control in every criterion that claims one.

### 20.4 v1.2.0 — the spectrum with structure, and the audit of it

Achieved. The full linear $C_\ell$: acoustic peaks, a Silk damping tail, the complete source assembly, and the integral solution (102) carried with a generic $u^a$ and specialised back. Criterion 5b closed at $9.5 \times 10^{-6}$ against a gate of $10^{-3}$. This shows that *Annals II*'s two writings of its own integral solution — (102) specialised, and (177)–(179) assembled — are the same calculation when evaluated, which the paper recovers symbolically and which nobody had ever computed.

Learned, three times over, and they are the same lesson in three places. *First*, the tolerance this criterion was written against could only ever be failed: a slow-decoupling solution with no polarisation hierarchy cannot meet two full Boltzmann codes, and a criterion that can only be failed is not a criterion (S4c). *Second*, the figure $10^{-5}$ that had stood in the record for the two routes' agreement was never measured — it came from a fixed grid on which both routes were *un-converged in the same direction* (T-45). *Third*, the gate written to catch all of this printed `PASS` on a `nan`, because `nan >= gate` is `False` (T-44). A gate is not tested by the run that passes it.

### 20.5 v1.3.0 — the structural audit, and the discrepancy localised

Two halves, in the order decision S37 fixed: the structural audit of the code first, the acoustic modes second. The audit produced the generated registers this supplement now rests on — the grid register, which walks the abstract syntax tree and *fails* when the code holds a discretisation the register does not name; the model table, derived from the `Background` objects the code integrates rather than transcribed from a parameter sheet; and the theory audit, generated from six self-contained instruments that import no project code.

The acoustic half did not go as planned, and the record says so. It began as a search for the missing peaks and became the localisation in §15, because the frame discrepancy turned out to be upstream of the peak question and to admit a cleaner instrument. Two defects were separated that had been one symptom: the frame gap, now reduced to a single site with both of its boundaries measured; and the peak positions, which sit above their own $\ell_A$ on all three backgrounds, in both frames, and which the acoustic scale does not explain — the signs are opposite.

What v1.3.0 has not done. It has not repaired the assignment, it has not recovered the peaks, and it has not bounded the envelope approximations A-3, A-5 and the damping inputs, which are deferred to v1.4.0 precisely because changing an envelope term mid-measurement moves the number being measured for reasons unrelated to it.

What the arc did not do, and says so. No release has bounded the accuracy of the *source* as distinct from the consistency of its assembly. The recombination grid moves the primary term by 1.2% and both routes of criterion 5b share it, so the criterion cannot see it. That is recorded as a stated limit of v1.2.0 and derived immediately after the tag.

**Planned end of the implementation sequence.** The physics-facing objective of v1.5.0 is to bring the *full in-scope physics introduced in Paper 1* into the implementation: the $O(\varepsilon^2\ell)$ kinematic couplings, their frame relabelling and lensing/aberration interpretation, after v1.3.0 has closed the linear assembly and v1.4.0 has supplied the radiation-era and tight-coupling ingredients. Every antecedent relation must be used with the corrections and provenance qualifications recorded in Paper 1 and this supplement. This is the *planned endpoint*, not a claim that v1.5.0 has been implemented, that all components have been independently certified, or that Paper 1 computes out-of-scope tensor amplitudes or polarisation.

# 21 The scattering-strength expansion, re-derived

*Annals II* Appendix C writes the almost-Friedmann–Lemaître integrated Boltzmann equation for Thomson scattering as

$$\mathcal{B} + \dot{\tau} + e^a \mathrm{D}_a \tau \;\simeq\; t_c^{-1}\big[v^a e_a - \tau\big], \tag{72}$$

inverts it, and approximates the result *"systematically ... in terms of the smallness parameter $t_c$"* by the recursion

$$\tau_{(n)} \;\simeq\; v^a e_a - t_c\Big[\mathcal{B} + \dot{\tau}_{(n-1)} + e^a \mathrm{D}_a \tau_{(n-1)}\Big], \tag{73}$$

printing $\tau_{(0)}$, $\tau_{(1)}$ and $\tau_{(2)}$. Section 5.5.3, *Expansion in Thompson Scattering Time*, states that the scheme carries *"terms up to $\mathcal{O}(t_c^3)$"*, so the ceiling is the antecedent's and the printed orders stop one short of it.

## 21.1 The recursion is geometric

Write $\mathcal{D} \equiv \partial_t + e^a \mathrm{D}_a$ for the derivative along the photon path — which is exactly the pair of terms (73) applies to $\tau_{(n-1)}$ — and $V \equiv v^a e_a$. Then (72) reads $\tau = V - t_c(\mathcal{B} + \mathcal{D}\tau)$, so

$$(1 + t_c\mathcal{D})\tau = V - t_c\mathcal{B}, \qquad \tau = (1 + t_c\mathcal{D})^{-1}(V - t_c\mathcal{B}), \tag{74}$$

and expanding the inverse as a Neumann series in $(-t_c\mathcal{D})$ gives a closed form at every order:

$$\boxed{\tau_{(n)} \;=\; V \;+\; \sum_{m=1}^{n} (-1)^m\, t_c^m \Big[\mathcal{D}^m V + \mathcal{D}^{m-1}\mathcal{B}\Big].} \tag{75}$$

This shows that Appendix C's recursion is not an open-ended approximation scheme but a geometric series with an explicit sum, and that $\tau_{(1)}$ and $\tau_{(2)}$ as printed are its first two partial sums. Equation (75) is implemented in `functions/spectra/scattering_expansion.py` and checked against the printed $\tau_{(1)}$ and $\tau_{(2)}$ term for term, with a deliberately failing control, in `tests/test_scattering_expansion.py`. The closed form was derived from (73) and *then* compared with the page, which is the only order in which the comparison carries information.

**The closed form holds the scattering time fixed along the path, and the antecedent's scheme is what fixes it.** The Neumann series of $(1 + t_c\mathcal{D})^{-1}$ runs in powers of the *operator* $t_c\mathcal{D}$, and $(t_c\mathcal{D})^2 = t_c(\mathcal{D}t_c)\mathcal{D} + t_c^2\mathcal{D}^2$. Iterating (73) directly returns $t_c(\mathcal{D}t_c)\mathcal{B}$ and $t_c(\mathcal{D}t_c)(\mathcal{D}V)$ at second order; neither appears in (75), which carries scalar powers $t_c^m$ against operator powers $\mathcal{D}^m$. The term counts say whose omission it is. (75) holds $2^m + 2^{m-1}$ terms and Appendix C prints three at first order and six at second, which it could not do if the $\mathcal{D}t_c$ terms were carried. The reconstruction is faithful to the page; the scheme on the page sets $\mathcal{D}t_c = 0$ without saying so.

**At the peak of the visibility the dropped term is the size of the term one order below it.** In conformal time $t_c = 1/\kappa'$, so along the time direction $\mathcal{D}t_c = -\kappa''/\kappa'^2$. The visibility $\mathcal{V} = \kappa' e^{-\kappa}$ is stationary where $\mathcal{V}' = (\kappa'' + \kappa'^2)e^{-\kappa}$ vanishes, and $e^{-\kappa}$ never does, so the peak sits exactly at $\kappa'' = -\kappa'^2$ — measured to $7.4 \times 10^{-15}$ in `tests/frame/c12_sampling_bound.py`. There $\mathcal{D}t_c = 1$, and the dropped $t_c(\mathcal{D}t_c)(\mathcal{D}V)$ is $t_c\mathcal{D}V$, which is the retained first-order term. The second-order remainder is the size of the first order at $\eta_*$, so the series is not asymptotic at the one epoch it is applied to. Two parameters control it and only one is small: $t_c k$ for the gradients, and $t_c\mathcal{D}\ln t_c$ for the time derivative, which is unity at the peak whatever $k$ is.

## 21.2 Two consequences the closed form makes immediate

**The term counts are not chosen.** $\mathcal{D}^m$ expands into a sum over words in $\{\partial_t,\ e^a\mathrm{D}_a\}$, so the $t_c^m$ bracket of (75) holds exactly $2^m + 2^{m-1}$ terms. Appendix C prints three at first order and six at second; those are $2+1$ and $4+2$. Table 9 lists every term to $\mathcal{O}(t_c^3)$, generated from the same module rather than transcribed.

**Order $m$ sources multipoles up to $\ell = m+1$.** Each $e^a\mathrm{D}_a$ carries one direction vector and $V = v^a e_a$ carries one already, so a word with $g$ gradients seeded by $V$ has $g+1$ factors of $e^a$; reduced by Appendix C's own solid-angle identities, which lower the rank two at a time, its highest PSTF part has rank $g+1$. At second order that reaches rank three, in the single term $e^c e^a e^b \mathrm{D}_c\mathrm{D}_a v_b$.

Equivalently, and more usefully for the high-$\ell$ question: each gradient contributes one power of $k$, so a term of rank $r$ carries $k^{r-1}$. The expansion parameter is therefore $t_c k$ — the ratio of the mean free path to the wavelength — and not $t_c$ alone. This shows that the truncation is $\ell$-dependent by construction: a cut in rank is a cut in $t_c k$, and the order at which the expansion may be stopped falls as $\ell$ rises.

## 21.3 What this costs the oscillator

The near-tight-coupling set of §7.1 is introduced *"rewritten to first order"* and closes with $\tau^{A_\ell} \simeq 0$ for all $\ell > 2$. At first order the expansion reaches rank two, so the closure costs nothing and the antecedent is consistent. What the rank counting fixes is the price of going further: second order reaches the octupole and $\mathcal{O}(t_c^3)$ the hexadecapole, and the closure carried with (152) and (153) reaches neither.

That is approximation A-4 of `provenance/SOURCE-APPROXIMATIONS-v1.md` stated as a rank rather than as a dropped term. The term the oscillator derivation discards, $-\frac{2}{5}t_c(1+R)^{-1}\mathrm{D}^b\mathrm{D}_{\langle a}u^m_{b\rangle}$, is dropped because *"the matter shear is already at least first order"* — a cross term of $\mathcal{O}(t_c\epsilon)$ between §5.5.1's almost-Friedmann–Lemaître ordering and §5.5.3's scattering-time ordering. *Annals II*'s Appendix E, after Ellis–Treciokas–Matravers, states what the discarded harmonics hold: if any four consecutive ones vanish exactly then $\sigma_{ab} = 0$ exactly. This shows that the truncation and the shear are not independent — the residual in the truncated harmonics is where the shear lives, and a treatment that sets them to zero has set the shear to zero with them.

## 21.4 What appears at second order that does not appear at first

Appendix C's own solid-angle reduction gives the rank decomposition of the source,

$$\int_{4\pi} \mathrm{d}\Omega\, O^{A_\ell}\mathcal{B} \simeq \delta^\ell_0\Big[-\tfrac{4\pi}{3}\mathrm{D}_a\tau^a\Big] + \delta^\ell_1\Big[\tfrac{4\pi}{3}\big(\tfrac{1}{4}\mathrm{D}^{a_1}\ln\rho_R + A^{a_1}\big)\Big] + \delta^\ell_2\Big[\tfrac{8\pi}{15}\sigma^{a_1a_2}\Big], \tag{76}$$

so $\mathcal{B}$ carries the shear in its rank-two part. At first order $-t_c\mathcal{B}$ therefore reproduces the quadrupole $\tau_{ab} \simeq -\dot\kappa^{-1}[\sigma_{ab} + \mathrm{D}_{\langle a}v^B_{b\rangle}]$ of §7.1, as it must.

At second order the closed form contributes $+t_c^2\,\mathcal{D}\mathcal{B}$, and through (76) that is $\dot{\sigma}_{ab}$ at rank two and $e^c\mathrm{D}_c\sigma_{ab}$ at rank three. Appendix G's $\dot{\sigma}_{ab} + 2H\sigma_{ab} + E_{ab} \approx 0$ then gives

$$\dot{\sigma}_{ab} = -2H\sigma_{ab} - E_{ab}, \tag{77}$$

so the electric Weyl tensor enters the radiation quadrupole at second order in the scattering time, and does not at first. This shows that the second-order expansion is not merely a smaller correction of the same kind: it introduces a gravitational tidal driving of the quadrupole that the first-order treatment has no term for.

**An open question, stated as a mechanism rather than a result.** *Annals II* flags the regime itself, in a footnote to the sentence about removing the radiation quadrupole: *"If there are any anisotropic contributions such as anisotropic scattering (in the matter frame) or large shear (from gravitational waves) at that time this sort of approximation should be considered with care — such phenomena would break tight-coupling."* For a tensor mode both $\sigma_{ab}^{(T)}$ and $E_{ab}^{(T)}$ oscillate at the gravitational-wave frequency, so (77) drives the second-order quadrupole oscillatorily on a scale set by the wave and not by the sound horizon. Whether that survives projection as a distinguishable *ringing* is open; it is recorded as question Q10 rather than claimed, because nothing here computes it.

**What this expansion cannot produce.** (73) is linear in $V$ and $\mathcal{B}$, so (75) yields terms of order $t_c^m\epsilon$ and never $\epsilon^2$. Terms of $\mathcal{O}(\varepsilon^2\ell)$ require the quadratic parts of the Boltzmann equation that §5.5.1's almost-Friedmann–Lemaître linearisation discards, and are the subject of a later release rather than of this appendix. The high-$\ell$ sensitivity established above is a separate statement and rests on $t_c k$, not on $\epsilon^2$.

Table 9: *Annals II* Appendix C's scattering-strength expansion to $\mathcal{O}(t_c^3)$, generated from `functions/spectra/scattering_expansion.py`. **Rank** is the highest PSTF multipole the term can source: one per $e^a\mathrm{D}_a$, plus one for $V = v^a e_a$. The closure (L0082) keeps $\ell \le 2$.

| **Order** | | **Term** | **Rank** |
|---|---|---|---|
| $t_c^0$ | + | $v_a e^a$ | 1 |
| $t_c^1$ | − | $\mathcal{B}$ | 0 |
| $t_c^1$ | − | $\left(v_a e^a\right)^{\cdot}$ | 1 |
| $t_c^1$ | − | $e^c\mathrm{D}_c\left(v_a e^a\right)$ | 2 |
| $t_c^2$ | + | $\left(\mathcal{B}\right)^{\cdot}$ | 0 |
| $t_c^2$ | + | $e^c\mathrm{D}_c\left(\mathcal{B}\right)$ | 1 |
| $t_c^2$ | + | $\left(\left(v_a e^a\right)^{\cdot}\right)^{\cdot}$ | 1 |
| $t_c^2$ | + | $e^c\mathrm{D}_c\left(\left(v_a e^a\right)^{\cdot}\right)$ | 2 |
| $t_c^2$ | + | $\left(e^c\mathrm{D}_c\left(v_a e^a\right)\right)^{\cdot}$ | 2 |
| $t_c^2$ | + | $e^c\mathrm{D}_c\left(e^c\mathrm{D}_c\left(v_a e^a\right)\right)$ | 3 |
| $t_c^3$ | − | $\left(\left(\mathcal{B}\right)^{\cdot}\right)^{\cdot}$ | 0 |
| $t_c^3$ | − | $e^c\mathrm{D}_c\left(\left(\mathcal{B}\right)^{\cdot}\right)$ | 1 |
| $t_c^3$ | − | $\left(e^c\mathrm{D}_c\left(\mathcal{B}\right)\right)^{\cdot}$ | 1 |
| $t_c^3$ | − | $e^c\mathrm{D}_c\left(e^c\mathrm{D}_c\left(\mathcal{B}\right)\right)$ | 2 |
| $t_c^3$ | − | $\left(\left(\left(v_a e^a\right)^{\cdot}\right)^{\cdot}\right)^{\cdot}$ | 1 |
| $t_c^3$ | − | $e^c\mathrm{D}_c\left(\left(\left(v_a e^a\right)^{\cdot}\right)^{\cdot}\right)$ | 2 |
| $t_c^3$ | − | $\left(e^c\mathrm{D}_c\left(\left(v_a e^a\right)^{\cdot}\right)\right)^{\cdot}$ | 2 |

| Order | | Term | Rank |
|---|---|---|---|
| $t_c^3$ | – | $e^c\mathrm{D}_c\big(e^c\mathrm{D}_c\big(\big(v_a e^a\big)^{\cdot}\big)\big)$ | 3 |
| $t_c^3$ | – | $\big(\big(e^c\mathrm{D}_c\big(v_a e^a\big)\big)^{\cdot}\big)^{\cdot}$ | 2 |
| $t_c^3$ | – | $e^c\mathrm{D}_c\big(\big(e^c\mathrm{D}_c\big(v_a e^a\big)\big)^{\cdot}\big)$ | 3 |
| $t_c^3$ | – | $\big(e^c\mathrm{D}_c\big(e^c\mathrm{D}_c\big(v_a e^a\big)\big)\big)^{\cdot}$ | 3 |
| $t_c^3$ | – | $e^c\mathrm{D}_c\big(e^c\mathrm{D}_c\big(e^c\mathrm{D}_c\big(v_a e^a\big)\big)\big)$ | 4 |

**One transcription note.** The fifth term of Appendix C's $t_c^2$ bracket is printed as $e^c e^d \mathrm{D}_c \dot{v}_b$, which leaves both $d$ and $b$ free in a scalar. It is $e^c e^d \mathrm{D}_c \dot{v}_d$ — the time derivative taken first and the gradient outside — forced independently by the four words of $\mathcal{D}^2$, by the bracket's count of six with the other five unambiguous, and by the two direction vectors having nothing else to contract with. Recorded as B-15; no computed quantity depends on it.

# 22 The CDM-frame division of the integral solution, term by term

The integral solution (102) is derived for a general $u^a$ — the antecedent says so immediately after (90), *"Note that these are CGI with respect to $u^a$"* — and then evaluated once, in the Newtonian frame. This appendix carries it into the CDM frame instead, coefficient by coefficient, and finds that the two frames do not agree. The failure is isolated to one coefficient at one power of $k$.

What this shows is narrower than it looks, and the conventions section exists because the first draft of this appendix did not have one.

## 22.1 Conventions, stated

Every relation below is in one convention and this subsection fixes it. The need for it was raised by the Paper 1 stream, whose check could not reproduce two of the four frame-table entries — correctly, because the entries are in the bundle's normalisation and the check was in Paper 1's.

**Wavenumber.** The bundle carries $k \equiv k_{\mathrm{com}}$ throughout and converts at the point of use, $k_{\mathrm{phys}} = k_{\mathrm{com}}/a$. Paper 1's C3.1 declares $k$ *physical*. So a relation printed there as $X = k\,Y$ is carried here as $aX = k_{\mathrm{com}}Y$, and the scale factor is the conversion, not a transcription error.

**The frame velocity.** $v^a$ is the relative velocity of the two frames, $\tilde{u}^a \approx u^a + v^a$ with $v_a u^a = 0$, expanded as $v_a = v\,Q_a$. So $v$ is a *velocity amplitude*: it is the $v$ of the Euler equation $v' + \mathcal{H}v = -k\Phi_A$, which is the frame condition $\tilde{A}_a = 0$, and it is the $v$ of $\sigma = -k_{\mathrm{phys}}v$. It is not a velocity potential; $v_a = \mathcal{D}_a V$ would put $k$ in the numerator.

**The dipole.** The bundle carries the dipole in the *covariant* normalisation, $\tilde{\tau}_1 \equiv \tilde{v}_B = v_B/k$ — the normalisation (109) needs in order to read $-\tilde{C}_1 = (\delta\tilde{T} + \Phi_A) + \kappa'\tilde{v}_B$. GDE99 (37), $\tilde{\tau}_a \approx \tau_a - v_a$, therefore reads $\tilde{\tau}_1 \to \tilde{\tau}_1 - v/k$ *here* and $\tau_1 \to \tau_1 - v$ in the un-reduced normalisation. The two are the same statement. This is finding T-27 in the bundle's record, where the un-reduced form was used by mistake and the symptom was a first-order frame change 186 times the quantity it corrects at $k = 300$.

**Consequently, two of the four entries reconcile and one does not.** $a\sigma = -k_{\mathrm{com}}v$ *is* $\sigma = -k_{\mathrm{phys}}v$, so n-t1 agrees with Paper 1's $\sigma = -kv$. $\tilde{\tau}_1 \to \tilde{\tau}_1 - v/k$ *is* $\tau_a \to \tau_a - v_a$ in the reduced normalisation, so n-t4 agrees. What does not reconcile from outside the bundle is the reduced normalisation itself: Paper 1 carries no $C_1$, so it cannot adjudicate $\tilde{\tau}_1 = v_B/k$, and $\Delta C_1$ below rests on it.

## 22.2 What has to hold

A first-order change of frame is a boost, and a boost changes the observed brightness by $\tau \to \tau - v_a e^a$: a pure dipole. In PSTF multipoles,

$$\tau_{A_1}^{(u)} = \tau_{A_1}^{(n)} - v_{A_1}, \qquad \tau_{A_\ell}^{(u)} = \tau_{A_\ell}^{(n)} \quad \text{for all } \ell \geq 2. \tag{78}$$

These relations are shorthand at first order in the frame tilt $v^a$. They do not assert exact equality of the raw multipoles at finite boost: Paper 1, Secs. II E and VI B, distinguishes the small-tilt multipole law from the exact transformation of the photon distribution. At the specified order the predicted $\ell \geq 2$ observable must agree after both results have been expressed in the same emitter and observer frames. The displayed equality signs have not been changed here pending the source-convention audit.

## 22.3 The three shifts

The next displayed coefficients reproduce the historical printed form attributed to published *Annals II* Eq. (90), not the corrected rank-zero result used in the live comparison below. They are retained to make the original calculation and the correction traceable.

The printed form is

$$-C_0 \simeq \tfrac{5}{2}\, a\mathcal{B}_2, \qquad -C_1 \simeq \tfrac{1}{k}(a\mathcal{B}_1 + \kappa' v_B), \qquad -C_2 \simeq \tfrac{1}{k^2}\, a\mathcal{B}_2, \tag{79}$$

with $a\mathcal{B}_1 = k\,\delta T + aA$ and $v_B = k\tilde{\tau}_1$. The frame carries four entries, each in §22.1's convention:

| | Newtonian $n^a$ | CDM $u^a$ |
|---|---|---|
| $aA$ | $k\Phi_A$ | $0$ (the frame condition $A_a = 0$) |
| $\delta T$ | $\delta\tilde{T}$ | $\delta\tilde{T} - \mathcal{H}v/k$ (n-t3) |
| $\tilde{\tau}_1$ | $\tilde{\tau}_1$ | $\tilde{\tau}_1 - v/k$ (n-t4, reduced) |
| $a\mathcal{B}_2$ | $0$ | $-kv$ (n-t1, $\sigma = -k_{\rm phys}v$) |

With $\Delta X \equiv X^{(u)} - X^{(n)}$: $\Delta(a\mathcal{B}_1) = -\mathcal{H}v - k\Phi_A = v'$ by the Euler equation, and $\Delta(\kappa' v_B) = \kappa' k\,\Delta\tilde{\tau}_1 = -\kappa' v$. Hence

$$\boxed{\Delta C_0 = +\tfrac{5}{2}\,k\,v, \qquad \Delta C_1 = -\tfrac{1}{k}(v' - \kappa' v), \qquad \Delta C_2 = +\tfrac{v}{k}} \tag{80}$$

all three in one $v$ and one $k$.

The first entry is superseded, and so is the title of the subsection that follows. Corrected 2026-10-23, findings T-104 and T-107. $\Delta C_0 = +\frac{5}{2}kv$ is the shift of (90)'s *printed* $C_0$. (89)'s $\delta_{\ell 0}$ row carries $\alpha_2$ where $\alpha_0$ belongs — a factor $\frac{15}{2}$, finding T-104 — and solved at its own normalisation the row gives $C_0 = -\frac{1}{3}a\mathcal{B}_2$, whence

$$\Delta C_0 = +\tfrac{kv}{3} \qquad \text{(corrected)}, \tag{81}$$

smaller by $\frac{15}{2}$. Every scaling below survives with its prefactor divided by $\frac{15}{2}$: (86) becomes $\frac{1}{6}(k/\mathcal{H})^2$ in place of $\frac{5}{4}(k/\mathcal{H})^2$, and the conclusion that the ratio grows as $(k/\mathcal{H})^2$ is unchanged, because $\Delta C_0$ and $\frac{1}{k}2v''$ are compared as powers of $k/\mathcal{H}$ and a constant does not move that.

What does not survive is "$\Delta C_0$ has no partner". $a\mathcal{B}_0 = -\frac{k}{3}\tau_1$ by identity, and $-C_1'$ carries $+\frac{k}{3}\tau_1$, so the partner exists and is *cross-coefficient* — finding T-107. $\gamma$, which pairs $\Delta C_0$ against $k^2\Delta C_2$, therefore asks the wrong pairing and cannot see it: M15, *a gate built from one coefficient cannot see a cancellation distributed across the bracket.* `tests/frame/c8_rank0_partner.py` scores the whole rank-0 content of the bracket and is the gate for this question.

### 22.4 The shift of (102), including the cross-coefficient partner

(102) reads

$$\frac{\beta_\ell \tau_\ell(\eta_0)}{2\ell+1} \simeq [C_2'(\eta_*) - C_1(\eta_*)]\mathcal{D}\, j_\ell(x_*) + kC_2(\eta_*)\mathcal{D}\, j_\ell'(x_*) + \int_{\eta_*}^{\eta_0} d\eta\; \mathcal{V} e^{-(k/k_D)^2} [C_0 - C_1' + C_2''] j_\ell(x), \quad (82)$$

and (80) gives

$$\Delta\big[C_2' - C_1\big]_{\eta_*} = \tfrac{1}{k}\big(2v' - \kappa' v\big)\Big|_{\eta_*}, \quad (83)$$

$$\Delta\big[kC_2\big]_{\eta_*} = v(\eta_*), \quad (84)$$

$$\Delta[C_0 - C_1' + C_2''] = \tfrac{5}{2}kv + \tfrac{1}{k}\big(2v'' - (\kappa' v)'\big). \quad (85)$$

*Historical printed-coefficient result.*[28] $v$ obeys $v' = -\mathcal{H}v - k\Phi_A$, so on sub-horizon scales it changes on the Hubble rate: $v' \sim \mathcal{H}v$, $v'' \sim \mathcal{H}^2 v$. Then

$$\frac{\frac{5}{2}kv}{\frac{1}{k}\,2v''} \;\sim\; \frac{5}{4}\left(\frac{k}{\mathcal{H}}\right)^2, \quad (86)$$

and the two boundary shifts are smaller still. For the printed coefficient alone this estimate produces a term of order $(k/\mathcal{H})^2$ relative to $v''$, but it does not establish a failure of the corrected frame transformation. The relevant comparison is the complete rank-zero bracket, including its cross-coefficient partner.[29]

**Corrected coefficient shift and its differentiated partners.** For the *stipulated first-order* frame relations of §22.1, the source-amplitude shifts read $\Delta\mathsf{A} = -kv$ and $\Delta\mathsf{F} = v' - \kappa' v$. Substitution into the corrected coefficient definitions (7) gives

$$\Delta C_0 = \frac{kv}{3}, \qquad \Delta C_1 = -\frac{v' - \kappa' v}{k}, \qquad \Delta C_2 = \frac{v}{k}. \quad (87)$$

The three terms carried by the line-of-sight expression follow by *differentiating these complete shifts*, at fixed comoving $k$:

$$\Delta(C_2' - C_1) = \frac{v'}{k} + \frac{v' - \kappa' v}{k} = \frac{2v' - \kappa' v}{k}, \quad (88)$$

$$\Delta(kC_2) = v, \quad (89)$$

$$\Delta(C_0 - C_1' + C_2'') = \frac{kv}{3} + \frac{v'' - (\kappa' v)'}{k} + \frac{v''}{k} \quad (90)$$

$$= \frac{kv}{3} + \frac{2v'' - \kappa'' v - \kappa' v'}{k}. \quad (91)$$

The Leibniz expansion on the last line is retained explicitly because $\kappa'$ is time-dependent. Equations (87)–(91) replace the *printed-coefficient algebra* as the conditional shift of the corrected matching; they do not prove that the assembled observable is frame invariant. Such a claim requires the frame transformation of all sources, weights, projected boundary terms and the integration variable in one expression. In

[28] This is the shift of the historical printed coefficient $C_0 = -5a\mathcal{B}_2/2$ and therefore carries $+5kv/2$. If the corrected monopole row of Eq. (49) is adopted, its contribution is $+kv/3$ instead. The required cross-coefficient cancellation involving $-C_1'$ remains unverified. The historical equation is retained for audit; it must not be used as the current corrected frame-shift law.

[29] The estimate in Eq. (86) uses the superseded coefficient $C_0 = -5a\mathcal{B}_2/2$. In the corrected matching its prefactor changes, and $-C_1'$ can provide an additional rank-zero term. Neither the printed coefficient estimate nor its scaling alone is a complete frame-covariance test.

particular, a possible cross-coefficient cancellation must be exhibited on the complete bracket and tested in the implementation. The source of the frame relations is Paper 1, Secs. II E and VI B, with the first-order limitation of its Eq. (74), together with the *published* Annals II matching equations as cited provisionally here. The latter still require a journal-page comparison.[30]

### 22.5 The $k^2$ is (90)'s, not a mixed convention

A factor $k^2$ between $\Delta C_0$ and $\Delta C_2$ is exactly what a comoving/physical or amplitude/potential mismatch produces, and the Paper 1 stream was right to require that reading be excluded before the rest stands. It is excluded, and not by argument.

$C_0$ and $C_2$ are the same array. (79) divides one quantity, $a\mathcal{B}_2$, by $k^0$ and by $k^2$. Whatever $a\mathcal{B}_2$ is — physical or comoving, amplitude or potential, correctly transcribed or not — it is *the same object in both*, so their ratio carries no convention at all:

$$\frac{C_0}{k^2 C_2} = \frac{p_0}{p_2} = \frac{5}{3\beta_2} \qquad \text{(a pure rational).} \tag{92}$$

Measured at $\eta_*$ on 'CDM (Einstein–de Sitter)', 'LCDM' and 'PLANCK_MODEL', at $k = 1, 5, 20, 60, 140, 300$: **2.5000** at every entry. That measurement predates 21 October and is the ratio under (89)'s *printed* $\delta_{\ell 0}$ row. Under the released row today — `MONOPOLE_ROW = "corrected"`, (81) — the same quantity reads 0.333333 at $k = 1, 10, 60$ on Einstein–de Sitter and `PLANCK_MODEL` alike: exactly $\frac{1}{3}$. *Only the value moves.* What this subsection's argument needs is that the ratio is a pure rational, independent of $k$ and of background, and it still is. What the measurement shows is internal consistency, which is not correctness — that was true at $\frac{5}{2}$ and is true at $\frac{1}{3}$. A mismatch between two different objects does not return a background-independent, $k$-independent rational; it drifts. So the dominance is (90)'s own structure, and a spurious scale factor in $a\mathcal{B}_2$ — were there one — would multiply $C_0$ and $C_2$ equally and cancel from the ratio entirely.

What this does not exclude is $\Delta C_1$, which is the one shift that depends on the reduced normalisation $\tilde{\tau}_1 = v_B/k$ of §22.1. If that reading is wrong, $\Delta C_1$ is wrong by $k^2$ — and the bundle has made exactly that error once already, as T-27. Paper 1 cannot adjudicate it, since it carries no $C_1$. It is open, and it is the first thing to settle.

### 22.6 Measured

Both frames from one environment, one acoustic solution and one $k$ grid, differing only in the frame (110) specialises to, against the Newtonian total's scale over $\ell \in \{2, 5, 10, 50, 200\}$:

| background | $k$ | as assembled | with $C_0$ dropped |
|---|---|---|---|
| CDM (Einstein–de Sitter) | 20 | 53.5 | 0.032 |
| | 300 | 1719 | 0.035 |
| ΛCDM | 20 | 62.8 | 0.105 |
| | 300 | 6868 | 0.154 |
| `PLANCK_MODEL` | 20 | 72.2 | 0.111 |
| | 300 | 3687 | 0.058 |

Three to four orders of magnitude, to a residual of 1–15%, with the $k$ dependence of (86). The rank decomposition agrees: $C_0$ carries 98.6% of $C_\ell$ in the CDM frame and is identically zero in the Newtonian one.

[30] The $\Delta\mathsf{A}$ and $\Delta\mathsf{F}$ assumptions are the frame-convention input to this calculation. The code's corresponding field transforms and its complete projected line-of-sight frame equality have not yet passed an independent present-state repository test.

### 22.7 Why the antecedent never meets this

(102) is evaluated once, in the Newtonian frame, where *"the vanishing shear condition $\tilde{\sigma}_{ab} \approx 0$ implies that $\tilde{C}_0(\eta) \approx 0$ and $\tilde{C}_2(\eta) \approx 0$"*. Both members of the pair vanish together, so the imbalance between them cannot be seen. Nothing the antecedent publishes is affected. What is missing is a stated condition on a formula labelled frame-general.

### 22.8 Three repairs tested, and one of them is untested

**The dropped lines of (93) are not the counterweight.** (102) is (93) with its last two integrals removed. Restoring $\int(\kappa' e^{-\kappa})[C_1 - 2C_2]j_\ell$ and $\int((\kappa')^2 - \kappa'')e^{-\kappa}C_2 j_\ell$ moves the disagreement from 72.2 to 71.6 at $k = 20$ and from 3686.9 to 3686.8 at $k = 300$.

**(89)'s $\ell = 0$ line is not mis-sourced.** $a\mathcal{B}_0\delta_{\ell 0}$ is commented out of the equation the integral solution solves and restored in the open-case extension, which looks like a dropped source. It is not: the ansatz must give $\tau_0 = 0$, and differentiating $\int A_0\, d\eta' = 0$ yields (89)'s $\ell = 0$ line with zero on the right. $C_0$ is correctly derived.

**Undoing the integration by parts: untested, not refuted.** (93) comes from a kernel in which the two derivatives act on $\tau_\ell^{(0)}$ and bring down $k$, not on the coefficients where they bring down $\mathcal{H}$. The numerical test of this was invalid — it failed its own control, reproducing the by-parts form in the Newtonian frame only to within $10^{-3}$ to 2.5 — because (102) does not differentiate the weight $\mathcal{V}$ that (93) carries inside the integral, so the two are not an exact by-parts pair once a weight is present. It remains the most promising repair and it needs doing analytically.

### 22.9 What is open

1. The reduced normalisation $\tilde{\tau}_1 = v_B/k$, on which $\Delta C_1$ rests and which nothing outside the bundle can adjudicate. The bundle has erred here once (T-27). Settle it by index algebra on (109).
2. The by-parts step, the one untested repair.
3. Whether $C_0$ belongs in the integrand, which the measurement answers empirically and the algebra does not.
4. The 1–15% residual after $C_0$ is removed: whether it is $O(\varepsilon^2)$ or an artefact of the grids.

What stands independently of all four: $C_0$ and $C_2$ are one $a\mathcal{B}_2$ over $k^0$ and $k^2$, that ratio is a measured pure rational carrying no convention, and (102) in a sheared frame is dominated by the member of the pair that the Newtonian specialisation sets to zero.

## 23 Discussion

### 23.1 Suppression, not truncation

Thomson scattering damps every multipole of rank two and above at the rate $\kappa'$, leaves the monopole untouched because scattering conserves photon number, and couples the dipole to the baryons. The free-streaming operator moves power between neighbouring ranks at the rate $k$. When $\kappa' \gg k$ the hierarchy therefore slaves itself: each rank is driven by the one below it and damped before it can act, so that

$$\tau_\ell \;\sim\; \left(\frac{k}{\kappa'}\right)^{\ell-1} \tau_1, \qquad \ell \geq 2. \tag{93}$$

**The monopole and the dipole carry the dynamics, and the quadrupole is not negligible.** Figure 1 states the same thing for the tight-coupling panel. At $O(k/\kappa')$ the quadrupole is the smallest rank that survives tight coupling, and it is what produces diffusion damping and the whole of the polarisation. Ranks three and above are smaller again by further powers of $k/\kappa'$.

**The distinction between a suppressed rank and a truncated one is not cosmetic.** A truncation is a modelling choice, made outside the equations, which can be made wrongly and which must be justified. A suppression is a limit the hierarchy takes on its own, with (93) as its statement, and nothing is set to zero. The exact hierarchy is carried throughout this bundle for that reason: the tight-coupling behaviour is a consequence to be derived rather than an approximation to be imposed.

### 23.2 The kernel is the cascade

Once $\kappa'$ falls below $k$ nothing damps the higher ranks and power cascades upward without limit. That cascade is not computed by integrating the hierarchy to high $\ell$. The free-streaming operator is linear and collisionless in that regime, and its solution is known in closed form:

$$j_\ell(k(\eta_0 - \eta)) \quad \textit{is} \text{ the free-streaming cascade, solved exactly.} \tag{94}$$

**A source computed at rank two suffices for every observed multipole.** The line-of-sight assembly evaluates the source where the hierarchy is slaved, at ranks zero, one and two, and hands it to (94). The kernel performs the transfer to $\ell = 2000$ and beyond. The depth of the hierarchy and the range of the spectrum are therefore unrelated quantities, and extending one does not extend the other.

**A coefficient in the wrong kernel slot lands at the wrong angular scale.** A coefficient placed against $j_\ell$, against $j'_\ell$ or against $j''_\ell$ enters the cascade at a different rank and produces a different angular dependence. §14.1 is an accounting of which coefficient belongs at which slot, and the $\ell$-leverage of the three slots differs: at fixed argument the multipole maximising $|j'_\ell|$ sits about 37% below the one maximising $|j_\ell|$. A term in the wrong slot is therefore not a small error of magnitude but a contribution at the wrong angular scale.

### 23.3 Two weights, two jobs

The transport carries two weights and they are not interchangeable.

| | **what it is** | **what it controls** |
|---|---|---|
| $\mathcal{V} = \kappa' e^{-\kappa}$ | the visibility, sharply peaked at recombination | *when* a source is handed to the kernel |
| $W = e^{-\kappa}$ | the optical-depth transfer, rising to unity and staying there | *what continues to accumulate* along the ray afterwards |

**Scattering-generated sources ride at the visibility and ray-long gravitational sources ride at the transfer.** The Sachs–Wolfe monopole and the Doppler dipole exist because scattering put them there, so they are deposited in the narrow window where $\mathcal{V}$ is non-zero. The integrated Sachs–Wolfe contribution exists because the potentials evolve along the whole ray, so it accumulates wherever $W \simeq 1$, which is the 82% of the conformal span lying after recombination.

**The two weights are related by an identity and are not alternatives.** $W\kappa' = \mathcal{V}$ holds everywhere, so a term already carrying an explicit factor of the scattering rate is at visibility weight whatever it is multiplied by. Reading that identity as a licence to exchange the weights removes the late-time contribution entirely, because $\mathcal{V}$ vanishes once recombination is complete and $W$ does not.

## 23.4 One equation, three regimes, and without matching

Metric-perturbation treatments commonly solve the tight-coupling problem and the free-streaming problem separately and join the two across recombination. *The* $1+3$ *covariant formulation never joins anything.* One hierarchy holds at every epoch, and the three regimes differ only in which of its terms dominates.

Figure 1 draws them along $\eta$, together with the two weights. Its second, third and fourth panels are the three regimes below; its first, the radiation-era evolution of $\Phi$, is upstream of all three and supplies the forcing the oscillator inherits.

| **regime** | **what dominates** | **what the hierarchy becomes** |
|---|---|---|
| $\kappa' \gg k$ | the collision term | the acoustic oscillator, obtained by eliminating the dipole between the monopole and dipole equations, with $c_s^2 = 1/[3(1+R)]$ exactly and the baryon drag $R'/(1+R)$ |
| $\kappa' \sim k$ | neither | recombination. The visibility peaks and the sources are deposited |
| $\kappa' \ll k$ | free streaming | the projection (94) |

**The line-of-sight integral performs the matching exactly, rather than by joining.** Its two weights select, for each source, the regime that source belongs to, and the integral is taken once over the whole span. No boundary conditions are imposed at recombination and no solutions are continued across it, so the questions that attend a junction condition — how sharp it is, which quantities are continuous there, what is discarded in the transition — do not arise. What replaces them is the requirement that each coefficient carry its own weight and its own kernel slot, which is a requirement on the assembly rather than on a boundary.

**The acoustic oscillator and the projection are therefore one calculation and not two.** The oscillator supplies the boundary value $(\delta\tilde{T} + \Phi_A)(\eta_*)$ and the kernel supplies its angular dependence; the phase of the first carries into the position of the features of the second, and the width of the visibility sets how much of that phase survives the deposit. An acoustic feature of order $n$ requires the phase $kr_s$ to be coherent across the deposit window, so successive features are lost from the highest order downwards as that window broadens — a statement about the transport, and one that holds whatever the oscillator does.

**Two widths bound how many acoustic features survive, and neither of them can bind.** The first is the deposit window itself, drawn in Figure 1 and recorded there at $\Delta\eta = 0.00972$ in the code normalisation. The second is the resolution at which the transport is sampled, and it is not one resolution. The oscillator is integrated mode by mode on a grid of its own, and the projection is integrated on the shared conformal grid: different integrands, different code, and no grid in common. Each carries its own rule and each rule is a sampling rate per period of the thing that grid resolves.

| grid | what it resolves | bound | cleared by |
|---|---|---|---|
| oscillator, per mode, 24 samples of $2\pi\sqrt{3}/k$ | $\cos(kr_s)$ | $\pi/(kc_s)$ | 12 |
| projection, `ETA_POINTS_PER_PERIOD`= 12 per zero | $j_\ell(k(\eta_0-\eta))$ | $\pi/k$ | 12 |
| projection, same grid | the *product* | $\pi/[k(1+c_s)]$ | $12/(1+c_s)\simeq 7.61$ |

The first two rows are the rules the two grids were built from, so each is cleared by exactly its own sampling rate at every $k$, by construction and not by measurement. A quantity satisfied by construction cannot be what removes a feature, and naming either as a candidate is circular. Editions before v1.3.20 named the first of them.

The third row is the one neither document names and the only one of the three that is a statement rather than a restatement. The projection integrand is a *product* of the source and the kernel, so it carries their sum: the fastest component anywhere in it runs at $k(1+c_s)$, not at $k$, and the step that resolves it is $\pi/[k(1+c_s)]$. The released grid clears that by 7.61 — the tightest of the three margins and still not tight. All three are independent of $k$, because each step is itself proportional to $1/k$, and a floor on the sample count can only enlarge them. The order $n$ of a feature therefore does not enter either grid: a feature at $k\simeq nk_1$ is sampled exactly as well as the first.

So neither sampling width removes a feature at any order the calculation reaches. Editions before v1.3.20 offered the window and the sampling as alternatives and closed on the window; the sampling is two things rather than one, and the window does not carry the conclusion either.

**The deposit window is measured and it suppresses the second feature by nine parts in a thousand.** The window's effect on a feature is its own Fourier weight at that feature's phase, which needs no assumption about whether $\Delta\eta = 0.00972$ is a width or a standard deviation:

$$\mathcal{C}(k) = \frac{\left|\int \mathcal{V}(\eta)\, e^{\,\mathrm{i}kr_s(\eta)}\,\mathrm{d}\eta\right|}{\int \mathcal{V}(\eta)\,\mathrm{d}\eta}, \tag{95}$$

with $\mathcal{C}=1$ a delta-function deposit. Measured in the implementation stream on `PLANCK_MODEL` with the released history, where $r_s(\eta_*)=0.03682$ and $k_1=\pi/r_s=85.3$:

| feature order $n$ | 1 | 2 | 3 | 4 | 6 | 8 | 12 | 17 |
|---|---|---|---|---|---|---|---|---|
| $\mathcal{C}(k)$ | 0.998 | 0.991 | 0.980 | 0.965 | 0.926 | 0.877 | 0.773 | 0.673 |
| suppression | 0.2% | 0.9% | 2.0% | 3.5% | 7.4% | 12.3% | 22.7% | 32.7% |

Even the seventeenth feature retains two thirds of its coherence. That is a narrow window behaving as a narrow window, and it is not a quantity that removes a feature.

**What removes a feature is structure in $\ell$, and no discretisation measured here has any.** Five candidates have now been scored and all five are exonerated: the projection grid at 0.95 per cent and flat in $\ell$; the oscillator grid at 0.28 per cent, rising with feature order exactly as the sampling argument predicts and topping out there; diffusion damping at 9 per cent at the second peak with $k_D/k_1=6.5$; the recombination grid at 3.4 per cent by $\ell=900$, the largest of the five, and smooth through $\ell=539$ and 811 with no spike; and the deposit window above. Four were dismissed on magnitude and the fifth is the largest and is still not it, which says that magnitude was never the right test. A smooth multiplicative factor tilts an envelope and leaves the peak-to-trough ratio alone by construction, however large it grows, so a feature can only be removed by something carrying structure at that feature's position, and nothing measured in this bundle carries any. The absence of the secondaries is therefore a property of the assembly rather than of any grid, which is where §15 leaves two weightings indistinguishable, and it is open.

# 24 Implications

## 24.1 Coupling growth against the Rees–Sciama bound

Seljak (1996) places the Rees–Sciama contribution at $\Delta T/T \sim 10^{-7}$–$10^{-6}$, peaking at $\ell \sim 100$–$300$, at least an order of magnitude below the COBE-normalised primary signal, and dominant only near $\ell \approx 5000$. The basis is N-body simulation combined with second-order perturbation theory rather than a scaling argument alone. Sanz et al. (1996) reach a compatible bound by second-order perturbation theory in open and flat models, putting the secondary contribution at 0.1–10% of the radiation power spectrum in a flat universe, depending on reionisation history and normalisation.

The exact hierarchy carries couplings explicitly linear in $\ell$. Read directly as power, they would grow where those bounds say the secondary contribution does not, and the apparent conflict is what §17 resolves.

**The couplings are a remapping operator, and the response enters at fourth order.** Expanding the lensed spectrum in the deflection amplitude $L$, with $C_0$, $C_1$ and $C_2$ the unlensed spectrum and its first two derivatives, the flat-sky average of the order-$L^2$ coefficient is $\frac{1}{2}\ell^2 C_0$. The deflection response does not enter there: what survives at $L^2$ is the unlensed spectrum carried through the kernel, not a new term. $C_1$ and $C_2$ first appear at order $L^4$, where the residual is

$$\tfrac{3}{16}\,\ell^2 C_2 + \tfrac{17}{16}\,\ell\, C_1 + C_0, \tag{96}$$

which for a power-law spectrum of index $\nu$ factorises as $(\nu-2)(3\nu-8)/16$ and vanishes at $\nu = 2$ and $\nu = 8/3$. Both statements are asserted in `paper/checks/independent_checks_v1.1.6.py`, section H, in exact rationals.

**What cancels is the deflection response, and variance conservation is not established.** It establishes that the linear source does not produce a term growing as $\ell^2$ beyond the unlensed spectrum, which is what the coupling question asks. It does *not* establish that the remapping conserves the total variance. The gain-and-loss comparison that appeared to establish conservation was withdrawn on 2026-10-21 as a tautology — it compared $AB/2$ against $(B/2)A$ — and the instrument now carries the record in its place: *quartic scaling alone proves neither conservation nor redistribution.* Conservation needs the lensed spectrum's full second moment including the linear–quadratic cross term, which is not computed in this bundle.

## 24.2 The lensing efficiency kernel of a secondary source

A single remapping of the whole sky carries one efficiency kernel, $W(\chi;\chi_*) = 1 - \chi/\chi_*$, the kernel appropriate to a source at last scattering (Lewis and Challinor, 2006). A secondary anisotropy generated at $\chi_s < \chi_*$ — integrated Sachs–Wolfe, Rees–Sciama, Doppler — is lensed only by the matter between the observer and $\chi_s$, so its kernel is $W(\chi;\chi_s) = 1 - \chi/\chi_s$.

The ratio of the two is not a constant:

$$\frac{W(\chi;\chi_*)}{W(\chi;\chi_s)} = \frac{\chi_s(\chi-\chi_*)}{\chi_*(\chi-\chi_s)}, \tag{97}$$

which depends on $\chi$. Two reductions of it are well defined. At the peak of the efficiency $\chi(1-\chi/\chi_s)$, which is $\chi = \chi_s/2$, (97) is exactly $2 - \chi_s/\chi_*$. Weighted along the ray by that same efficiency and integrated over $[0,\chi_s]$, it is $3 - 2\chi_s/\chi_*$.

| source distance | at the efficiency peak | integrated along the ray |
|---|---|---|
| $\chi_s = \chi_*/2$ | 1.50 | 2.00 |
| $\chi_s = \chi_*/3$ | 1.67 | 2.33 |
| $\chi_s \to 0$ | 2.00 | 3.00 |

So a secondary generated halfway to last scattering is over-weighted by 50% at the peak of its own efficiency and by a factor of two when the weighting is carried along the ray. The two figures are not independent, and their ratio is exact: writing $x \equiv \chi_s/\chi_*$, the ray-integrated over-weighting $3-2x$ exceeds the peak over-weighting $2-x$ by $(3-2x)/(2-x)$, which is unity at $x=1$, $4/3$ at $x=\frac{1}{2}$, and rises to $3/2$ as $x \to 0$. *Quoting the peak ratio therefore understates the over-weighting, and understates it by no more than* $3/2$ *however near the source lies.*

**This is a statement about the kernels and not about any solver.** (97) follows from the two definitions and nothing else. Whether a particular Einstein–Boltzmann code inserts the single kernel is a property of that code's implementation, and this bundle does not score it: criterion 5b compares against CAMB and CLASS at the level of the spectrum and does not port either. The bearing is on cross-correlations where a secondary is lensed and the lensing weight is carried explicitly — integrated Sachs–Wolfe to lensing cross-power, and lensing tomography — rather than on the primary acoustic peaks, where $\chi_s = \chi_*$ and (97) is unity.

### 24.3 What the covariant route supplies, and what is not claimed

| | what it supplies | what it does not claim |
|---|---|---|
| Lensing inside the hierarchy | The deflection is obtained within the non-truncated 1+3 covariant Boltzmann hierarchy rather than imported as an external remapping (Lewis and Challinor, 2006) or derived in second-order cosmological perturbation theory (Beneke and Fidler, 2010; Pettinari et al., 2013). | Priority. The claim is that the route differs from the three named above, not that no other covariant derivation exists. |
| The second half of the deflection | The kinematic couplings supply $2GM/b$. The remaining $2GM/b$, which completes the Einstein value $4GM/b$, is carried by the perturbed connection inside the free-streaming operator $e^a \mathrm{D}_a \Pi$ rather than being absent. | That the split is new. What is supplied is the location of the second half in this formulation. |
| Frame covariance without second-order gauge machinery | Stewart–Walker invariance holds at the background level and the threading change is an exact pointwise relabelling of direction phase space, so no second-order gauge-invariant variable is constructed and no coordinate gauge is changed (Bruni et al., 1997). | That the metric route is wrong. The two are different routes to the same invariants; this one has fewer constructions in it. |
| Deflection and aberration as one operator | The bulk gravitational deflection and the observer's local kinematic aberration are the same operator evaluated at different points along the ray. | A measurement. The observer-end evaluation is the aberration dipole convention and is not fitted here. |

Rows one, two and four are statements about the derivation in Paper 1 and are carried by it. Row three is gated in this bundle: `tests/frame/c1` scores the threading invariant equal in both threadings, independent of $k$ and $\eta$, and `tests/frame/c4` and `c5` score the matching coefficients the invariance forces.

# 25 What is open

*Each row records an unresolved physical question and its proposed acceptance test. The supporting history is in* `provenance/OPEN-QUESTIONS-v1.0.0.md`.

| | the question | what would settle it |
|---|---|---|
| **Q5** | Does the full $1+3$ treatment predict excess power at low $\ell$? | A rank-0/2 share that rises at low $\ell$ rather than with it. **F2**'s third panel measures it and finds the share flat at the quadrupole, which parks the question rather than answering it |
| **Q9** | The discretisations are audited; the *physical* ranges are not | A stated validity range per approximation, with the measurement that bounds it |
| **Q10** | Does the second-order quadrupole ring for tensor modes? | The second-order source carried at rank two with a tensor initial condition — recorded as a mechanism, not claimed |
| **Q11** | The acoustic source has the right phase and the spectrum does not | Downgraded from blocker to residual, *not closed*: the oscillator tracks $\cos(kr_s)$ in exact phase and period, so the phase is lost between the source and the transfer |
| **Q12** | Two implementations of (102) agree at $k=1$ and diverge above $k=3$ | Which of the two reproduces the integral (102) is derived from — §14.1 answers this for the bulk and not for the boundary |
| **Q13** | Which construction of $(\kappa'\tilde{v}_B)'$ is right? | The by-parts identity, term by term, against each construction |
| **Q14** | Should $C_0$ enter the line-of-sight integrand at all? | §14.2 settles the $\ell=0$ row; whether the resulting $C_0$ belongs in the bulk is the remaining half |
| **Q15** | **What does omitting helium recombination cost?** The recombination history here integrates the hydrogen Peebles equation. Helium recombines earlier and shifts $x_e$, hence $\kappa'$, $\eta_*$, the visibility width and the damping scale. | $\eta_*$ and $k_D$ recomputed with helium, against the percent-level differences the peak comparison turns on. *Not measured* |
| **Q16** | **Is the tight-coupling source frozen late enough?** $\delta T$ and $\tau_1$ are held at $\eta_{\rm couple}=6\eta_*$, which is correct physics — after decoupling nothing drives the monopole — on the condition that the visibility has closed by then. | $\mathcal{V}(\eta_{\rm couple})/\mathcal{V}(\eta_*)$ on every background. *Nothing currently measures it*, and the recombination history differs between the three |
| **Q17** | **Is the frame-shift coefficient right on the background where it is predicted in closed form?** On Einstein–de Sitter the shift is constant, (51), so the *derivative* comparison is zero on both sides and tests nothing there. | Score $\Delta(a\mathcal{B}_1)$ against $-\frac{1}{3}k\Phi_A$ at the coefficient, not at its derivative, on Einstein–de Sitter. A measured ratio is then $\Phi_A$, which the background already fixes, so the row has an answer before it is run and is a real test for the first time |

**Q16 freezes a pair that has nothing left to drive it, so it cannot move a peak.** What is frozen is the tight-coupling pair, which after decoupling has no scattering to drive it; the line-of-sight integral itself runs to $\eta_0$ on a grid whose density is set by the Bessel kernel's own period in $\eta$ and gated against aliasing — finding T-3. A freeze that was too early would cut source while the visibility was still live, which is a boundary effect and not a projection one.

**One finding that stood beside these is now closed.** T-67 asked whether $(a\mathcal{B}_1)'$ should carry $-a\mathcal{H}\,\delta T - 3a\mathcal{H}\,\Phi_A$ beyond the terms quoted in §14.1. **Refuted by measurement, 2026-10-25**: the terms move $(a\mathcal{B}_1)'$ by 42% and move *both frames identically*, so they are frame-neutral in the difference and cannot restore frame content to a decomposition that lacks it. An earlier form of this paragraph said the reduced form was "unavailable in exactly the regime the low-$\ell$ plateau occupies"; that was a costing

by magnitude, the test it pointed at came back the other way, and it is withdrawn. Every statement of $(a\mathcal{B}_1)'$ in this supplement remains the expansion-terms-off form and says so at its own site.

**And the late ISW is a weight question before it is a physics one.** The effect is already in the formalism: $\Phi'$ enters at rank one through $(a\mathcal{B}_1)'$, hence through the bulk bracket. In any line-of-sight assembly it rides on $W = e^{-\kappa}$ — the photon accumulates $\int \Phi'$ along the whole path, weighted only by the probability it has not scattered — and (35) agrees. (102) carries that bracket at $\mathcal{V}\mathcal{D}$, and $\mathcal{V}$ has closed by the time the late ISW acts. So discrepancy (i) of §14.1 does not merely insert a spurious $\kappa'$; it removes the late ISW. *This is testable on the background axis*: $\Phi$ is constant for Einstein–de Sitter ($K = 0$ dust, Appendix G of the antecedent), so EdS has no late ISW to lose, while ΛCDM and `PLANCK_MODEL` do. Correcting the weight should lift the plateau on the latter two and leave EdS alone.

That test has now run and the first half of it is confirmed. The late-time share of $C_\ell$ — the fraction accumulated after the visibility closes — is 0.0000 in all thirty-six curves: six multipoles, three backgrounds, both frames. Einstein–de Sitter is the control and 0.0000 is correct there; ΛCDM and `PLANCK_MODEL` require $\mathcal{O}(0.1)$ and return the same zero. The accumulation reproduces the released quadrature at $1.15 \times 10^{-12}$, so it is that integral decomposed and not a second definition of it. A fault in the accumulation would annihilate late deposition on every background including the one where zero is correct, and a fault manufacturing it would show on EdS; neither signature is present. The paragraph above was written before the measurement and is confirmed by it.

What remains open is the second half, and it is one run on an existing control: with the bracket moved to $W$, the late share at $\ell = 2$ must become non-zero on both Λ backgrounds *and must stay exactly* 0.0000 *on Einstein–de Sitter*. If it stays zero on the Λ backgrounds, discrepancy (i) is not the mechanism and a second effect shares its signature. That is the discriminator, and it is what turns the reading below from consistent-with into established.

# 26 The almost-EGS bounds

The Ehlers–Geren–Sachs theorem states that exact CMB isotropy for all fundamental observers in an expanding dust universe forces exact Friedmann–Lemaître–Robertson–Walker geometry. The *almost*-EGS theorem of Stoeger, Maartens & Ellis, *Ap. J.* 443, 1 (1995) relaxes that to bounds: limits on the CMB multipoles $\tau_{A_\ell}$ become limits on $\sigma_{ab}$, $\omega_{ab}$, $\dot{u}_a$, $E_{ab}$ and $H_{ab}$.

Those bounds are stated in the *covariant multipoles* — not in mode coefficients — which is what this bundle computes and what a gauge-fixed code cannot supply without a translation. Applying them is therefore natural work for this formalism, and it is v2.0.0 work, because it is a calibration against data rather than an analytical reconstruction.

## 26.1 The bounds require four objects, and three of them are derivatives

The step from a bound on the multipoles to a bound on the geometry is one row of the hierarchy already printed at (4), and it can be written out. The rank-two gravitational source is $\mathcal{B}_2 = \sigma_{ab}$, and $(\ell+1)/(2\ell+3)$ is 3/7 at $\ell = 2$, so that row rearranges to

$$a\sigma_{ab} = \dot{\tau}_{ab} + \tfrac{3}{7}\mathrm{D}^c\tau_{abc} - \mathrm{D}_{\langle a}\tau_{b\rangle} + \kappa'(\tau_{ab} - \mathcal{S}_{ab}). \tag{98}$$

Nothing is approximated between the printed row and (98): the shear has been moved to the left and nothing else has been done. Whatever an almost-EGS argument is in general, in these variables it has to pass through this identity, and the hypotheses it needs can be read off it.

**A bound on the quadrupole alone does not bound the shear.** Four objects stand on the right of (98) and $\tau_{ab}$ is one of them. Of the other three, $\dot{\tau}_{ab}$ is a time derivative at the same rank, $\mathrm{D}^c\tau_{abc}$ is a spatial divergence one rank up, and $\mathrm{D}_{\langle a}\tau_{b\rangle}$ is a spatial gradient one rank down. What the argument needs is therefore a bound on a band of three consecutive ranks together with their first derivatives, and that is a different assumption from a bound on each multipole separately: a sequence can be uniformly small and have large derivatives. This document records no bound on any of the four, and §25 is where that stands.

**The collision residual is largest where the quadrupole is made.** This release takes $\bar{f}_2 = 0$, so $\mathcal{S}_2 = 0$ by approximation A-5, and the final term of (98) is $\kappa'\tau_{ab}$ exactly. $\kappa'$ is not small near last scattering; it is the rate whose exponential integral defines the visibility, and §14.3 places the visibility maximum at $\kappa'' = -\kappa'^2$. The one term carrying no derivative is therefore weighted by the largest rate in the problem over exactly the interval in which $\tau_{ab}$ is generated. A bound stated for the free-streaming era and a bound stated through recombination are different statements, and only the second has to carry $\kappa'\tau_{ab}$ where $\kappa'$ is at its largest.

**$\mathrm{D}_a$ is fixed by the threading, so these hypotheses are not threading-invariant.** Every spatial derivative in (98) is the projected derivative built from $h_{ab} = g_{ab} + u_a u_b$, and $\tau_{A_\ell}$ is a PSTF tensor about the same $u^a$ by (3). Both are properties of a threading in the sense of §9 — a smooth congruence over a region, not a unit vector at an event. Two threadings related by a tilt do not share $h_{ab}$, so they do not share $\mathrm{D}_a$, and the statement that the multipoles and their spatial gradients are small is indexed by the congruence it is made about. An almost-EGS cosmology is not specified until its threading is named. That is what makes the CDM and the Newtonian threadings two settings for one theorem rather than two descriptions of one setting, and §15 is the measurement of how far apart they are.

**The identity inherits one uncertified placement of $a$, and that does not change which objects enter.** (4) combines an overdot with $\kappa'$, and its factors of $a$ are held for audit rather than certified. (98) inherits that, and the inheritance is confined to the coefficients: rescaling a row multiplies its terms and can neither remove one nor introduce one. Which objects an almost-EGS argument must bound is settled by the identity as it stands. How large each contribution is is not, and waits on the same reconciliation.

**The exact hierarchy adds terms to this identity and removes none.** (4) is the reduced transport equation. The exact brightness hierarchy of Paper 1, Eq. (16) carries acceleration, shear and vorticity multiplying the anisotropy multipoles, and this row omits those products; restoring them puts further terms on the right of (98) beside the four already there. Four is therefore a floor on the number of objects to be bounded, and the three statements above are made in the direction that survives the restoration.

**The $H_0$ posteriors are where this would show, and the question is not posable until the threading is named.** In an explicitly defined almost-EGS cosmology — one whose threading is named, whose anisotropy bound is stated with the derivative bounds (98) requires, and whose collision term is declared — can a unified $1+3$ covariant treatment of collisionless matter beyond dust at low redshift and of source-resolved CMB propagation at high redshift produce a measurable shift or broadening of the corresponding $H_0$ posteriors, once both are computed for the same observer congruence and the same observable endpoints? Nothing in this release answers it. It is recorded because the identity above is what would have to be evaluated to begin, and because the question is not well posed until the threading is named, which is the content of this subsection.

### 26.2 The table v2.0.0 fills

**Why the table needs both columns.** The original bounds were derived for a CDM model, not for ΛCDM, and the difference is not cosmetic. The argument converts multipole limits into limits on $\sigma/H$

Table 10: The structure v2.0.0 fills. No cell is computed at v1.0.0, and none is quoted here from memory: observational columns are filled from the mission papers at v2.0.0 under the same rule that equation numbers are resolved from the source.

| **Mission** | $\ell$ **reach** | **bound, standard CDM** | **bound, ΛCDM** |
|---|---|---|---|
| COBE-DMR (1992) | $\lesssim 20$ | *the original* | *not in the original* |
| WMAP (2003–2013) | $\lesssim 1000$ | pending | pending |
| Planck (2013–2018) | $\lesssim 2500$ | pending | pending |

using the background expansion, and $\Lambda$ changes $H(z)$ at low redshift. More sharply, $\Lambda$ adds a source the original did not have: the late integrated Sachs–Wolfe effect contributes to exactly the low-$\ell$ multipoles the COBE bounds rest on. In standard CDM it is absent — $\Phi$ is constant, which this bundle's growth factor reproduces to six figures — and in ΛCDM it is not, $\Phi$ falling to 0.779 of its early value by today. A low-$\ell$ limit therefore constrains a *different combination* in the two models, so a bound quoted for one is not a bound for the other. The CDM column has a published precedent; the ΛCDM column is new.

**Beyond v2.0.0.** The published bounds are first order in the anisotropy parameter. The natural extension is the next order, and this project is placed to make it because v1.5.0 supplies the $O(\varepsilon^2\ell)$ couplings a second-order argument needs and no external code carries them. Stated as a direction and not a plan: it is gated on v1.5.0, and the correction's sign and size are unknown until it is done. It is recorded so the v2.0.0 table is built in a shape that can take it. See `provenance/ALMOST-EGS-BOUNDS-v1.md`.

## 27 Source-Distance Lensing Biases of Secondary CMB Anisotropies

Lensing of the microwave background is ordinarily applied as a single remapping of the observed sky, in which the entire unlensed temperature field is displaced by the gradient of one lensing potential, and that potential is the one appropriate to a source lying on the last-scattering surface (Lewis and Challinor, 2006). For the primary anisotropy the construction is exact up to the thickness of the visibility, because the primary is generated in a shell whose width is small against the distance to it. For anything generated after that shell it is not exact, and what is lost is not a small correction to the primary calculation but a quantity the primary calculation has no slot for. A secondary contribution arising at a comoving distance $\chi_s < \chi_*$ — the late integrated Sachs–Wolfe effect, the Rees–Sciama term, the Doppler contribution from reionised gas — can be deflected only by matter lying between the observer and its own source, so the potential that lenses it is

$$\psi(\hat{n};\chi_s) = -2\int_0^{\chi_s} \mathrm{d}\chi \left(\frac{1}{\chi} - \frac{1}{\chi_s}\right) \Phi_{\mathrm{L}}(\chi\hat{n}, \eta_0 - \chi), \tag{99}$$

in which $\Phi_{\mathrm{L}}$ is the lensing source built from the combination $\Phi_A - \Phi_H$ shown frame-invariant in §9, and the source distance enters the upper limit and the kernel alike. The overall normalisation of $\Phi_{\mathrm{L}}$ is common to every potential written in this section and cancels from the comparison that follows, which is a statement about kernels and not about amplitudes.

Writing the same expression with $\chi_*$ in place of $\chi_s$ and subtracting isolates exactly what a single-kernel treatment adds to a secondary, and the excess separates into two terms of different character:

$$\psi(\hat{n};\chi_*) - \psi(\hat{n};\chi_s) = -2\left(\frac{1}{\chi_s} - \frac{1}{\chi_*}\right)\int_0^{\chi_s} \mathrm{d}\chi\, \Phi_{\mathrm{L}} \;-\; 2\int_{\chi_s}^{\chi_*} \mathrm{d}\chi \left(\frac{1}{\chi} - \frac{1}{\chi_*}\right) \Phi_{\mathrm{L}}. \tag{100}$$

The first term re-weights the deflectors that do lie in front of the source, and it does so by an amount constant along the ray, the two kernels differing by $\chi_s^{-1} - \chi_*^{-1}$ at every $\chi$ in $(0, \chi_s)$. The second term is the whole contribution of an interval in which no deflector can lens this source at all, every point of it

lying behind the source. Both excess kernels are positive throughout their ranges for every $\chi_s < \chi_*$, so a single-kernel treatment weights every deflector in front of a secondary more heavily than the correct kernel does and then adds a range of deflectors behind it; the departure has a definite direction and is not a scatter about the right answer. Its size is the ratio obtained in §24.2: carried along the ray it is $3 - 2\chi_s/\chi_*$, unity for a source at last scattering and rising to three in the limit of a source at the observer. Both terms vanish identically at $\chi_s = \chi_*$, which is why the acoustic peaks are untouched by any of this.

What makes the bias structural rather than a defect of implementation is where the source distance is lost. A line-of-sight solution delivers the transfer function as one integral over conformal time against the radial kernel, and once that integral has been performed the contributions have been summed and the distance at which each was generated is not recoverable from the result. The spectrum handed to a lensing convolution is therefore already a sum over source distances, and a convolution applied to a sum cannot distinguish its terms; a single lensing-potential spectrum is then the only object the convolution can be given. Recovering what was lost requires the integrand partitioned by source distance before it is squared, together with a lensing operation carrying a potential spectrum, or a set of cross-spectra, for each partition. That is a rearrangement of the order in which the assembly is performed, not a correction applied at the end of it, and it is the same non-commutativity between source integration and propagation recorded in §2. The bearing is on observables in which a secondary and the lensing field are correlated deliberately — integrated Sachs–Wolfe to lensing cross-power, and lensing tomography — rather than on the primary spectrum. Nothing in this release measures the size of the effect in any such observable, and nothing in this release is arranged to.

# Part III

# Qualification evidence

## 28 Parameter and numerical evidence

Table 11 lists every number the numerics use. Its *provenance* column distinguishes values taken from the source paper, from standard values that are not, from values derived here, and from values stipulated for this release. The distinction matters: $\Omega_\gamma h^2$ is a standard value and not one the antecedent supplies, and it enters only through $R(\eta_*)$ and hence the sound speed.

Table 11: Every number the numerics use. *Provenance* distinguishes values taken from the source paper, standard values that are not, values derived here, and values stipulated for this release.

| Symbol | Code | Value | Provenance | Set in |
|---|---|---|---|---|
| $k_{\mathrm{com}}$ | `K_COM` | 1.0 | stipulated | `scripts/figure_f1_appendix_f.py` |
| $\ell_{\max}$ | `ELL_MAX` | 80 | stipulated | `scripts/figure_f1_appendix_f.py` |
| $\ell_{\mathrm{compared}}$ | `ELL_COMPARED` | 25 | stipulated | `scripts/figure_f1_appendix_f.py` |
| rtol, atol | `rtol, atol` | $10^{-10}$, $10^{-12}$ | stipulated | `functions/harmonics/free_streaming.py` |
| $z_*$ | `Z_STAR` | 1100 | standard | `scripts/derive_ell_range.py` |
| $h$ | `H` | 0.5 | AnnalsII §8.3.2 | `scripts/derive_ell_range.py` |
| $\Omega_B$ | `OMEGA_B` | 0.05 | AnnalsII §8.3.2 | `scripts/derive_ell_range.py` |
| $\Omega_\gamma h^2$ | `OMEGA_GAMMA_H2` | $2.47 \times 10^{-5}$ | standard | `scripts/derive_ell_range.py` |
| tolerance | `ADOPTED_TOLERANCE` | 5% | derived | `scripts/derive_ell_range.py` |
| acceptance threshold | `--` | $10^{-11}$ | stipulated | `scripts/figure_f1_appendix_f.py` |

### 28.1 The three backgrounds, side by side

Table 12 is the one place every derived scale of the three backgrounds can be read against the others. It exists because a question the PI asked — *is the difference between two panels a parameter difference?* — had no cheap answer while the models were defined in three places and compared nowhere.

The last three rows distinguish peak-placement errors from other spectral-shape discrepancies. Two models whose spectra differ *in the position of the peaks* differ in $\ell_A = \pi r_\theta / r_s$. Two whose peaks sit at the same $\ell_A$ and still look different do not differ by a parameter, and the difference is in `provenance/BUNDLE-DEFECTS-v1.md`.

Two rows carry a known defect and say so rather than being quietly correct for a different question. $\Omega_r = 0$ is the *control* on Einstein–de Sitter, which is what makes that background the one where A-3 and $T(k)$ are identically inapplicable — and on LCDM it is neither a control nor intended, worth 37% in $\ell_A$ through $\eta_*$ at 1.571× and $r_s$ at 1.610× (findings T-81c, T-83). $k_D$ is the $R = 0$ value, because `diffusion_scale` is called with $R$ zeroed while the oscillator beside it runs at $R(\eta_*) \simeq 0.61$ — finding T-21, measured at $\ell_D = 1783$ against 1985.

Table 12: The three backgrounds side by side, with every derived scale. **Generated by `scripts/make_model_table.py`** from the `Background` objects the code integrates, not transcribed. The last three rows are the ones that answer a shape question: two models whose peaks differ in position differ in $\ell_A$, and two whose peaks sit at the same $\ell_A$ and still differ do not differ by a parameter. $\Omega_r = 0$ is the control on CDM and finding **T-81c** on LCDM; $k_D$ is the $R = 0$ value, finding **T-21**.

| | CDM (Einstein–de Sitter) | LCDM | PLANCK |
|---|---|---|---|
| $\Omega_m$ | 1 | 0.3 | 0.313772 |
| $\Omega_\Lambda$ | 0 | 0.7 | 0.686173 |
| $\Omega_r$ | 0 | 0 | $5.4502 \times 10^{-5}$ |
| $\omega_b h^2$ | 0.0224 | 0.0224 | 0.02237 |
| $h$ | 0.5 | 0.7 | 0.6736 |
| $n_s$ | 1 | 1 | 0.9649 |
| $\eta_0$ | 2 | 3.30508 | 3.19871 |
| $\eta_*$ | 0.0595884 | 0.109581 | 0.0697426 |
| $a_*$ | $8.8769 \times 10^{-4}$ | $9.0059 \times 10^{-4}$ | $8.9643 \times 10^{-4}$ |
| $z_*$ | 1125.51 | 1109.38 | 1114.54 |
| $R(\eta_*)$ | 0.603776 | 0.612548 | 0.608901 |
| $r_s$ | 0.0316406 | 0.0581262 | 0.0361159 |
| $r_\theta$ | 1.94041 | 3.1955 | 3.12897 |
| $\ell_A = \pi r_\theta / r_s$ | 192.663 | 172.71 | 272.177 |
| $k_D$ (at $R = 0$, T-21) | 754.639 | 483.11 | 536.866 |

## 29 Numerical resolution and reproducibility

The hierarchy is integrated with an explicit Runge–Kutta method of order 8 (`DOP853`) at relative tolerance $10^{-10}$ and absolute tolerance $10^{-12}$, truncated at $\ell_{\max} = 80$ with comparisons restricted to $\ell \leq 25$ so that reflection from the truncation does not contaminate them. The acceptance threshold used in the figures, $10^{-11}$, sits an order above the integrator floor.

The release route is

```
python scripts/run_all.py --strict
```

which regenerates every output, figure and table, recomputes the SHA-256 manifests, and reports any artefact that differs from the recorded hash.

Three manifests serve distinct purposes; the release fingerprint is binding. `PORTABLE-MANIFEST-SHA256.txt` is the release fingerprint: it *excludes the renderings* — everything under `figures/` and `supplementary-materials/`, and every `.png` and `.pdf` — and canonicalises numeric output to nine significant figures, so a machine that did not generate the artefacts can still satisfy it. A mismatch there is release-blocking. `FILE-MANIFEST-SHA256.txt` is then a *within-machine* drift check, reported by artefact class, where only an `exact`-class mismatch blocks. `IMMUTABLE-MANIFEST-SHA256.txt` covers frozen reference material, where any change is release-blocking. The corrections record is append-guarded rather than frozen, because it must take new findings: appending is permitted and rewriting recorded bytes is not.

**Why nine PNGs always report as drifted.** Every `.png` written into a working copy through the Claude desktop bridge gains a 5 770-byte `caBX` provenance chunk, inserted immediately after `IHDR` by that application and not by anything in this bundle. The pixels are bit-identical — `IHDR`, `pHYs` and every `IDAT` chunk match exactly — and the `.pdf` of the same figure round-trips byte-for-byte, as does every `.csv` and every source file. These nine therefore sit in the `rendered` class, are reported and do not

block, which is the rule finding T-15 wrote for a different cause with the same shape: fonts, freetype and TEX Live versions, where *a PNG that differs in hinting carries no scientific difference*. The PDF is the authoritative artefact of every figure in this bundle; the PNG is a convenience copy. Finding T-36. The drift is explained here rather than silenced by canonicalising PNGs, because a manifest that stopped reporting a class of change for tidiness would stop reporting real ones too.

Weights are carried as exact rationals rather than floating-point numbers wherever a sign or an $\ell$-weight is asserted.

## 30 Interpretive limits

This is an analytical reproducibility bundle for the associated paper and its supplement. It is not an empirical calibration, not a parameter-estimation pipeline, and not a Boltzmann code for general use. No new effect is claimed and none is reported.

*Annals II*'s own §8.3.2 model is standard CDM — $h = 0.5$, $\Omega_B = 0.05$, flat and matter dominated — and the $C_\ell$ it prints there is a closed-form Sachs–Wolfe result on large scales with the transfer function set to unity. This release goes past that in one respect and one only: it computes the spectrum from the hierarchy rather than from the closed form, and it computes it for the ΛCDM model as well, which *Annals II* does not treat. The marginal cost is the expansion law and the late-time integrated term; every other ingredient — recombination, tight coupling, diffusion damping — is already required by the CDM case and is not additional. No new *physics* is introduced by Λ, and none is claimed.

The bundle does not calibrate to observations. It uses external CAMB and CLASS reference spectra, as described in the scientific scope and in Figure F9. These comparisons test the reconstruction and do not constitute a fit. Parameter estimation and calibration against data remain outside this release.

The nonlinear transport term $\delta\tau_{NL} \propto \Phi^2$ is the lensing and aberration coupling identified in Paper 1. Its contribution to the power is of fourth order in the potential under the statistical assumptions used there. Quartic order alone does not establish conservation of ensemble variance: that conclusion follows for a statistically homogeneous source independent of the deflection, when the complete remapping terms are kept (Paper 1, Sec. VII, especially Eqs. (88)–(90)). The contribution redistributes angular power under those assumptions.

## AI-use disclosure

AI-assisted tools were used for editorial review, LaTeX consolidation and limited algebraic checks. These tools are not sources of physical or mathematical authority. The derivations, numerical results and scientific interpretations require the independent tests specified in this supplement; the author retains responsibility for the scientific content.

## References


Tim Gebbie. Frame covariance of CMBR lensing in the exact 1+3 covariant radiation hierarchy, 2026. URL `https://arxiv.org/abs/astro-ph/9912072v4`. Paper 1, version 4; version 3 is the correction, and versions 1–2 are historical papers.

Tim Gebbie and George F. R. Ellis. 1+3 covariant cosmic microwave background anisotropies I: Algebraic relations for mode and multipole expansions. *Annals of Physics*, 282:285–320, 2000. doi: 10.1006/aphy.2000.6033.

Tim Gebbie, Peter K. S. Dunsby, and George F. R. Ellis. 1+3 covariant cosmic microwave background anisotropies II: The almost friedmann–lemaître model. *Annals of Physics*, 282:321–394, 2000. doi: 10.1006/aphy.2000.6034. Citations to equations use the published journal numbering.

Roy Maartens, Tim Gebbie, and George F. R. Ellis. Cosmic microwave background anisotropies: Nonlinear dynamics. *Physical Review D*, 59:083506, 1999. doi: 10.1103/PhysRevD.59.083506. Equation references use the published journal version.

R. K. Sachs and A. M. Wolfe. Perturbations of a cosmological model and angular variations of the microwave background. *Astrophysical Journal*, 147:73–90, 1967. doi: 10.1086/148982.

Uroš Seljak and Matias Zaldarriaga. A line-of-sight approach to cosmic microwave background anisotropies. *Astrophysical Journal*, 469:437–444, 1996. doi: 10.1086/177793.

Uroš Seljak. Rees-Sciama effect in a cold dark matter universe. *Astrophysical Journal*, 460:549–555, 1996. doi: 10.1086/176991. arXiv:astro-ph/9506048.

José L. Sanz, Enrique Martínez-González, Laura Cayón, Joseph Silk, and Naoshi Sugiyama. CMB radiation power spectrum in CDM open universes up to second order perturbations. arXiv:astro-ph/9602148, 1996.

Antony Lewis and Anthony Challinor. Weak gravitational lensing of the CMB. *Physics Reports*, 429: 1–65, 2006. doi: 10.1016/j.physrep.2006.03.002. arXiv:astro-ph/0601594.

Marco Bruni, Sabino Matarrese, Silvia Mollerach, and Sebastiano Sonego. Perturbations of spacetime: gauge transformations and gauge invariance at second order and beyond. *Classical and Quantum Gravity*, 14:2585–2606, 1997. doi: 10.1088/0264-9381/14/9/014. arXiv:gr-qc/9609040.

Martin Beneke and Christian Fidler. Boltzmann hierarchy for the cosmic microwave background at second order including photon polarization. *Physical Review D*, 82:063509, 2010. doi: 10.1103/PhysRevD.82.063509. arXiv:1003.1834.

Guido W. Pettinari, Christian Fidler, Robert Crittenden, Kazuya Koyama, and David Wands. The intrinsic bispectrum of the cosmic microwave background. *Journal of Cosmology and Astroparticle Physics*, 2013(04):003, 2013. doi: 10.1088/1475-7516/2013/04/003. arXiv:1302.0832.